Ministry Approved
2013

National Textbook Award

Compulsory Education Textbook

Mathematics

SHUXUE

Grade Four Volume Two

北京师范大学出版社

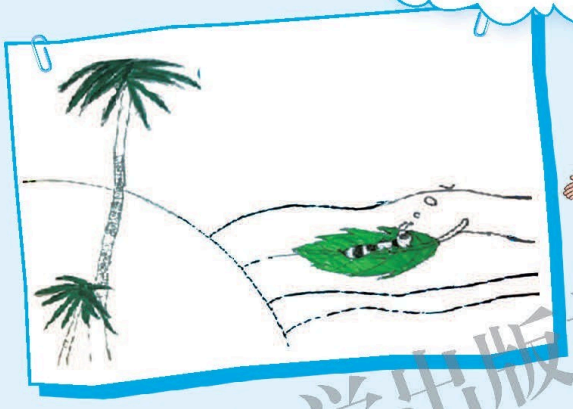
Beijing Normal University Press
BEIJING NORMAL UNIVERSITY PRESS

Dear children,

Learning mathematics is like setting out on a magical journey of exploration. Let's go!

Wise Old Man

An ant is so light, only 0.1 grams! My weight is 35.35 kilograms. We really do need decimals in everyday life.



Taoqi

The Kingdom of Maths even has expressions with letters!

Xiaoxiao

What fun! Woof woof!

Clever Dog

Children, mathematics is everywhere in life, and it can help us solve many problems. Let's experience the joy of learning mathematics together!

Your friends, the editors

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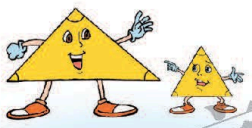

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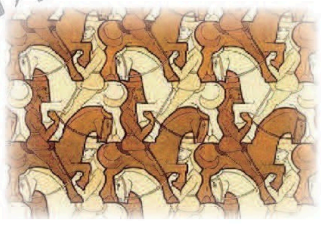

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car	正正正正正
bicycle	正正正正正正正正正正
bus	正正
truck	正正正正

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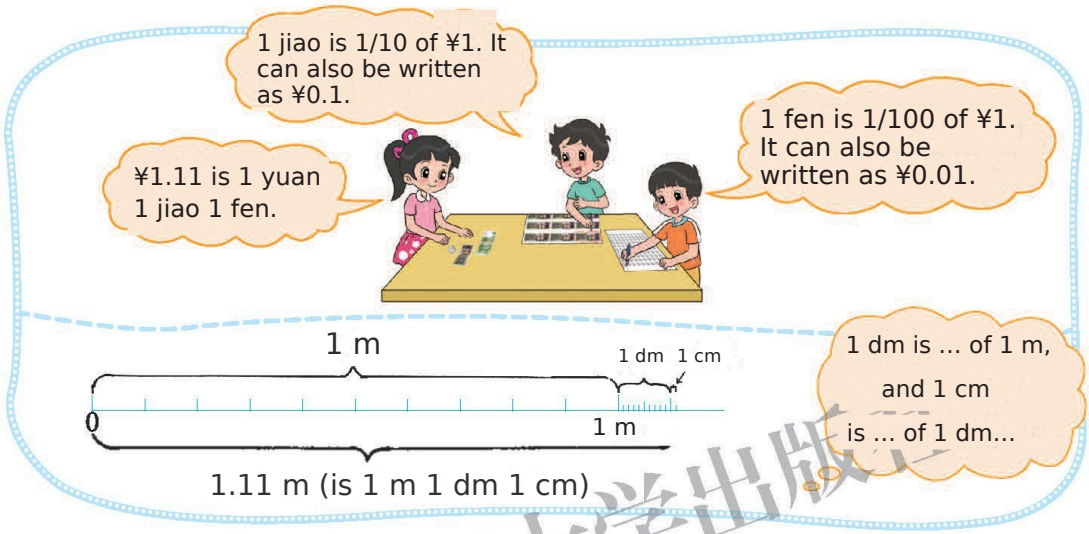
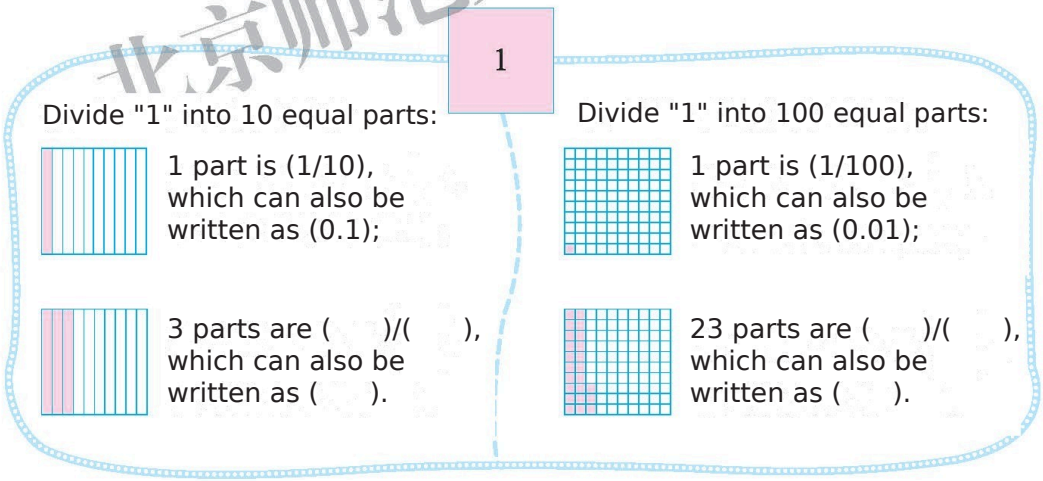
Unit 1 The Meaning of Decimals, Addition and Subtraction

Meaning of Decimals (1)

Use Appendix 1: try it and say what ¥1.11 means. What about 1.11 m?



Write each shaded part as a number.



Think and fill in.

Divide "1" into 1000 equal parts. 1 part is ( )/( ), which can also be written as ( ); 59 parts are ( )/( ), which can also be written as ( ).

Find decimals in everyday life and talk about them with your partner.

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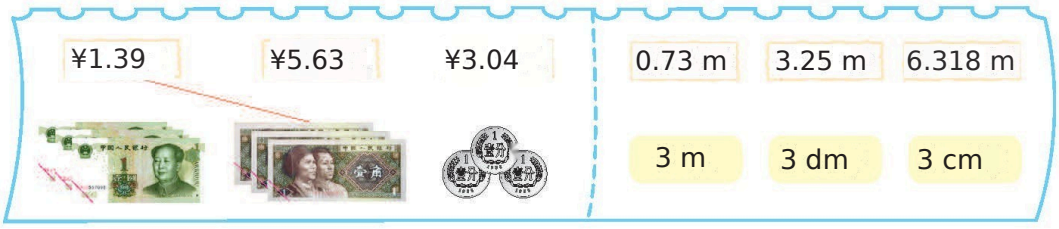
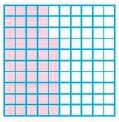
Practice

1. First say what "3" means in each number below, then draw lines to match.

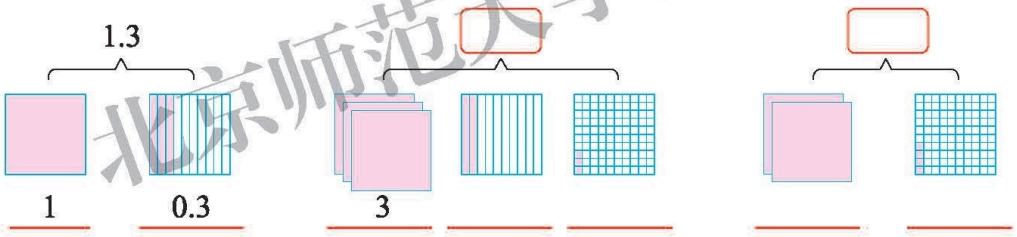


2. Think and fill in.

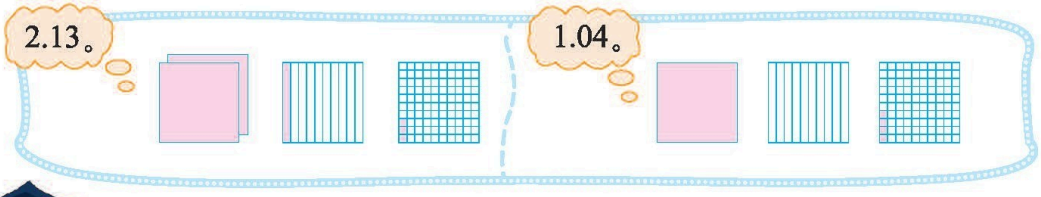
(1)
7 parts are $\frac{(\quad)}{(\quad)}$, also written as ( )

47 parts are $\frac{(\quad)}{(\quad)}$, also written as ( )

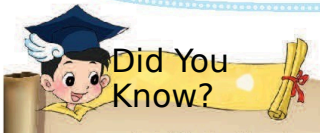
(2) Divide "1" into 1000 equal parts. 61 parts are $\frac{(\quad)}{(\quad)}$, which can also be written as ( )

3. Write each shaded part as a number.



4. Cut out the figures in Appendix 2 and colour them.





Liu Hui wrote in his Commentary on The Nine Chapters on the Mathematical Art in 263 AD: "Tiny numbers with no name are taken as the numerator; one step down uses ten as the denominator, another step down uses a hundred as the denominator; the further down, the finer the division." Roughly: one place below the ones place is one tenth, two places below is one hundredth. The more places down, the finer the division. Here "tiny numbers" means "decimals".

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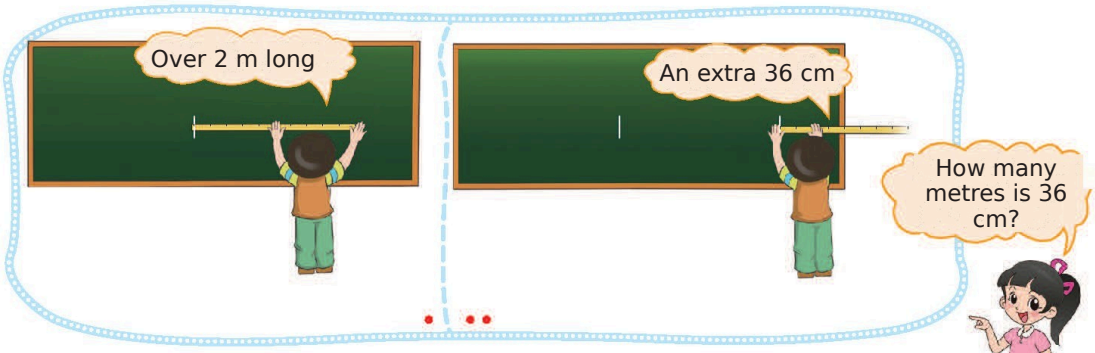
Meaning of Decimals (2)

How long is the blackboard?



1 cm

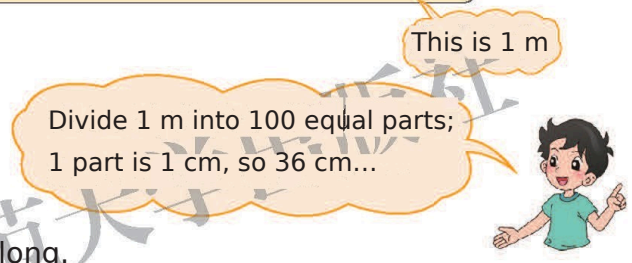
$\frac{1}{100}$ m

0.01 m

36 cm

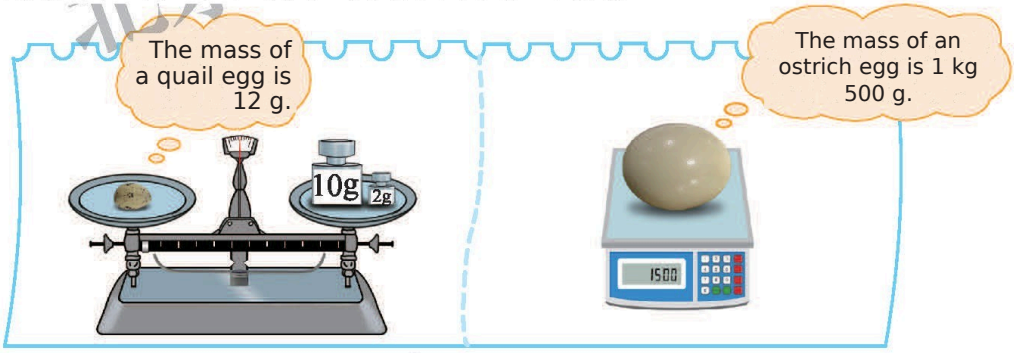
$\frac{36}{100}$ m

0.36 m



The blackboard is ( ) m long.

What is the mass of each egg in kilograms?



1 kg =1000 g, 1 g = $\frac{1}{1000}$ kg =0.001 kg.

12 g = $\frac{(\quad)}{(\quad)}$ kg = ( ) kg,

500 g = $\frac{(\quad)}{(\quad)}$ kg = ( ) kg.

The mass of the quail egg is ( ) kg, and the mass of the ostrich egg is ( ) kg.

Practice

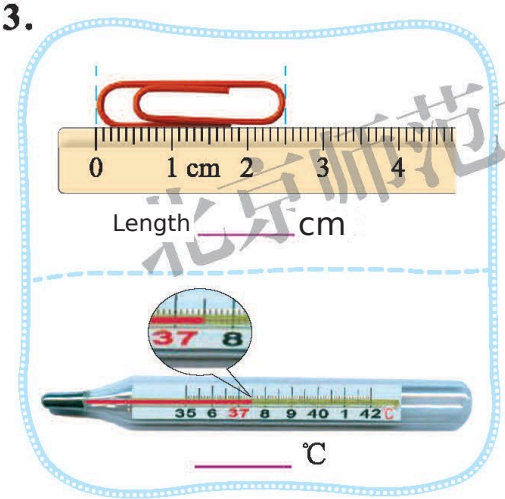
1. Measure and fill in.

Name		Measurement	In metres	Name		Measurement	In metres
Storybook cover	l	23 cm		Classroom door (back)	l	2 m 3 dm	
	w	17 cm			w	80 cm	
Blackboard	l	2 m 85 cm					
	w	1 m 1 dm					

2. A frigate bird weighs about 1 kg 600 g, has a wingspan of 2 m 1 dm, and its bones weigh 113 g. How do you write these as decimals?

1 kg 600 g = ( ) kg
2 m 1 dm = ( ) m
113 g = ( ) kg

3.



Baby Growth Record Card

At birth:
Height 51 cm, weight 3 kg 600 g.
……
At age 1:
Height 79 cm, weight 10 kg 500 g.
……

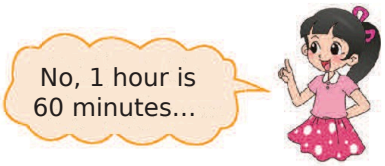
	At birth	Age 1
Height/m		
Weight/kg		

4.

52 cm = ( ) m 5.6 m = ( ) m ( ) dm

5 yuan 6 jiao 7 fen = ¥( ) ¥1.35 = ( ) yuan ( ) jiao ( ) fen

5. How many minutes are there in 0.3 hours? Think about it.



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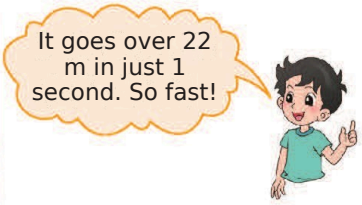
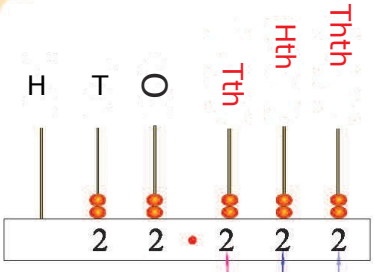
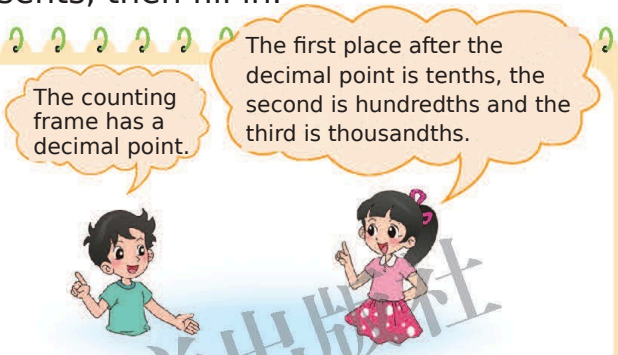
Meaning of Decimals (3)

The top speed of a Beijing Subway Line 10 train is 80 km/h, about 22.222 m/s.



Move the beads on the counting frame to show 22.222, and say what each "2" represents, then fill in.





means 2 × 1/10, or 2 × 0.1.

means 2 × 1/100, or ( ) × ( ).

means ( ) × ( )/( ), or ( ) × ( ).

Let's learn.

	Whole-number part					Decimal point	Decimal part				
Place	……	TTh	Th	H	T	O	•	Tth	Hth	Thth	TTth	……
Counting unit	……	10000	1000	100	10	1		1/10	1/100	1/1000	1/10000	……

The counting units of the decimal part are:

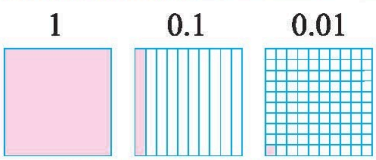
$\frac{1}{10}$, $\frac{1}{100}$, $\frac{1}{1000}$, …

They can also be written as 0.1, 0.01, 0.001, …

It's "carry 1 when you reach 10" too!

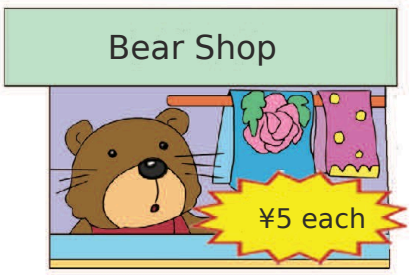
Decimal counting units also follow "carry 1 when you reach 10". Look and discuss.

10 × ¥0.1 is ¥1. 10 × ¥0.01 is ¥0.1.

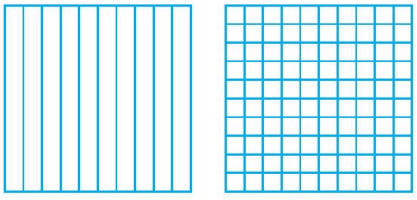


Try It

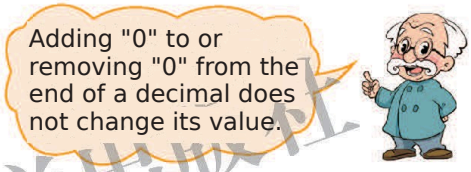
Are the prices of the towels in the two shops below the same?



Colour in. What do you notice?

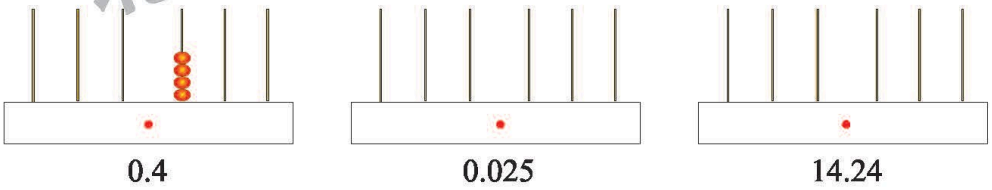


0.6 0.60



Practice

1. Draw on the counting frames, then fill in.

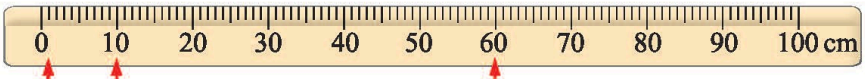


0.4 0.025 14.24

(1) There are ( ) 0.1s in 0.4, and ( ) 0.001s in 0.025.

(2) In 14.24, the "4" in the ones place means ( ) × ( ), and the "4" in the hundredths place means ( ) × ( ).

2. Look, fill in, and say.



$\frac{1}{100}$ m ( ) m ( ) m

0.01 m ( ) m ( ) m

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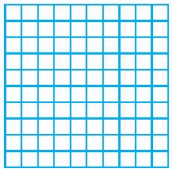
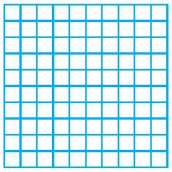
3. Which "0"s below can be removed without changing the value? Colour first, then share with your partner.



0.03

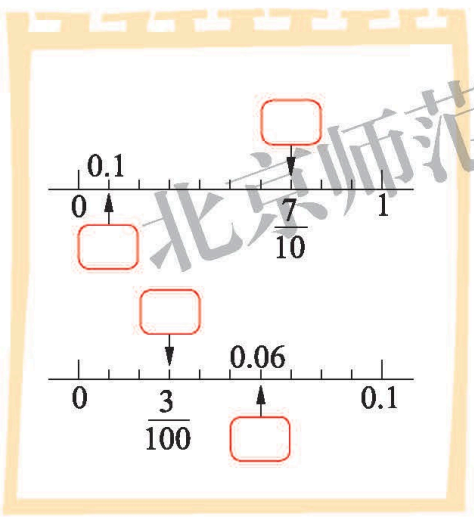


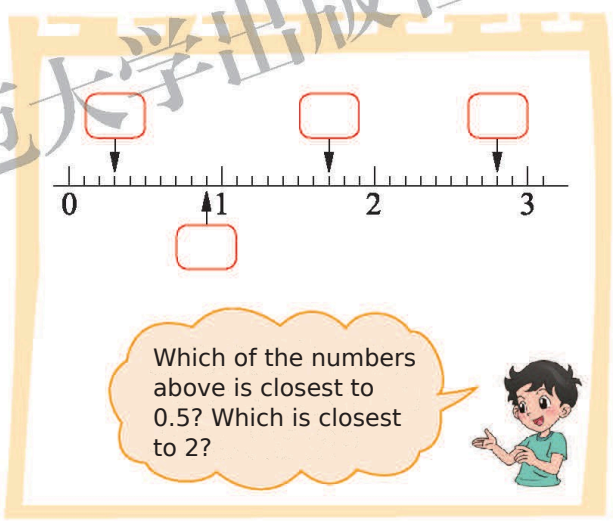
0.30

4. Match the equal numbers.

3.90	42.040	34.0800	0.0600	3.20
3.200	0.06	3.09	42.04	34.8

5. Fill ☐ with a suitable fraction or decimal.





6. Try to complete the calculations below.

0.1×10	0.01×10	0.001×10
0.1×100	0.01×100	0.001×100
0.1×1000	0.01×1000	0.001×1000

What do you notice?

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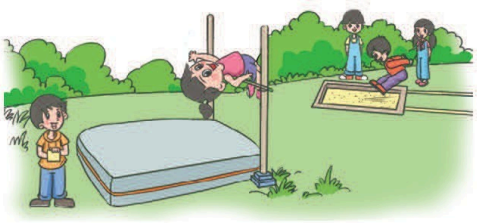
Compare Numbers



High jump

Name	Result (m)
Wang Hong	0.69
Li Na	0.8

Long jump

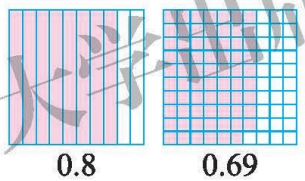
Name	Result (m)
Zheng Qiang	2.97
Li Ming	3.13
Zhang Hua	3.08

Who jumped higher?

0.69 m is 6 dm 9 cm,
0.8 m is 8 dm,
so 0.8 > 0.69.

0.69 is 69 × 0.01,
0.8 is 80 × 0.01,
so 0.8 > 0.69.

Draw a picture to see 0.8 > 0.69.



Li Na jumped higher.

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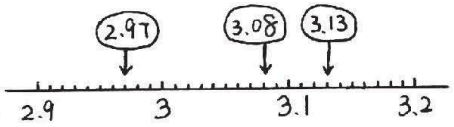
Who jumped the farthest? Do you understand the method below?

To compare whole numbers, first look at the highest place...

O		Tth	Hth
2		9	7
3	.	1	3
3		0	8



Discuss how to compare decimals.

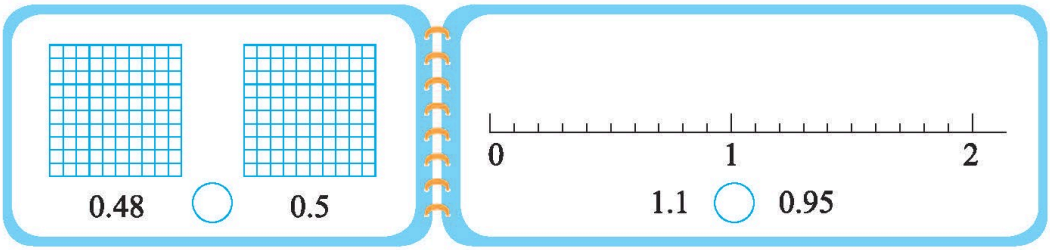
Look at the whole-number part first...

When comparing the decimal part, first compare the digits in the tenths place, then compare...

Practice

1. At different stages of growing up, who is taller, Qisi or Taoqi? Colour, fill in, then share your ideas with your partner.

	Qisi	Taoqi
At birth/m	0.48	0.5
At age 3/m	1.1	0.95



2. Compare.

0.839 ◯ 0.9　　6.07 ◯ 6.7　　5.45 ◯ 5.63

4.03 ◯ 4.009　　7.217 ◯ 7.22　　7.19 ◯ 71.9

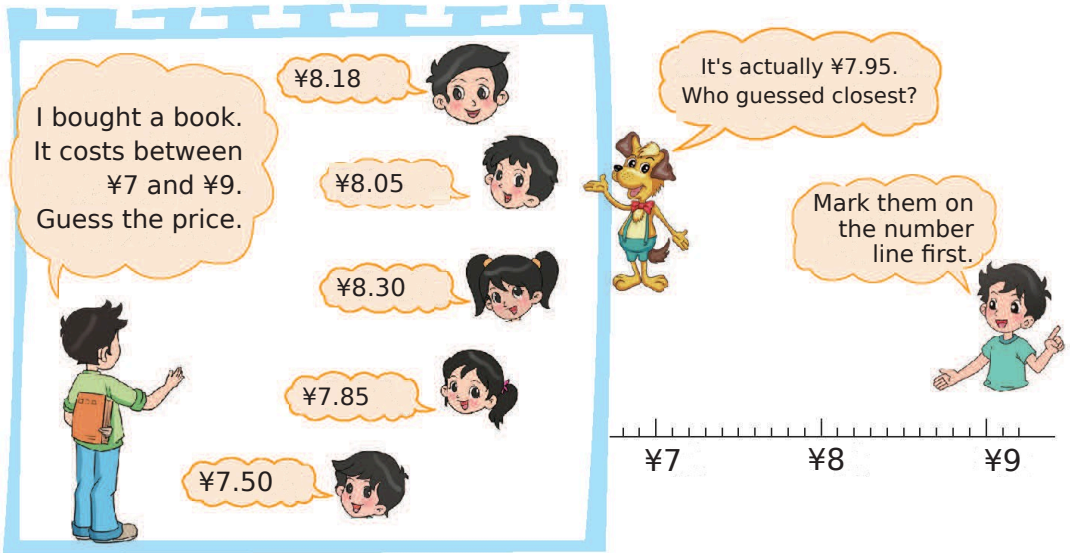
3.

In □ write digits to meet each condition:

(1) The largest possible number is (　　).

(2) The number closest to 31 is (　　).

4. Maths game.



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Buying Food

Estimate: about how much should I pay?

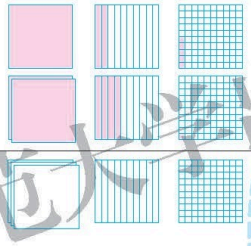
The salesperson charged ¥3.66. Is that right? Draw, calculate and explain.

$1.25+2.41=\square$ ( )

¥1 + ¥2 = ☐

¥0.25 + ¥0.41 = ☐

¥1.25 + ¥2.41 = ☐



	O	Tth	Hth
	1	2	5
+	2	4	1
	☐	. ☐	6

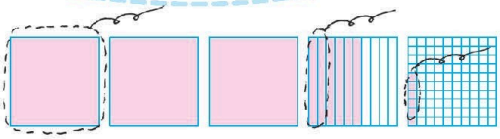
5 × 0.01 plus 1 × 0.01 is...

$3.66-1.25=\square$ ( )

¥3 − ¥1 = ☐

¥0.66 − ¥0.25 = ☐

¥3.66 − ¥1.25 = ☐



	O	Tth	Hth
	3	. 6	6
−	1	. 2	5
	☐	. ☐	1

6 × 0.01 minus 5 × 0.01 is...

Use the column method to calculate, and say how to add and subtract decimals.

$1.25+0.74$

$2.76-2.1$

Add or subtract the digits in the same place.

Just line up the decimal points; the rest is the same as adding and subtracting whole numbers.

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Practice

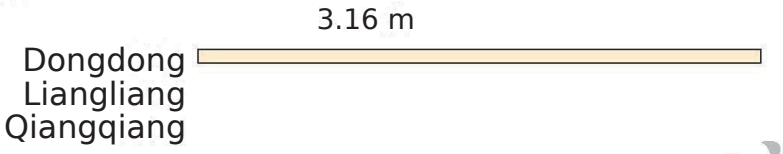
1. The table shows Taoqi's family's water bills for two months.

Month	Water bill (¥)
7	25.2
8	21.6

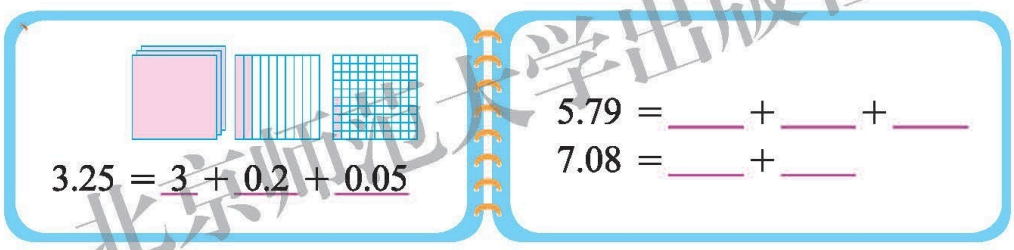
(1) Estimate: is ¥50 enough for the water bills? Share your ideas with a partner.

(2) What is the total water bill for the two months?

2. The students have a long jump competition. Dongdong jumped 3.16 m, Liangliang jumped 0.23 m more than Dongdong, and Qiangqiang jumped 0.12 m less than Liangliang. How far did Qiangqiang jump? Complete the diagram below, then calculate.



3. Fill in like the example.



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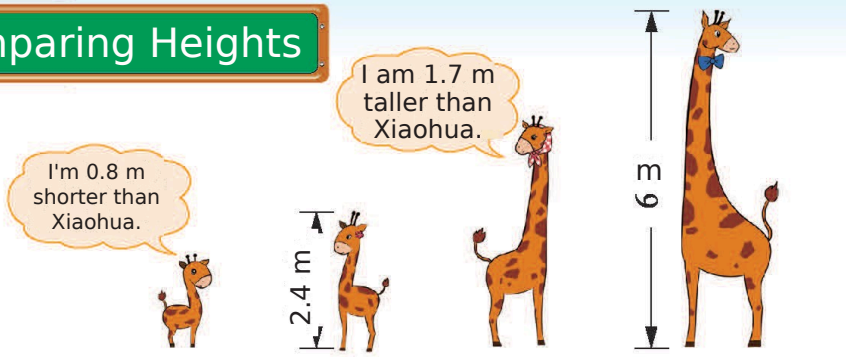
4.

$4.8-2.8$	$1.3+2.5$	$3.1+2.22$
$26.2-13.1$	$12.3+11.2$	$35.82-13.3$

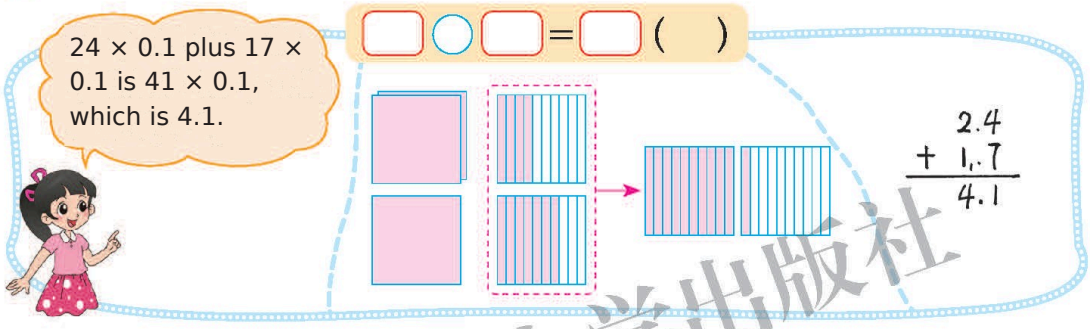
5. Liangliang is 1.45 m tall. Standing on a stool 0.4 m high, he is 0.05 m taller than his father. How tall is his father in metres?

6. A box of apples (including the box) has a mass of 46.6 kg. After half of the apples are taken out, the mass is 24.3 kg. What is the mass of all the apples? What is the mass of the box?

Comparing Heights

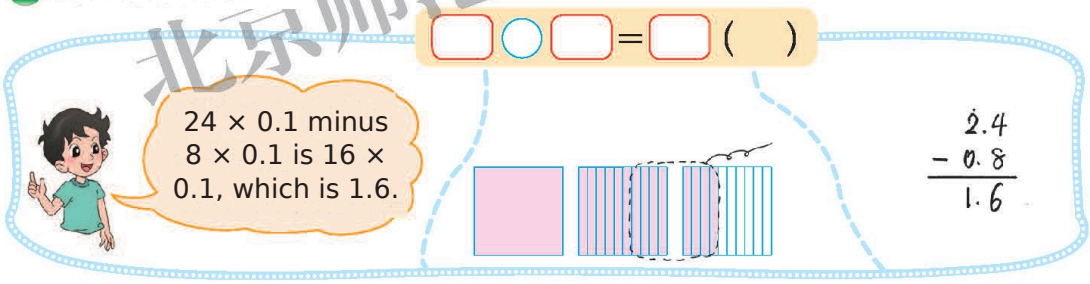


How tall is Mother Giraffe?



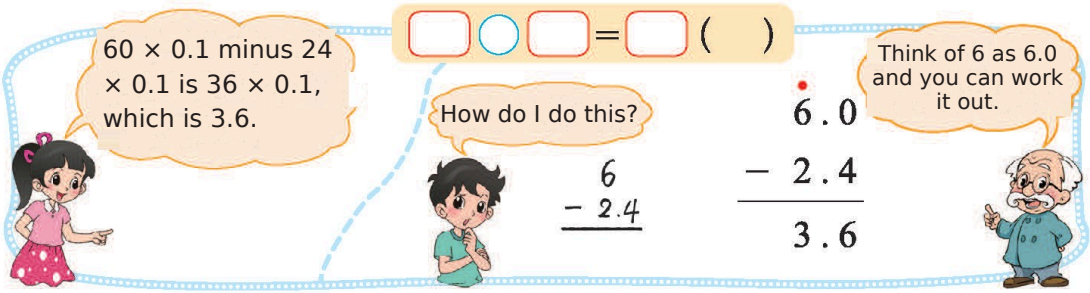
Answer: _____。

How tall is Xiaohei?



Answer: _____。

How much taller is Father Giraffe than Xiaohua?



Answer: _____。

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Try It

Weighing

(1) How many kg does Xiaoxiao weigh this time?

(2) How many kg did Taoqi weigh last time?

(3) How many kg did Clever Dog gain?

Think: what is wrong with the calculations below?

$$\begin{array}{r} 2.25 \\ +\ 7.3 \\ \hline 2.98 \end{array}$$

$$\begin{array}{r} 13.6 \\ -\ \ 5 \\ \hline 13.1 \end{array}$$

$$\begin{array}{r} 8 \\ -\ 6.24 \\ \hline 2.24 \end{array}$$

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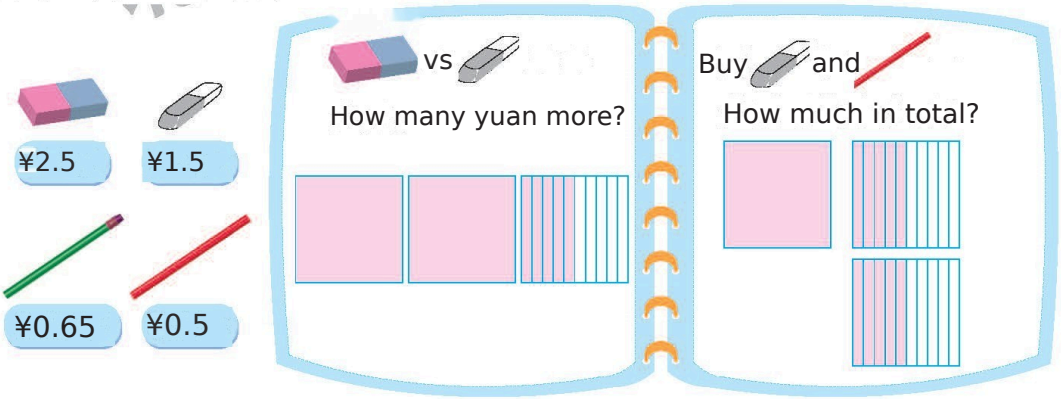
Practice

1. Draw and calculate.



2.

$$\begin{array}{r} 8.7 \\ +\ 1.5 \\ \hline \end{array} \qquad \begin{array}{r} 7.3 \\ -\ 2.5 \\ \hline \end{array} \qquad \begin{array}{r} 10 \\ -\ 2.4 \\ \hline \end{array}$$

3.

2008 Paralympics Men's Triple Jump

F11: Top Three Results

Place	Name	Result (m)
1st	Li Duan	13.71
2nd	Bilalov	12.80
3rd	Porras	12.71

(1) How many metres further did the first-place athlete jump than the second?

(2) Li Duan's result broke the old world record, which had stood for 10 years, by 0.24 m. How many metres was the old world record?

4. Forest Doctor

$$\begin{array}{r} 1\,1.7 \\ +\ 2.8\,6 \\ \hline 4\,0.3 \end{array}$$

$$\begin{array}{r} 3.15 \\ -\quad 2 \\ \hline 3.13 \end{array}$$

$$\begin{array}{r} 8 \\ -\ 7.23 \\ \hline 1.77 \end{array}$$

5. Use the column method.

$3.7+2.36$	$15.4+2.97$	$11.65-7.39$
$10-8.75$	$5.94+10.7$	$9.82-6.18$

6. Without changing the value of the numbers below, which 0s can be removed? Which 0s cannot? Why?

8.10　　15.00　　30.070　　0.090　　600

7.

(1) The bookshop has 4 new kinds of books, and Qisi likes them all. He has only ¥15. Which two books can he buy? How much money is left?

(2) Miaoxiang bought two of these books for exactly ¥16. Which two books did she buy?

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Singing Contest

Contestant 5
Professional score: 8.55
Quality score: 0.88
Total score: 9.43

Contestant 9
Professional score: 8.65
Quality score: 0.40
Total score:

Estimate: whose total score is higher?

Compare their professional scores and their quality scores.

I estimate contestant 5 has the higher total.

Calculate: how many points higher?

8.65 + 0.40 = _____ (points)

$$\begin{array}{r} 8.65 \\ +\ 0.40 \\ \hline 9.05 \end{array}$$

9.43 − 9.05 = _____ (points)

$$\begin{array}{r} 9.43 \\ -\ 9.05 \\ \hline 0.38 \end{array}$$

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9.43 − (8.65 + 0.40)
= 9.43 − 9.05
=0.38 (points)

Decimal addition and subtraction use the same order of operations as whole numbers.

Answer: _____

Xiaoxiao calculated it this way. Can you explain what each step means?

8.65 - 8.55 = 0.10 (points)
0.88 - 0.40 = 0.48 (points)
0.48 - 0.10 = 0.38 (points)

Practice

1. Two ropes are 1.18 m and 1.23 m long. Dad joins the two ropes together, and the joint uses up 0.25 m.

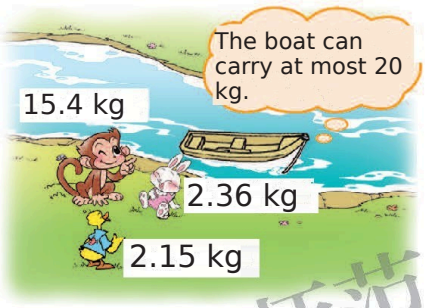
(1) Estimate: will the joined rope be at least 2 m long? Discuss with a partner.

(2) What is the actual length of the joined rope?

2.

$4.6-1.75+2.08$ $7.41-(2.96+3.04)$ $3.2+1.27-2.49$

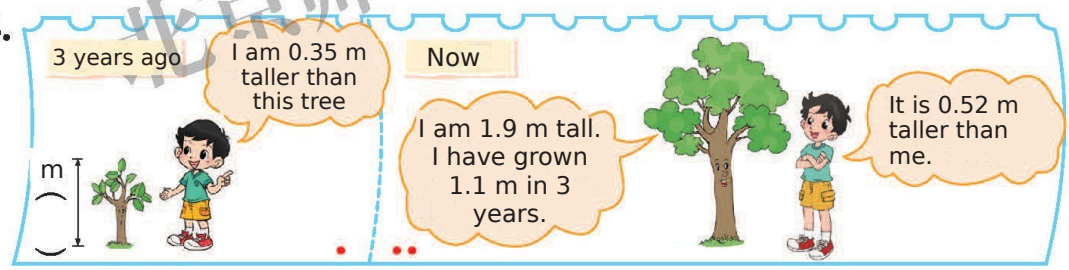
3.



(1) Can all these animals cross together?

(2) Another little deer comes along. It weighs 17.83 kg. If these animals cross the river in two groups, how could they be split up?

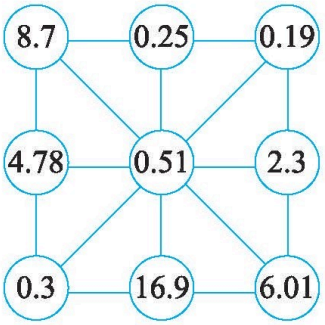
4.



(1) 3 years ago, how tall was Taoqi? How tall is he now?

(2) Ask another maths question and try to answer it.

5. Draw three lines in colour on the figure on the right so that the sum of the numbers on each line is:



(1) exactly 1.

(2) closest to 18.

(3) the greatest possible.

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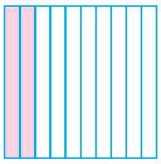
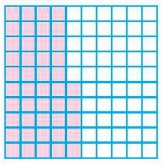
Exercise 1

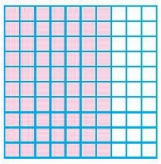
1. Write each shaded part as a fraction and as a decimal.



Fraction ( )
Decimal ( )

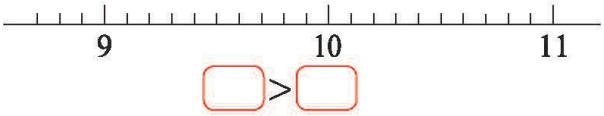


Fraction ( )
Decimal ( )

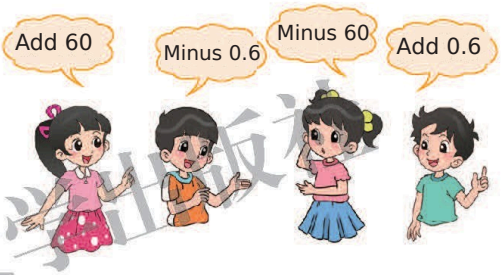


Fraction ( )
Decimal ( )

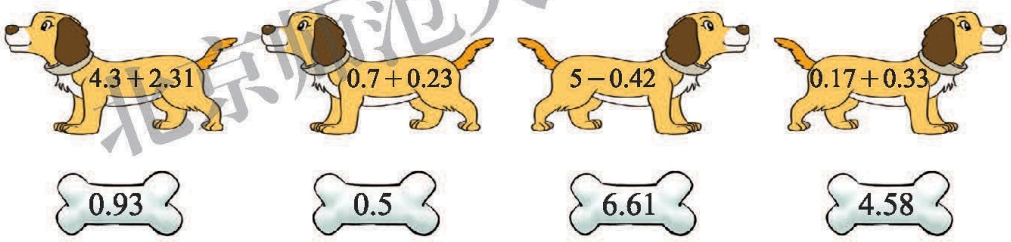
2. Find the positions of 9.8 and 10.1 on the number line below, and compare them.



3. Xiaoxiao used a calculator to work out 13.79 + 2.34, but she keyed in 13.19 + 2.34 by mistake. To correct the mistake, whose idea do you agree with? Talk about it with a partner.



4.



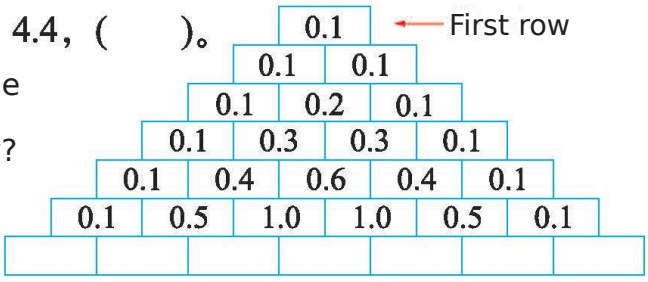
5. $5-(4.19-2.83)$ $4.28+0.53-3.37$ $1.54+2.08+3.62$

6. Find the pattern and fill in.

(1) ( ), 2.6, 2.8, ( ), 3.2, ( )。

(2) 8, 7.1 , 6.2 , ( ), 4.4, ( )。

(3) How would you fill in the seventh row?
What about the eighth row?



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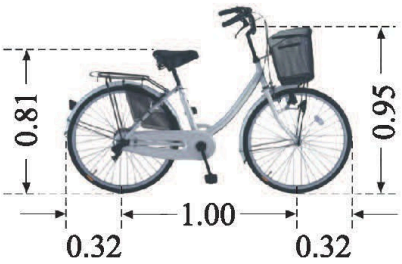
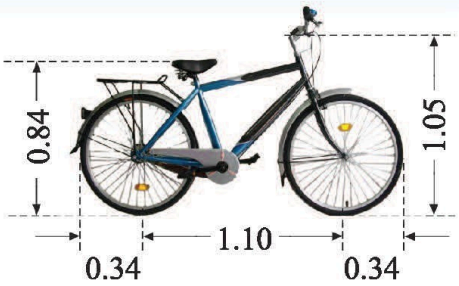
7. Compare and discuss. (Unit: m)





(1) Which bicycle's seat is closer to the ground?

(2) Which bicycle is longer?

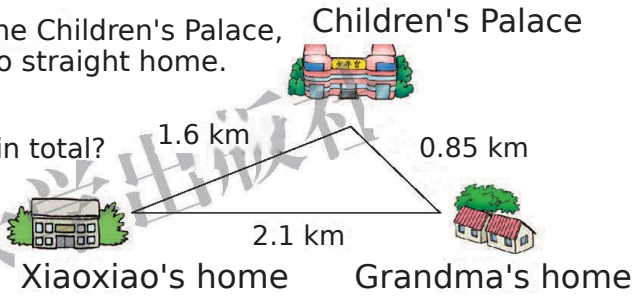
(3) Ask another maths question and try to answer it.

(4) With a parent, measure the seat height and the length of your bicycle at home.

8. On Sunday, Xiaoxiao will first go to the Children's Palace, then visit her grandmother, and then go straight home.

(1) How many km will Xiaoxiao walk in total?

(2) Ask another maths question and try to answer it.



9.

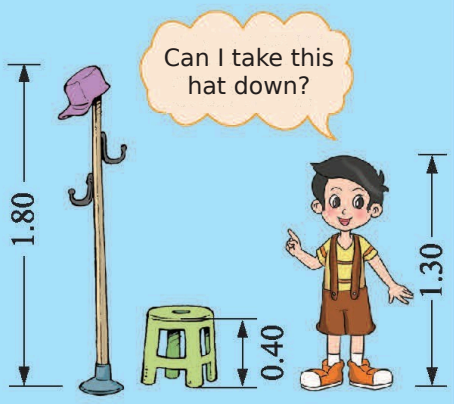
(1) How much does it cost to buy 1 pen and 1 ruler?

(2) I have ¥5. Is it enough to buy 1 eraser and 1 ballpoint pen?

(3) I paid ¥50 for 1 pencil case and 1 schoolbag and got ¥2.20 change. Is that right?

10. Think and discuss.

(Unit: m)



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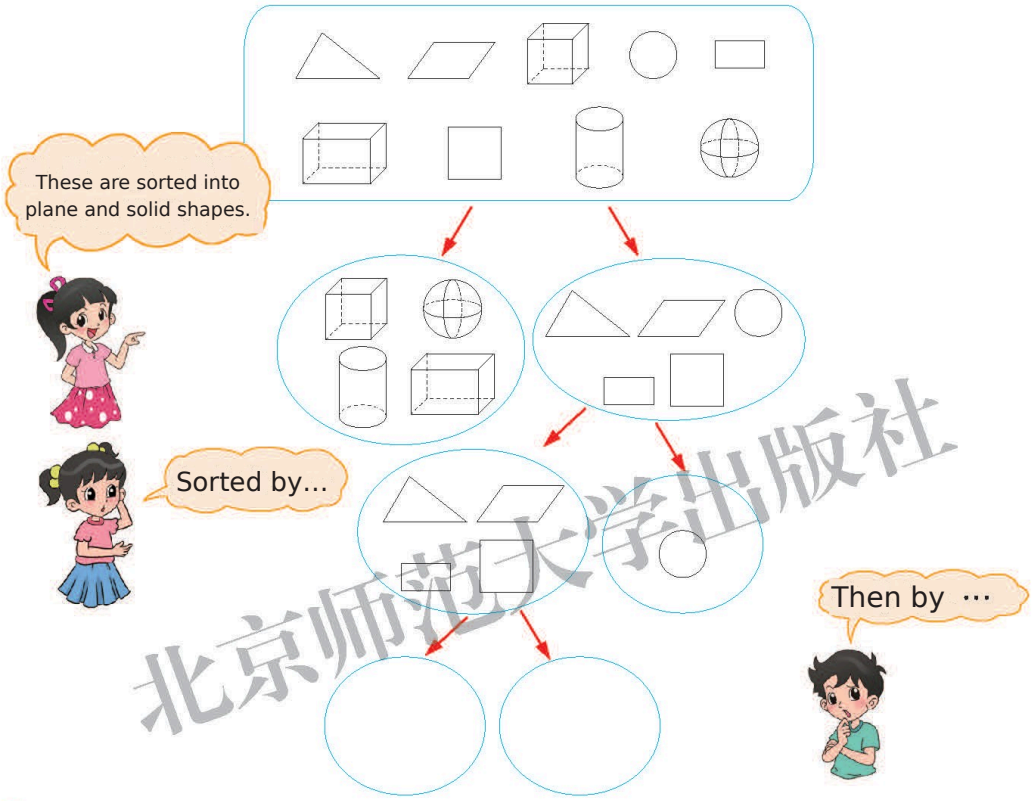
Unit 2 Understanding Triangles and Quadrilaterals

Sorting Shapes

What shapes have we learnt? Discuss and sort them.



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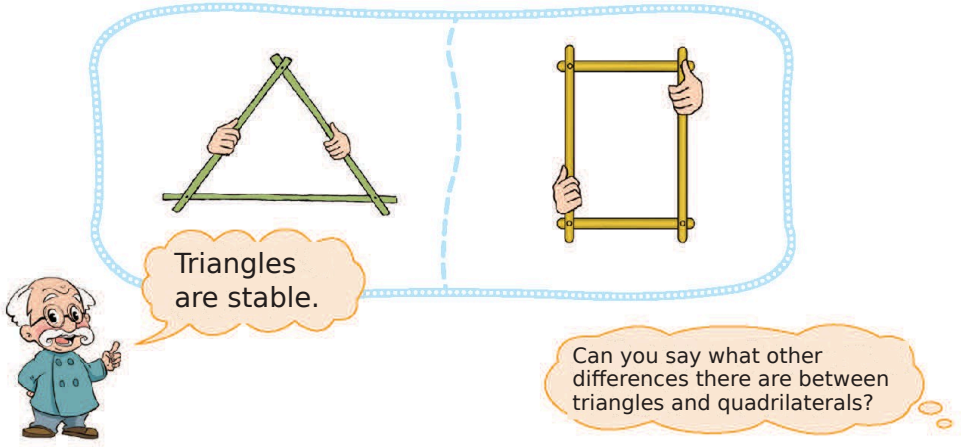
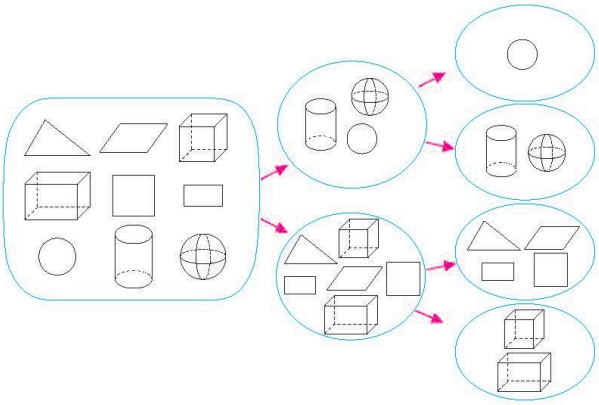
Use sticks to make a quadrilateral frame and a triangle frame. Pull on them. What do you notice?

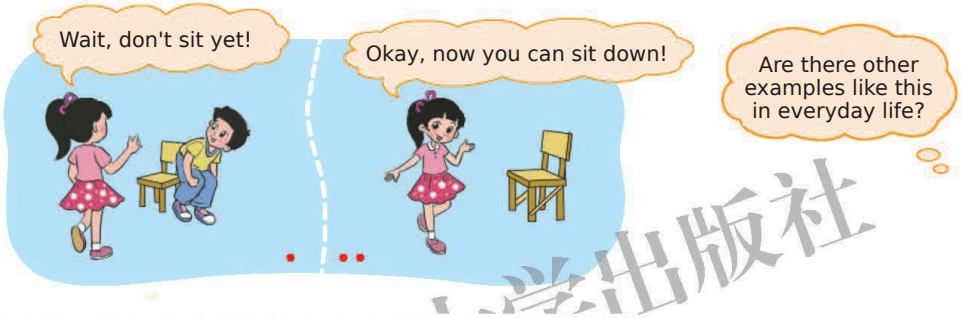


Practice

1. Taoqi sorted the shapes as shown on the right. Do you know how he was thinking? Discuss with a partner.

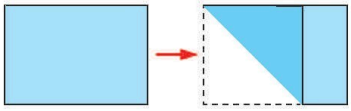


2. Can you explain why?



3. Taoqi folds a rectangular sheet of paper as shown on the right. Make a guess.

(1) How can Taoqi cut out a square and a rectangle with one cut?



(2) How could you cut it to get a triangle and a quadrilateral?

4. Challenge yourself: build a bridge with toothpicks.

Some "toothpick bridges" can even hold up 100 kg!

Did You Know?

Shapes in Buildings

What familiar shapes are there in buildings? Look at the buildings around you and you will find triangles and domes (half spheres); they are the "strongest" shapes.

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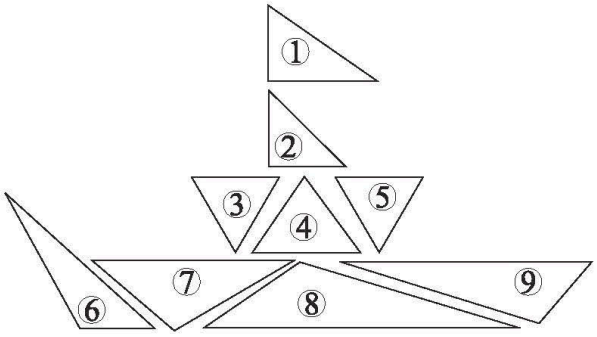
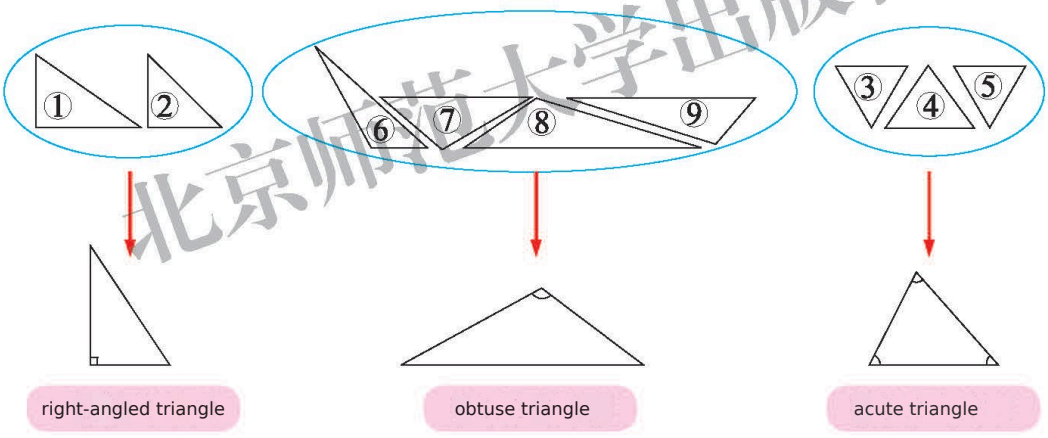
Classifying Triangles

Classify the triangles that make up the pattern below (use Figure 1 on Appendix page 3), and discuss your classification with your partner.



Xiaoxiao sorted them like this. Do you know why? Talk with a partner and name them.



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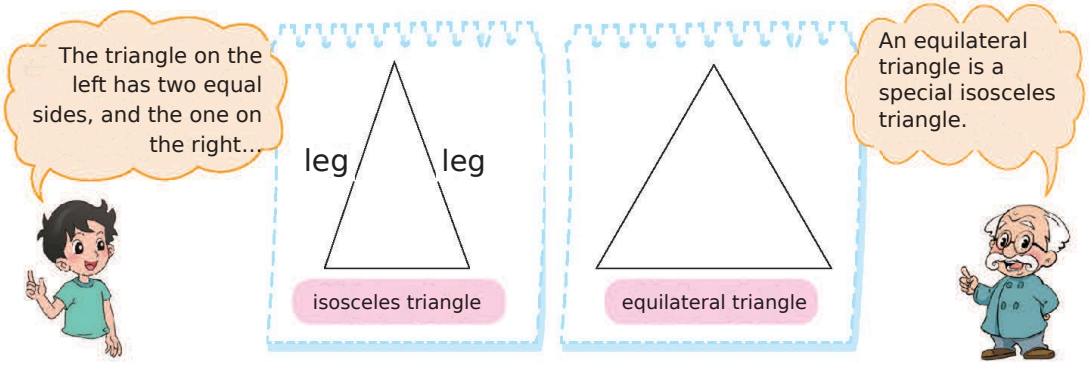
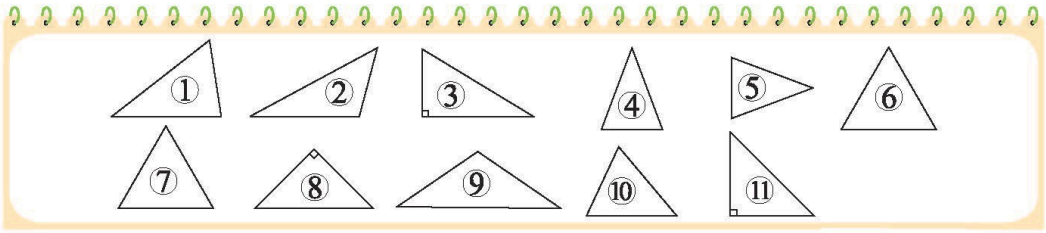
Taoqi finds the two triangles below special. Talk about them and name them.



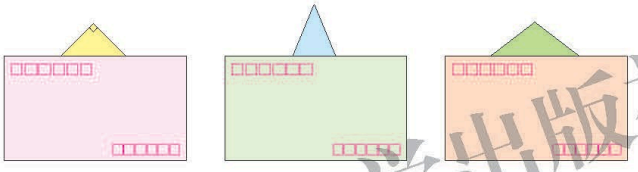
Practice

1. Find and fill in

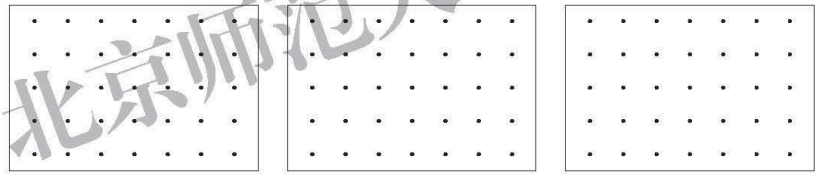


Figures ④ _____ are isosceles, figures _____ are equilateral,
Figures _____ are right-angled, figures _____ are obtuse,
Figures _____ are acute.

2. Guess what triangle each envelope might be hiding.



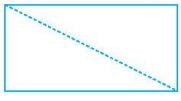
3. Draw each triangle on the dot grid.



isosceles triangle obtuse triangle right-angled triangle

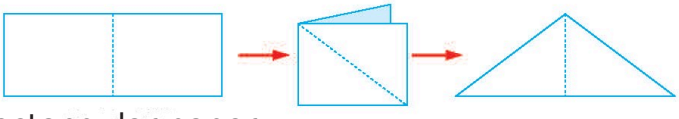
4. Cut and see.

(1) What triangles do you get by cutting along the dashed line?



Rectangular paper

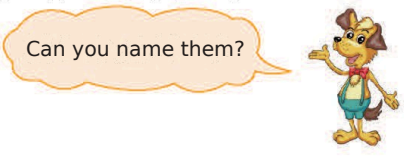
(2) How can you cut an isosceles triangle?



Rectangular paper

(3) What triangles do you get by cutting along the dashed line?

Square paper



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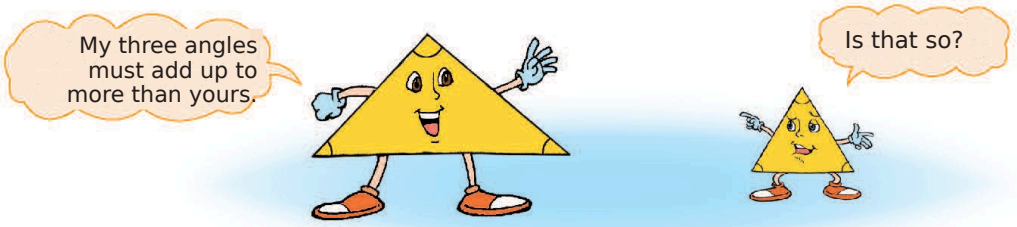
Explore and Discover: Angle Sum of a Triangle



Group activity: everyone prepares a triangle. Measure and fill in.

Group activity record sheet Group ___

Members' names	Triangle shape	Size of each angle			Angle sum

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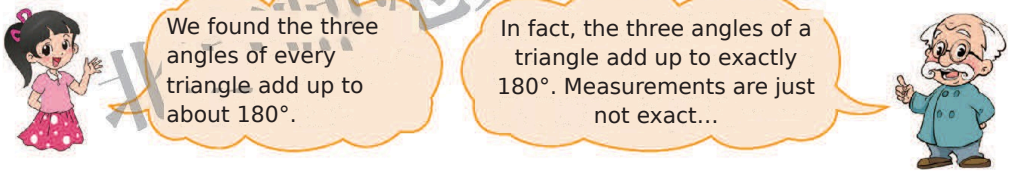
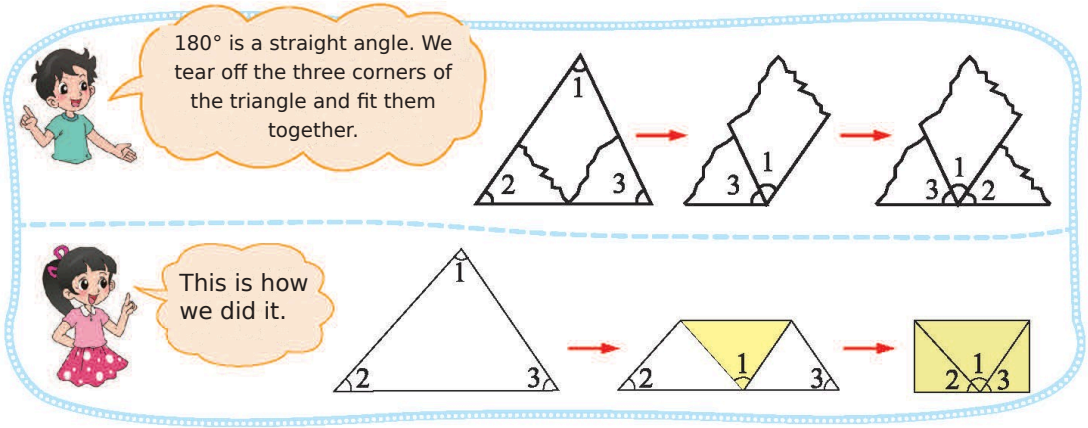
Group discussion: what did you find?



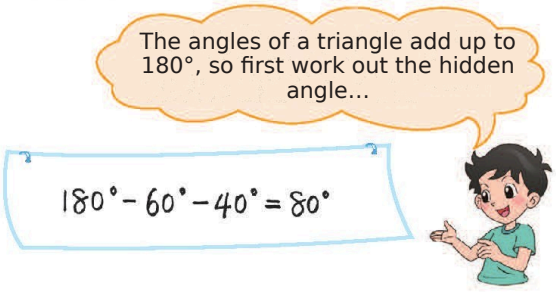
How can you check your idea? Talk about it and try it.



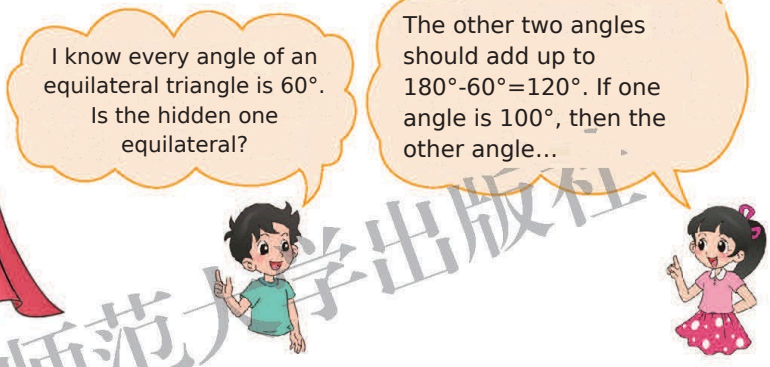
The angle sum of a triangle is 180°

Try It

Guess: what triangle could it be?



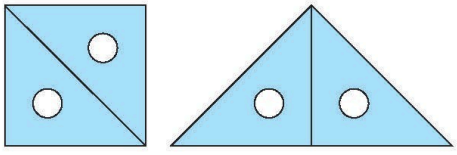
Can you still guess what triangle it is?



Practice

1. What is the angle sum of a triangle? Review how you explored and discussed it.

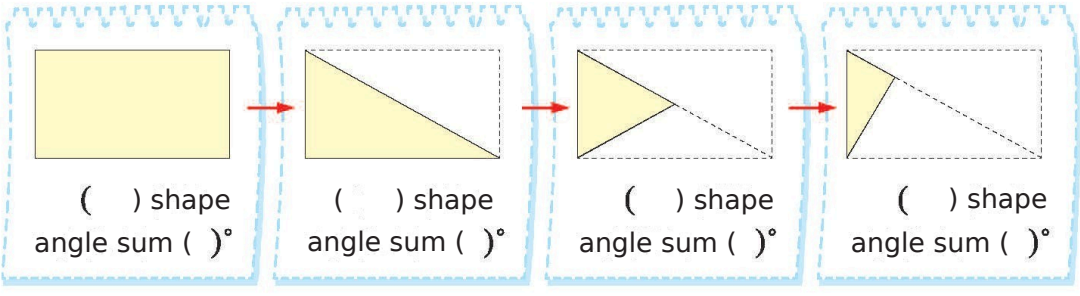
2. As shown, two identical set squares make a quadrilateral and a triangle.



(1) Think: what is the angle sum of each shape? Tell a partner how you worked it out.

(2) Measure, then work out each angle sum.

3. Cut a rectangular piece of paper, then fill in.



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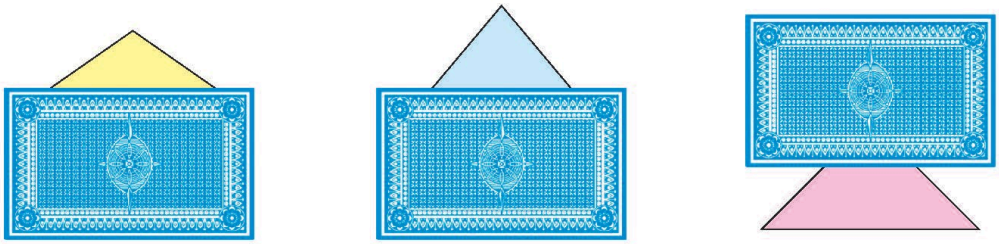
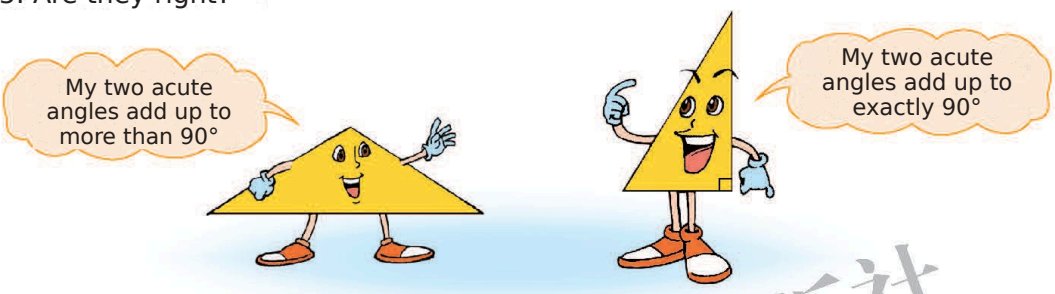
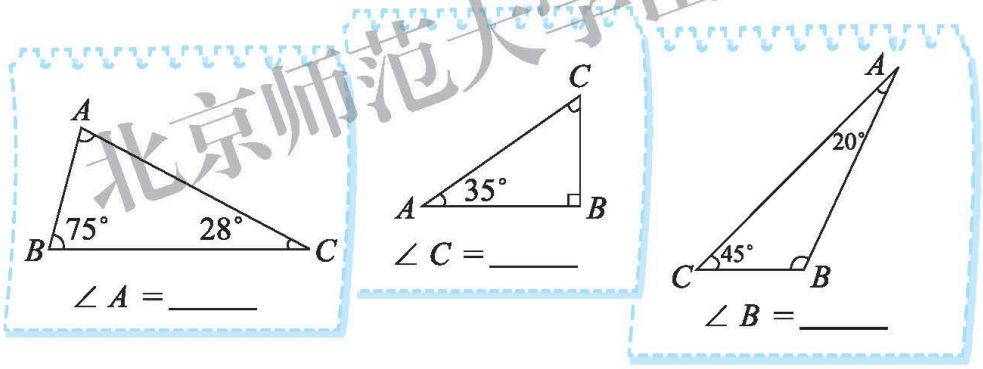
4. Measure and guess: what triangle could it be?



5. Are they right?



6. Find the size of each angle below.



7. Challenge yourself: explore the angle sum of a quadrilateral.

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Explore and Discover: Sides of a Triangle

Make triangles with sticks. Which sets below work? Which don't? Discuss with a partner.

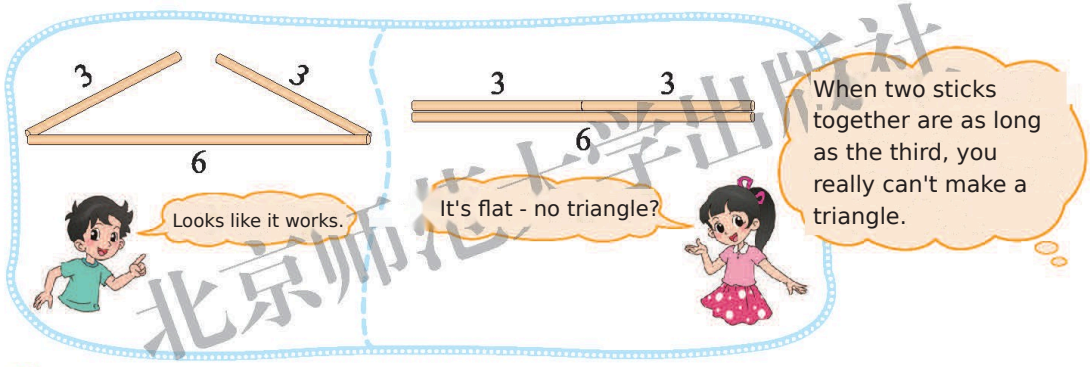
(Unit: cm)

(1) 3, 5, 6

(2) 3, 4, 6

(3) 3, 3, 6

(4) 3, 2, 6



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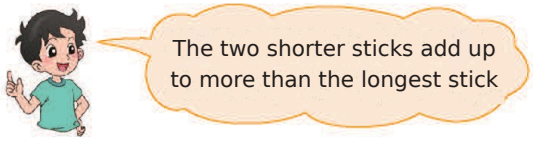
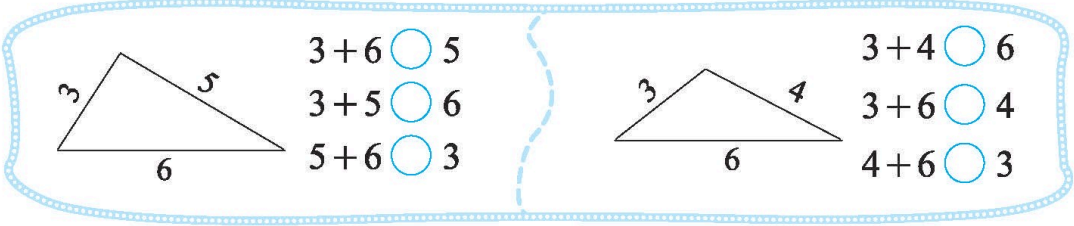
Think: what kind of 3 sticks can make a triangle? Tell a partner.



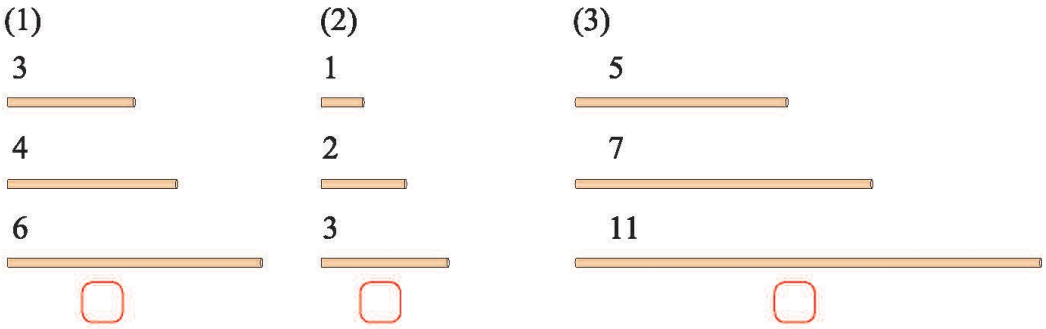
Calculate and compare: what is the relationship among the lengths of the 3 sticks that can make a triangle? (unit: cm)



Any two sides of a triangle add up to more than the third.

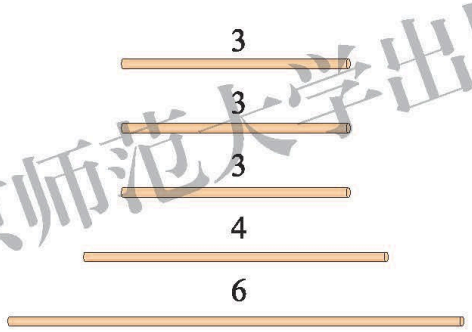
Practice

1. Draw "√" under the sticks that can make a triangle. (unit: cm)



2. Take any 3 of the 5 sticks below and make two different triangles.

(Unit: cm)



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3. Use sticks of the same length to make triangles and complete the table.

(1) 3 sticks: can they make a triangle? What kind of triangle is it?

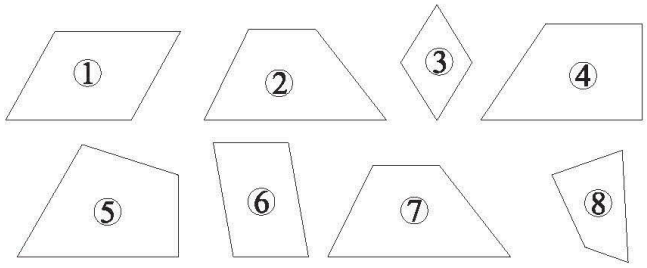
(2) Can 4 sticks make a triangle? What about 5 sticks or 6 sticks?

Number of sticks	3	4	5	6
Triangle possible?				
Kind of triangle				

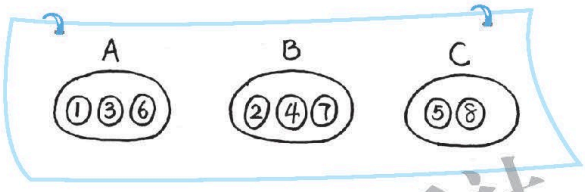
4. Two sides of a triangle are 5 cm and 8 cm long. How long could the third side be? Write two answers.

Classifying Quadrilaterals

Classify the quadrilaterals below and say how you classified them.



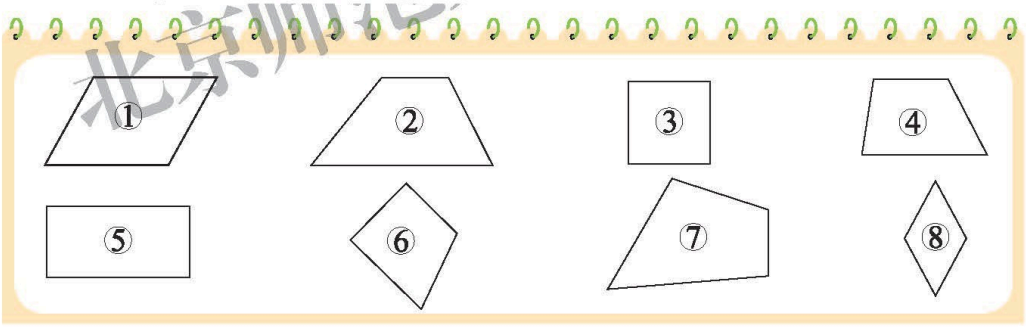
Xiaoxiao sorted them like this. Can you see how? Talk about it and name them.



Group A shapes have two pairs of parallel sides. They are all parallelograms.

Group B shapes have just one pair of parallel sides. They are all trapeziums.

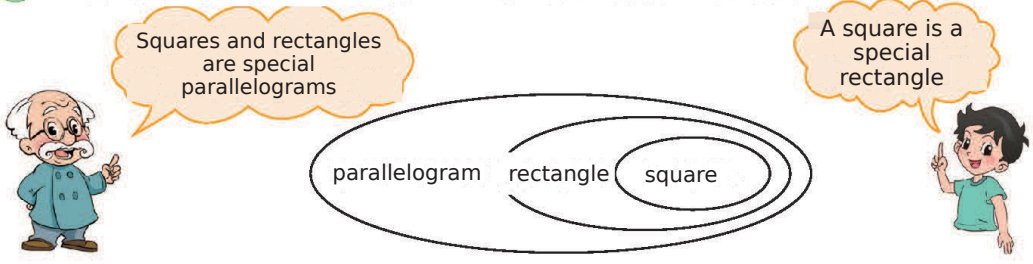
Which figures below are parallelograms? Which are trapeziums? Find and fill in.



Figures ① _____ are parallelograms.

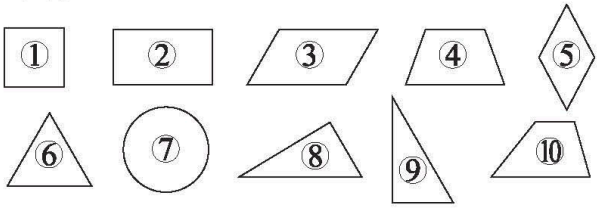
Figures _____ are trapeziums.

How are squares, rectangles and parallelograms related?

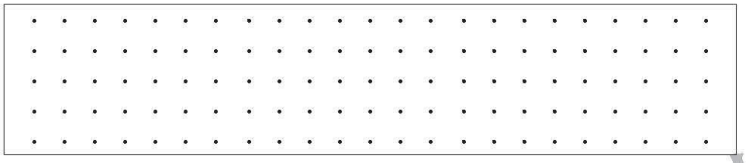


Practice

1. Sort the shapes below and share your method with a partner. Cut out the shapes in Appendix 3, Figure 2 and try it.

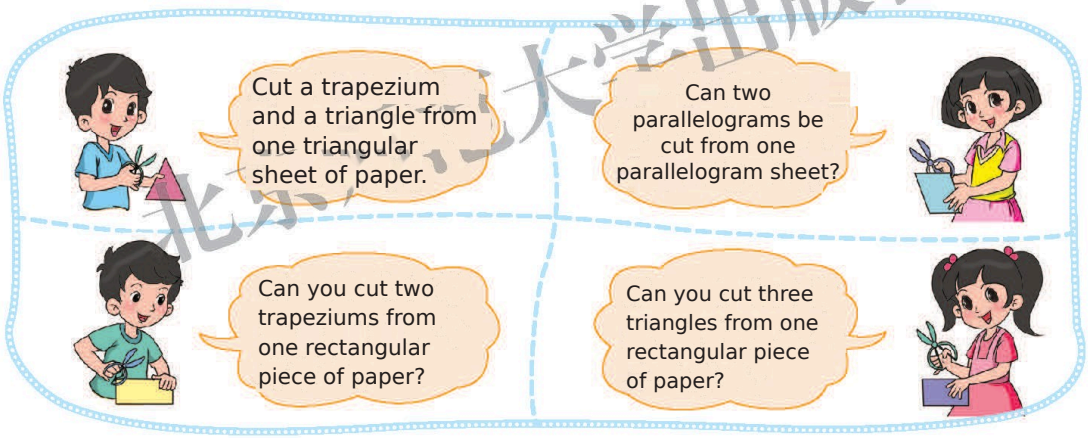


2. Draw as required on the dot grid.



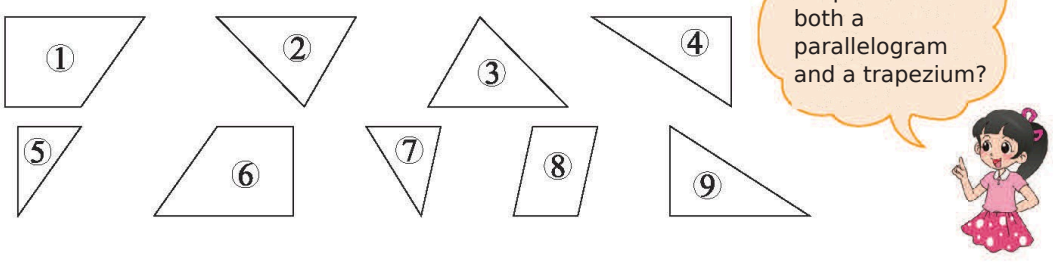
Parallelogram Trapezium Triangle

3. Make just one cut.



4. Combine shapes.

Which two shapes below can make a rectangle, a parallelogram or a trapezium? Cut out the shapes in Appendix 3, Figure 3 and try.



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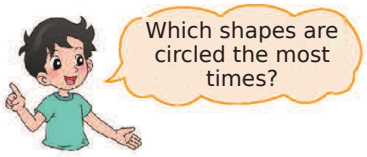
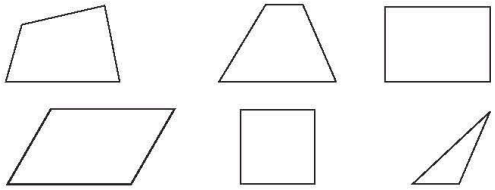
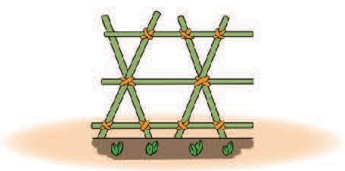
Exercise 2

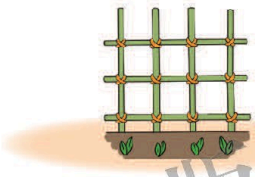
1. Use pens of different colours. First circle the shapes with right angles, then circle the shapes with two pairs of parallel opposite sides, and finally circle the quadrilaterals.



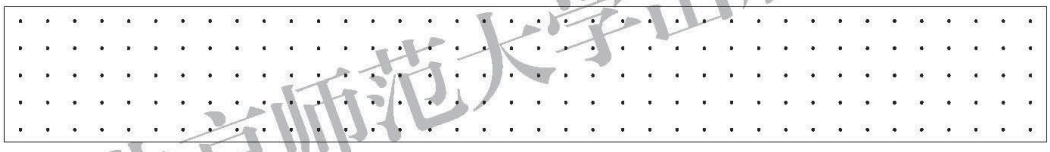


2. Choose: which structure does not change shape easily?



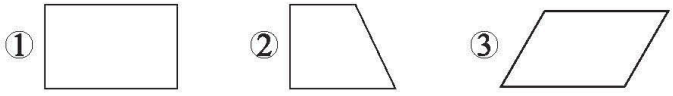


3. Draw the figures on the dot grid as required.



right-angled triangle isosceles triangle trapezium parallelogram

4. Draw a straight line in each figure below.

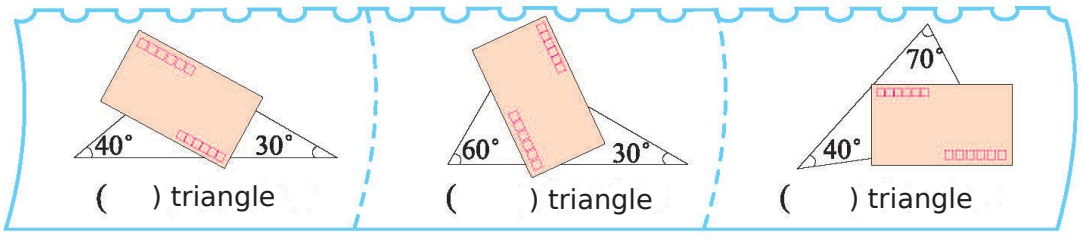


Two right-angled triangles: Figure _____。

Two obtuse triangles: Figure _____。

A right-angled and an acute triangle: Figure _____。

5. What kind of triangle is each one below?



( ) triangle ( ) triangle ( ) triangle

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6. Put a "√" under the group of sticks that can form a triangle. (Unit: cm)

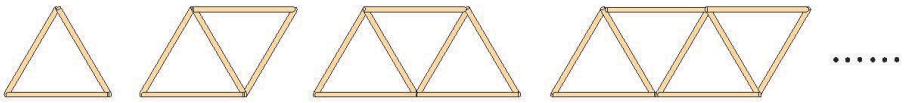
(1) 3, 4, 5 ☐

(2) 2, 2, 6 ☐

(3) 3, 3, 3 ☐

(4) 3, 3, 5 ☐

7. Arrange the sticks as shown.



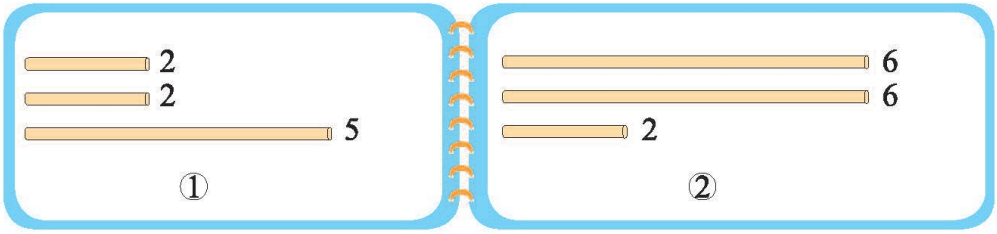
3 sticks 5 sticks 7 sticks 9 sticks .....

1 triangle needs 3 sticks, 2 triangles need 5 sticks, so 5 triangles need ( ) sticks.

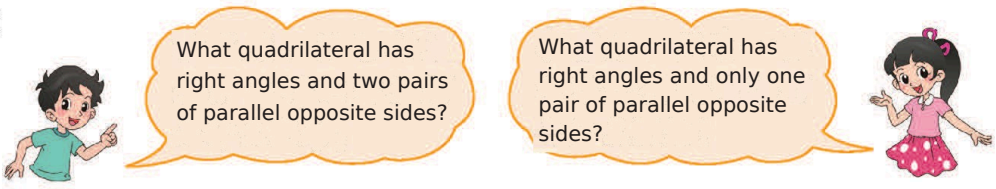
8. Guess.

(1)

(2) Which group of sticks below can form an isosceles triangle? (Unit: cm)



(3)



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Unit 3 Multiplication of Decimals

Buying Stationery

¥0.3

¥0.2

¥0.4

¥0.7

How much do 4 erasers cost? Share your thinking with a partner.

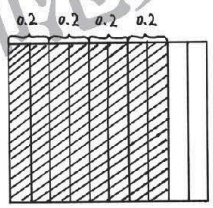
Adding 4 lots of ¥0.2 gives ¥0.8.

0.2 yuan =2 jiao,
2×4=8 (jiao),
8 jiao =0.8 yuan,
that is:
0.2×4=0.8(yuan).

Answer: _____.

Xiaoxiao did it this way. Do you understand?

$0.2 \times 4 = \square$ ( )

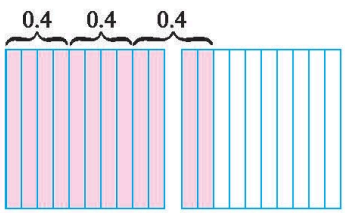


0.2 is 2 lots of 0.1, so 4 lots of 0.2 is 8 lots of 0.1, which is 0.8.

How much do 3 rulers cost? Share your thinking with a partner.

0.4 yuan =4 jiao,
3×4=12 (jiao),
12 jiao =1.2 yuan,
0.4×3= 1.2 (yuan).

□ ○ □ = □ ( )



0.4 is 4 lots of 0.1, so 3 lots of 0.4 is 12 lots of 0.1, which is 1.2.

Answer: _____.

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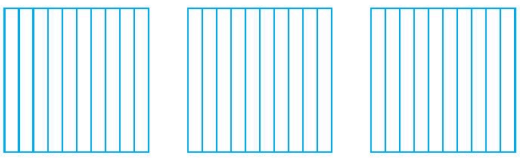
Practice

1. How much do 4 pencil sharpeners cost? Shade, calculate and share your thinking with a partner.

¥0.7



2. How much do 6 pencils cost? Share your thinking with a partner.

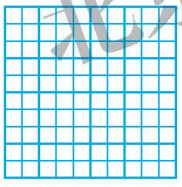
¥0.3

3.

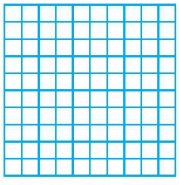
0.3×3	0.2×8	4×0.2
3×0.5	0.4×4	0.1×9
0.7×2	6×0.2	0.6×4

4. Colour and fill in.

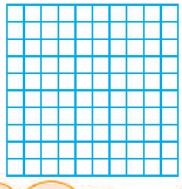
$0.01 \times 10 =$ ☐



$0.01 \times 50 =$ ☐

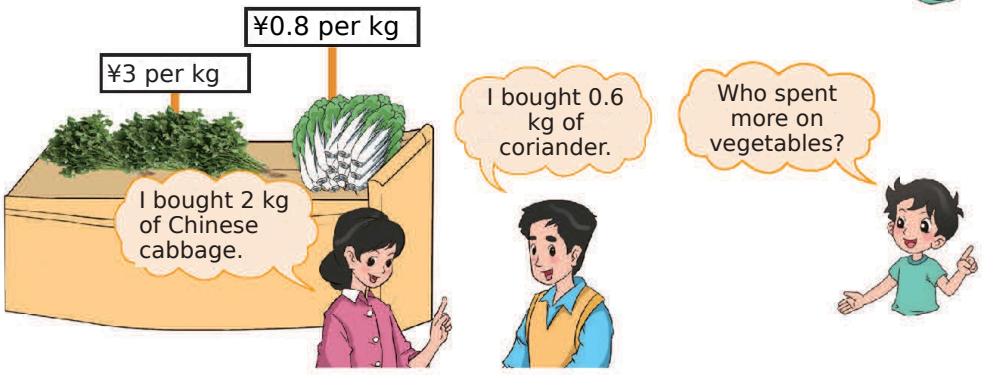


$0.01 \times 100 =$ ☐



What does 0.01 × 1000 equal?

5.



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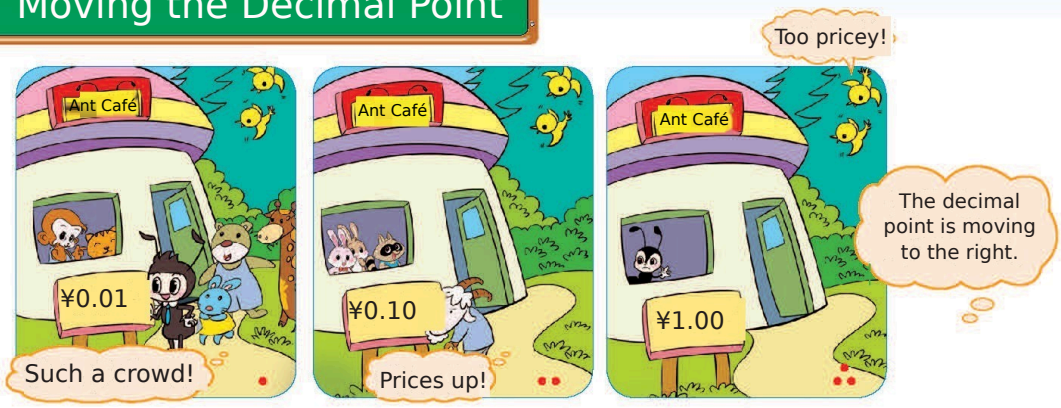
Moving the Decimal Point

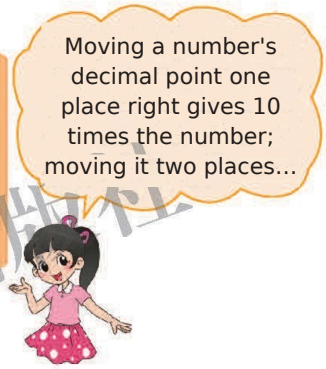


When the decimal point moves right, how does the size of the decimal change? Think and discuss.

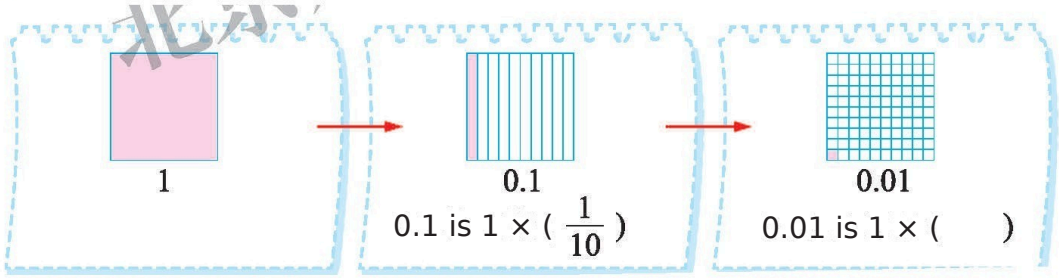
One place right

Two places right

¥0.01 =1 fen
¥0.10 =1 jiao
¥1.00 =1 yuan



When the decimal point moves left, how does the decimal change? Discuss and fill in.



Moving the decimal point one place left multiplies a number by ($\frac{1}{10}$).

Moving the decimal point two places left multiplies a number by ( ).

Give another example of how moving the decimal point changes a decimal.

0.04 m =4 cm,
0.40 m =4 dm,
4.00 m =4 m,
the point moves right...

¥8.00 =8 yuan,
¥0.80 =8 jiao,
¥0.08 =8 fen,
the point moves left...

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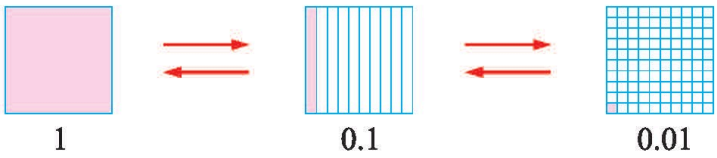
Try It

Look, think and discuss.



What are 10 times and 100 times 0.01?

10 times 0.01 means multiplying it by 10……

$0.01 \times 10 = 0.1$

$0.01 \times 100 = \square$

To get 10 times 0.01, just move the decimal point of 0.01 one place right...

$1 \times \frac{1}{10}$, $\frac{1}{100}$ — what are they?

$1 \times \frac{1}{10}$ means dividing it by 10...

$1 \div 10 = 0.1$

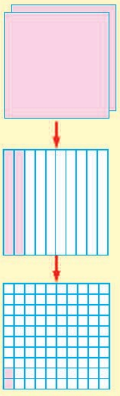
$1 \div 100 = \square$

$1 \times \frac{1}{10}$ just move the decimal point of 1 one place left...

Practice

1. Fill in.

(1)



2

0.2 0.2 is 2 × ( )

0.02 0.02 is 2 × ( )

(2)

¥0.05

¥0.5 ¥0.5 is ¥0.05 × ( )

¥5 ¥5 is ¥0.05 × ( )

2. Give an example like Taoqi's and explain it.

tens	ones	·	tenths	hundredths
	0	·	5	

When the decimal point of 0.5 moves one place right, 5 lots of 0.1 become 5 lots of 1, so 5 is 10 times 0.5.

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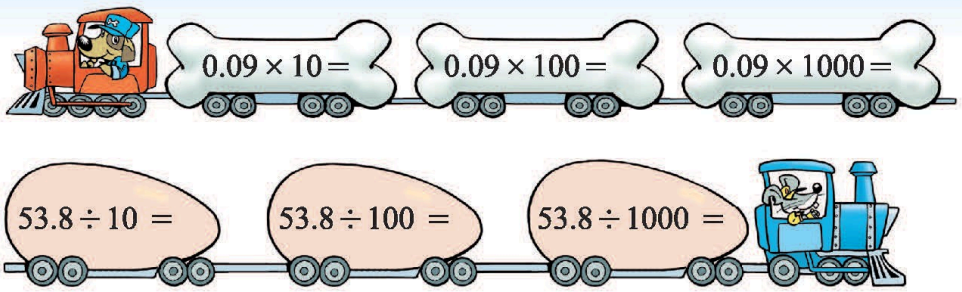
3.

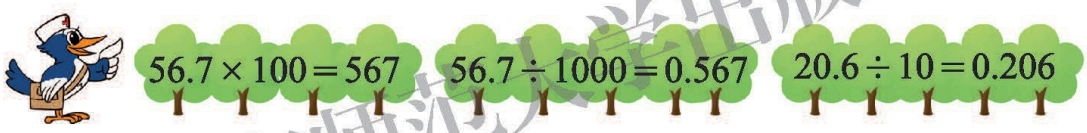


4. The shortest tree in the world is the dwarf willow. It is usually only 0.05 m tall.

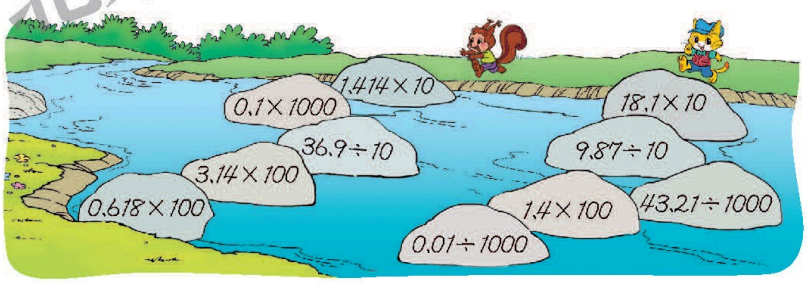
(1) A newborn baby is about 10 times as tall as the dwarf willow. How tall is a newborn baby, in metres?

(2) The King of Cypresses, in Bajie Village, Nyingchi County, Tibet, is about 2600 years old. It is 1000 times as tall as the dwarf willow. How tall is this cypress, in metres?

5. Forest Doctor.



6. Cross the river.



7. The school bought some prizes. Can you complete the table?

	Unit price (¥)	Quantity	Total price (¥)
Skipping rope		100	75
Pencil case	3.8	100	
School bag		10	225
Eraser	0.2	1000	

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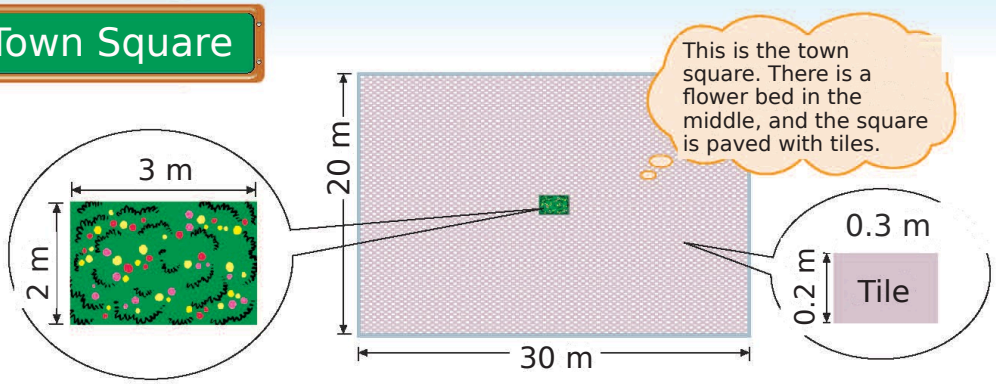
Town Square



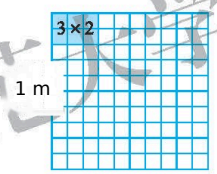
Talk about the mathematical information in the town square, then calculate and fill in.

	Town Square	Flower bed	Tile
Area (m^2)			

Taoqi is having trouble finding the area of a tile. Can you help him?

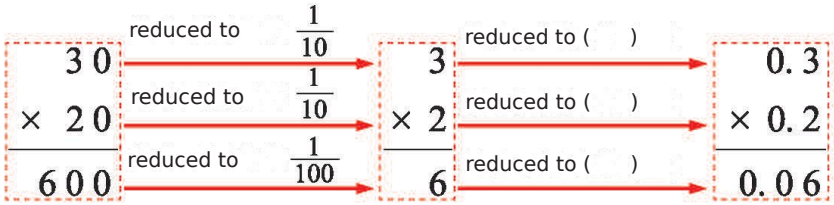
$0.3 \times 0.2 =$ ☐ ( )

0.3 m=3 dm,
0.2 m=2 dm,
3×2=6 (dm^2)
……



The length is 0.3 m, the width is 0.2 m, the area is 6 small squares, and each small square is 0.01 square metres…

Using the calculation above, think: how are the multipliers and the product related?



How is the number of decimal places in the product related to that in the multipliers? Calculate and think.

Number sentence	$4 \times 0.3 = 1.2$	$0.4 \times 0.3 =$	$0.13 \times 2 =$	$0.13 \times 0.2 =$
Decimal places in the first multiplier				
Decimal places in the second multiplier				
Decimal places in the product				

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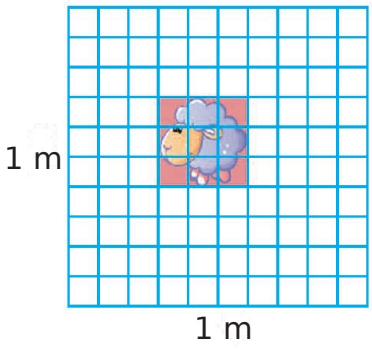
Practice

1. Can you work out the area of this jigsaw piece? Think and calculate.



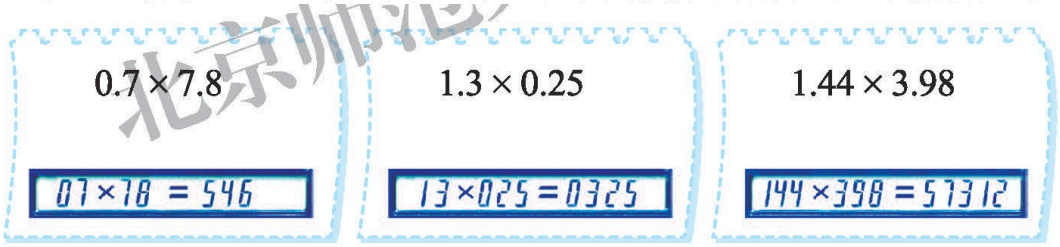
(1) Share your ideas with your partner.

(2) Using your calculation, say how the decimal places in the product are related to the decimal places in the multipliers.

2. Use the product in the first column to quickly write the products in the other columns.

Multiplier	36	3.6	36	3.6	0.36
Multiplier	24	2.4	2.4	24	2.4
Product	864				

3. Taoqi's calculator won't show the decimal point. Help him find the correct answers.



4. Mingming takes the bus once to school and once home every day, and each bus ride costs ¥0.2. His bus card has ¥3 left. Is that enough for 5 days of school in one week?

5. Using 13×12=156, fill each ☐ with a suitable number.

☐×☐=15.6
☐×☐=15.6
☐×☐=0.156
☐×☐=0.156

☐×☐=1.56
☐×☐=1.56
☐×☐=156
☐×☐=156

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Wrapping

Wrapping one gift box takes 0.8 m of wrapping paper and 2.4 m of ribbon.

Wrapping paper: ¥2.6 per metre; ribbon: ¥0.85 per metre

How much does the wrapping paper cost? Estimate and calculate.

$2.6 \times 0.8 = \square$ ( )

Less than 1 m, so no more than ¥2.6

I can calculate 26×8...

$2.6 \times 0.8 = 2.08$

multiplied by 10

multiplied by 10

reduced to $\frac{1}{100}$

$26 \times 8 = 208$

2.6 → 1 decimal place

× 0.8 → 1 decimal place

2.08 ← 2 decimal places

Answer: _____。

How much does the ribbon cost? Estimate and calculate.

$2.4 \times 0.85 = \square$ ( )

Less than ¥1 a metre, so no more than ¥2.4

First multiply as with whole numbers, then place the decimal point in the product. The result is 2.040.

```
  24
× 85
 120
1920
2040
```

Or write it as:

```
  2.4
×0.85
 120
1920
2.040
```

The final 0 can be removed.

Answer: _____。

Try these, then share with your partner.

0.82×0.03　　0.05×0.2

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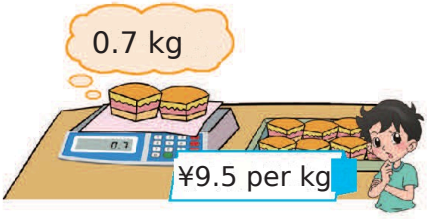
Practice

1. Taoqi took ¥10 to buy a cake for Grandma.



(1) Estimate. Does Taoqi have enough money?

(2) How much should the salesperson charge? Share how you worked it out with your partner.

2. Put the decimal point in each product below.

$$\begin{array}{r} 1.36 \\ \times\ 0.8 \\ \hline 1088 \end{array} \qquad \begin{array}{r} 0.78 \\ \times\ 0.04 \\ \hline 312 \end{array}$$

3. Use the column method.

4.8×0.25 $\quad$ 0.32×0.8 $\quad$ 9.8×0.5

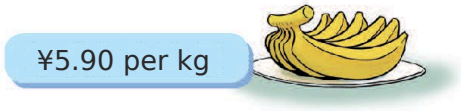
4.8×1.2 $\quad$ 0.32×1.2 $\quad$ 9.8×1.03

4. Compare.

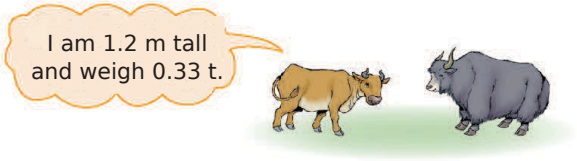
$6.5 \times 0.9 \bigcirc 6.5$ $\quad$ $1 \times 0.95 \bigcirc 1 \times 1.2$

$2.4 \times 1.1 \bigcirc 2.4$ $\quad$ $2.4 \times 1.5 \bigcirc 3.4 \times 0.55$

5. Mum wants to buy 3.8 kg of bananas. Is ¥25 enough?



6. A grassland cow is 1.2 times as tall as a Mongolian cow and 1.4 times as heavy. How tall and how heavy is it?



Mongolian cow $\quad$ Grassland cow

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Silk

Children, did you know? The silk spun by one autumn silkworm is about 1.2 km long and has a mass of about 0.35 g. The silk spun by a spring silkworm is longer, about 1.25 times that of an autumn silkworm.

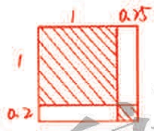
About how many kilometres long is the silk spun by one spring silkworm?

$1.2 \times 1.25 = \square$ ( )

$$\begin{array}{r} 1.25 \\ \times\ 1.2 \\ \hline 250 \\ 125 \\ \hline 1.500 \end{array}$$

1.2×1.25
$= 1 \times 1 + 0.2 \times 0.25$?
$= 1.05$ (km)

Partly right. Use the diagram.



This piece was missed.

Answer: _____.

A silk scarf takes the silk of about 300 autumn silkworms. About how many grams is its mass?

$0.35 \times 300 = \square$ ( )

$$\begin{array}{r} 0.35 \\ \times\ 300 \\ \hline 10500 \end{array}$$

Where does the decimal point go in the product? What is the result?

Answer: _____.

Think: what do we need to watch out for when multiplying decimals?

Calculate and fill in. What do you find? Can you give another example to show it?

$2.3 \times$		$=$	
	1.2		
	1.05		
	1		2.3
	0.9		2.07
	0.86		

Is the product always greater than the multiplier?

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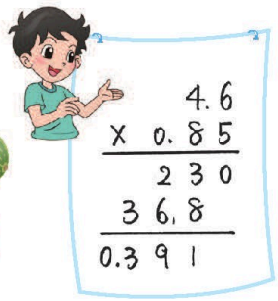
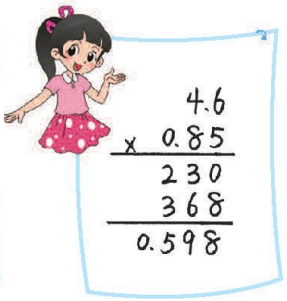
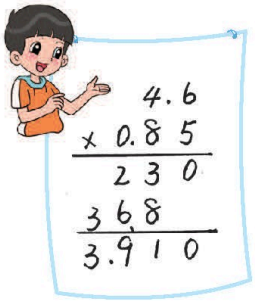
Practice

1. How much does a 4.6 kg watermelon cost? Say who got it right.







2. Without calculating, can you tell whether these are correct? Discuss with your partner.

$0.9 \times 1.5 = 13.5$ $9.9 \times 7.1 = 60.29$ $2.8 \times 1.5 = 2.4$

3. Put the decimal point in each answer below to make the calculation correct.

$4.8 \times 1.3 = 624$ $9.32 \times 1.4 = 13048$ $1.87 \times 2.3 = 4301$

$4.8 \times 0.9 = 432$ $0.7 \times 9.32 = 6524$ $0.15 \times 0.63 = 945$

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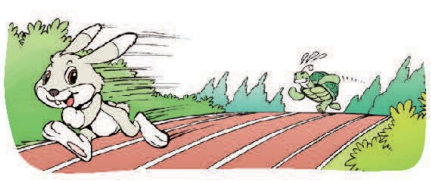
4. First think: is the product greater or less than the first multiplier? Then calculate.

0.12×50 2.6×0.5 1.3×2.2 9.6×0.8

5.



A tortoise crawls at about 0.05 m/s. A rabbit runs 280 times as fast as the tortoise. What is the rabbit's running speed?

6. The tallest bird in the world is the ostrich, and the tallest mammal is the giraffe. An ostrich is 2.75 m tall, and a giraffe is 2.2 times as tall as the ostrich. How tall is this giraffe in metres?

Hand in Hand

Item	Exercise book	Pencil case	Stories of Mathematicians
Price (¥)	2.8	6.1	7.2

Taoqi wants to buy 1 exercise book and 1 pencil case for each of 3 students at Hope Primary School. How much will it cost in total?

2.8×3=8.4(yuan)
6.1×3=18.3 (yuan)
8.4+18.3=26.7 (yuan)

$2.8\times3+6.1\times3$
$=8.4+18.3$
$=26.7$ (yuan)

$(2.8+6.1)\times3$
$=8.9\times3$
$=26.7$ (yuan)

Answer: _____。

Xiaoxiao wants to buy 1 exercise book and 1 copy of Stories of Mathematicians for each of 5 students at Hope Primary School. How much will it cost in total?

$2.8\times5+7.2\times5$
$=14+36$
$=50$ (yuan)

$2.8\times5+7.2\times5$
$=(2.8+7.2)\times5$
$=10\times5$
$=50$ (yuan)

Answer: _____。

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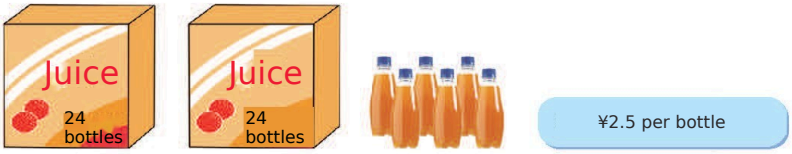
Practice

1. Do you remember the associative law of addition? Give an example to show that it also applies to decimals.

2. How much do these juices cost in total? Share your idea with your partner.



3. Calculate.

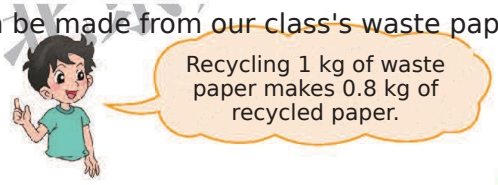
$2.5 \times 13 + 0.9$ $\quad$ $2.6 + 1.4 \times 3$

$0.25 \times 3.7 \times 0.4$ $\quad$ $0.89 \times 4.8 + 0.89 \times 5.2$

$1.83 + 3.79 + 0.17$ $\quad$ $10 - 0.34 - 0.66$

$2.5 \times 0.6 - 1.8 \times 0.5$ $\quad$ $(8 + 0.8) \times 1.25$

4. If each person recycles 1.5 kg of waste paper, how many kg of recycled paper can be made from our class's waste paper?



5. A residential area plans to build a lawn 10 m long and 6 m wide. The lawn actually built is 2.5 m wider than planned.

(1) What is the actual area of the lawn in square metres?

(2) By how many square metres is the actual area bigger than planned?

Draw a diagram.

(3) This kind of lawn can absorb about 0.04 kg of carbon dioxide per square metre per day. How many kg of carbon dioxide can this lawn absorb per day after it is built?

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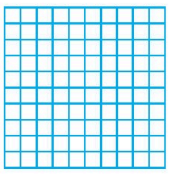
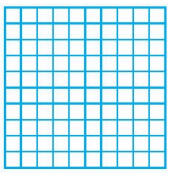
Exercise 3

1. Calculate the following, and use drawings to share your ideas with your partner.

0.06×4



2×0.4



2. Give an example to show that the associative law also works for decimal multiplication.

3. Using the first number sentence, fill in each ☐ with a suitable number, and share your ideas with your partner.

$1.6 \times 2.8 = 4.48$

☐ × ☐ = 4.48

☐ × ☐ = 4.48

☐ × ☐ = 4.48

$0.29 \times 3.5 = 1.015$

☐ × ☐ = 1.015

☐ × ☐ = 1.015

☐ × ☐ = 1.015

4. Pick one number from each group below and multiply them so the product is a number in the table. Write the number sentences you find. Group 1: 2, 5, 4, 9; Group 2: 4.6, 1.1, 1.9, 12.3.

5.5	41.4	18.4	9.2
61.5	9.5	24.6	4.4
9.9	110.7	7.6	23
3.8	49.2	2.2	17.1

5.

5.04×28	3.82×0.45	0.96×6.9
$(15.4-7.8) \times 3.5$	$4.15+3.24+9.85+6.76$	$3.2 \times 4.6+3.2 \times 5.4$

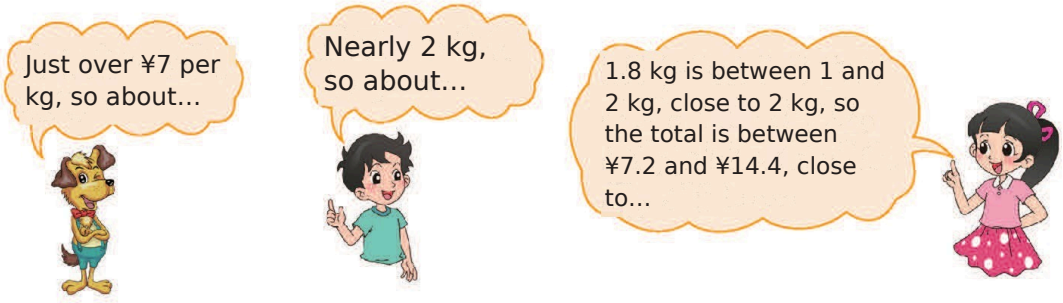
6. China's own Shenzhou-9 crewed spacecraft was launched successfully on 16/6/2012.

(1) The Shenzhou-9 spacecraft uses only 1.8 kWh of electricity to orbit the Earth for 1 hour, about as much as a 75-watt bulb uses in 1 day. How much electricity does it use in one day of flight?

(2) The Shenzhou-9 spacecraft orbited the Earth about 200 times in space, and 1 orbit of the Earth takes about 1.5 hours. About how long did it fly in space?

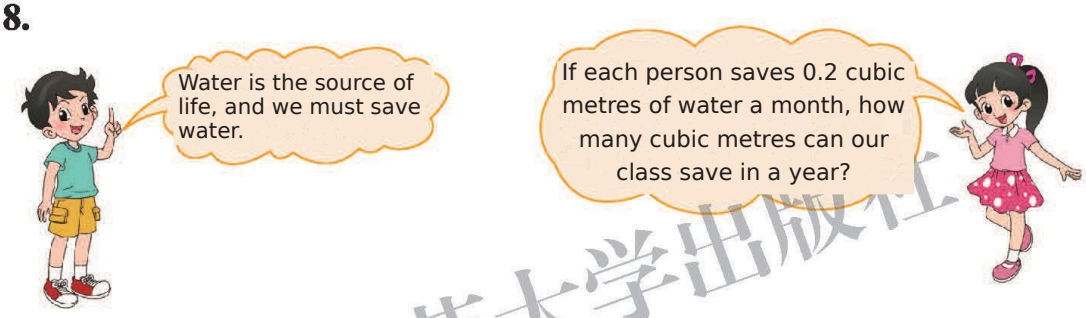
7. Estimate and calculate.

(1) Apples cost ¥7.2 per kg. How much should you pay for 1.8 kg of apples?

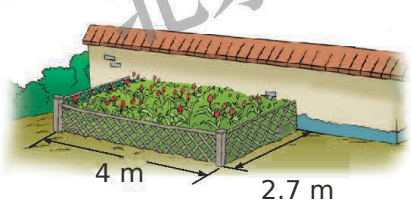


(2) Oranges cost ¥3.6 per kg. How much should you pay for 4.2 kg of oranges?

8.



9. A gardener used a fence against a wall to make a rectangular flower bed.



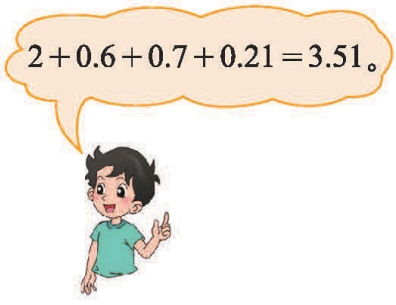
(1) How long does the fence need to be?

(2) How many square metres is the flower bed?

(3) If 0.6 kg of fertiliser is used per square metre, how many kg of fertiliser does this flower bed need?

10. Taoqi calculated 2.7×1.3 this way. Can you understand it? Find a problem and try it too.

	2	0.7
0.3	( )×( )	0.3×0.7
1	2×1	( )×( )



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Review

What I Learned

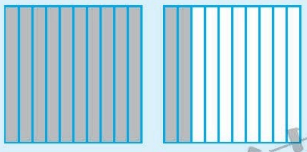
I can show 1.2 in many ways.

¥1 ¥0.2

¥1.2 is 1 yuan 2 jiao

Ones		Tenths
1	.	2

1 × 1 and 2 × 0.1



1.2 = 1 + 0.2

I have learned more about shapes.

Triangles { acute, right-angled, obtuse }

The sum of the interior angles of any triangle is 180°. The sum of any two sides of a triangle is greater than the third side.

Let me sum up decimal calculations.

$$\begin{array}{r} 0.15 \\ +\ 0.05 \\ \hline 0.20 \end{array} \quad 0.15 + 0.05 = 0.2$$

$$\begin{array}{r} 48 \\ \times\ 97 \\ \hline 336 \\ 432 \\ \hline 4656 \end{array} \quad \begin{array}{l} \xleftarrow{\times 10} \\ \xleftarrow{\times 10} \\ \\ \\ \xrightarrow{\div 100} \end{array} \quad \begin{array}{r} 4.8 \\ \times\ 9.7 \\ \hline 336 \\ 432 \\ \hline 46.56 \end{array}$$

To add or subtract decimals, line up the decimal points. Once they are lined up...

Multiply as with whole numbers. The product has as many decimal places as the two factors together.

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My Progress

I know what decimals mean, and I have learned to add, subtract and multiply decimals. I can solve more real-life problems now!

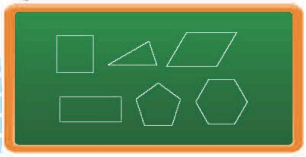
I can use drawings to help me understand and communicate.

The maths activity I remember best: tearing off the three corners of a triangle and putting them together to make a straight angle.

My Questions

Why is a triangle stable, but a quadrilateral is not?

The angle sum of a triangle is 180°. What is the angle sum of other shapes?



Can decimals be infinitely large?

Can decimal division also be turned into whole-number division?

My Maths Diary

Write a maths diary entry about the maths you have learned so far.

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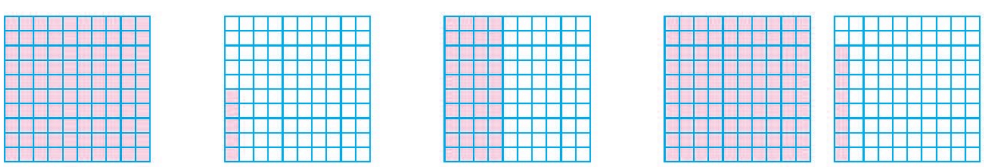
Consolidate and Apply

1. Write a decimal for the shaded part of each figure below.



1 ( ) ( ) ( )

2. Find the pattern and fill in the numbers.

(1) 1.5, 11.5, 21.5, _____, _____。

(2) 19.8, 18.6, 17.4, _____, _____。

(3) 1.2, 2.4, 4.8, _____, _____。

3. Think and fill in.

$7 \times 2 = \square$	$3 \times 6 = \square$	$15 \times 2 = \square$
$0.7 \times 2 = \square$	$0.3 \times 6 = \square$	$1.5 \times 2 = \square$
$0.7 \times 0.2 = \square$	$0.3 \times 0.6 = \square$	$1.5 \times 0.2 = \square$

4. Write each decimal below in the box where it fits.

28.4　7.59　0.85　80.6　29.01

$7 < \square < 8$　　$1 > \square > 0$

$80 < \square < 81$　　$29 > \square > 28$

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5. A diving competition usually has required dives and optional dives. After the required dives, divers A and B were first and second with 287.74 points and 262.65 points.

(1) At this point, what is the difference between their scores?

(2) The table shows the scores of divers A and B in the first three optional dives.

	Round 1	Round 2	Round 3	Total
A / points	51.09	70.48	65.52	
B / points	69.90	53.21	75.08	

After three rounds of optional dives, has the gap between the two divers' scores got smaller? In the remaining dives, at least how many more points than A must B score to have a chance of winning?

6.

3.8×4	48×0.15	0.125×8
1.8×0.85	0.86×1.2	12×0.16

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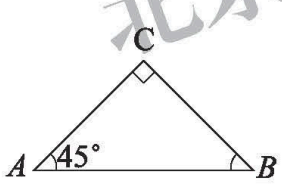
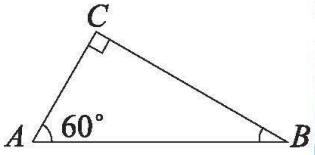
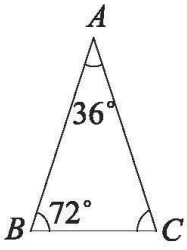
7. Fill in.



$\angle B =$ ____



$\angle B =$ ____



$\angle C =$ ____

8. Put a "√" under each set of sticks that can form a triangle. (Unit: cm)

(1) 6, 4, 5 ☐

(2) 2, 3, 6 ☐

(3) 4, 8, 4 ☐

(4) 4, 4, 4 ☐

9.

$12.8+2.04\times9.5$ $\qquad$ $4.5\times3.6+5.2\times0.35$

$6.15\times9.4-42$ $\qquad$ $4.5\times0.9+5.5\times0.9$

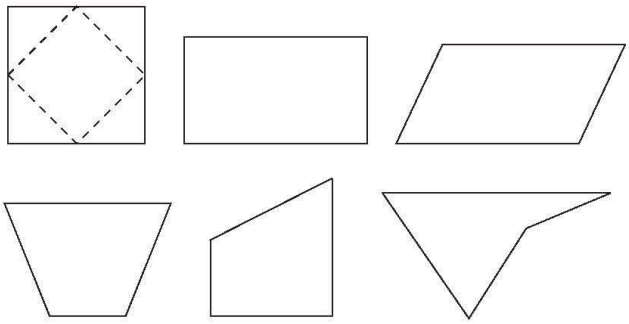
10. Xiaohong is going to the Great Wall with her mum and dad. How much will their return fares cost?

Single fare: ¥24.5 per person. Children half price.

11. Maths game: Guess the shape.

Play in groups of four. One person secretly draws a shape; the other three take turns asking questions to guess the shape. The first to guess it wins.

12. Mark the midpoint of each side of each quadrilateral below, then join the midpoints of adjacent sides to make a new shape. What do you notice?

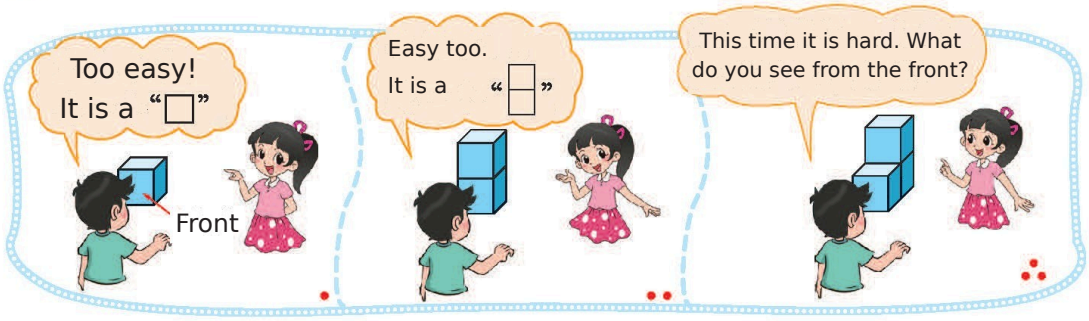


The point that divides a line segment into two equal parts is its midpoint.

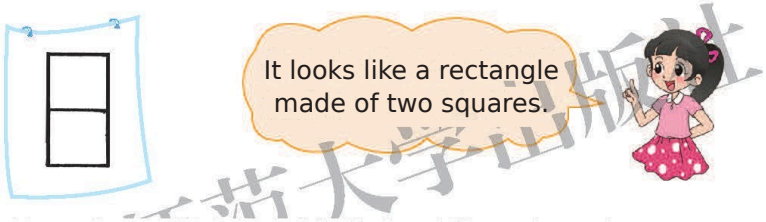
Unit 4 Observing Objects

Look

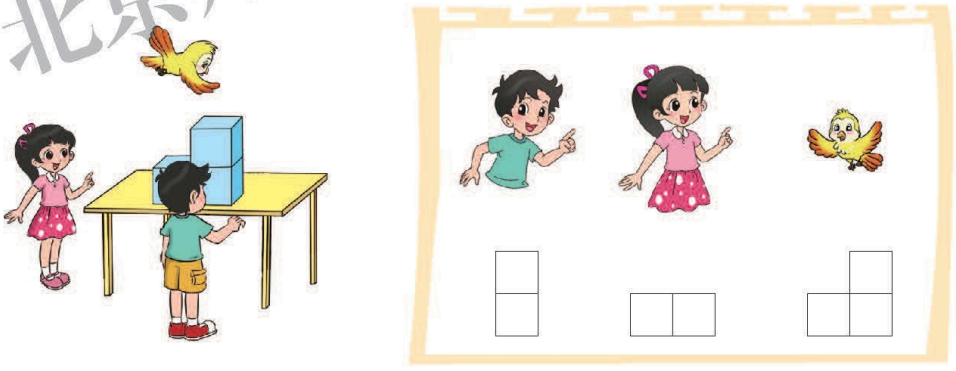
I build, you draw. What do you see from the front?



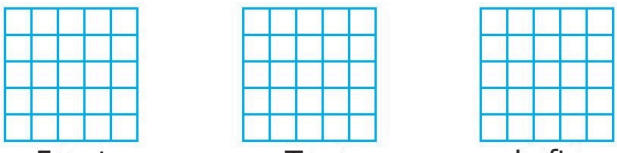
Look at the shape Taoqi drew, then think again.



What shapes do Taoqi, Xiaoxiao and the bird see? Match them.

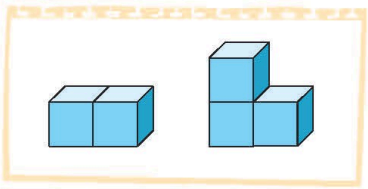


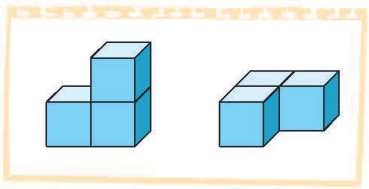
Build this , look at it, and draw the shapes you see from the front, the top and the left on grid paper.



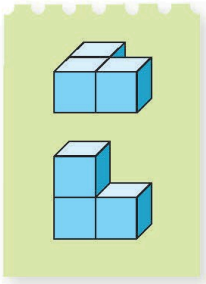
Practice

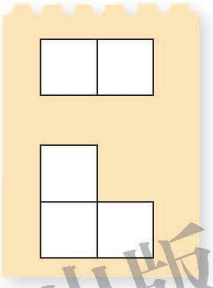
1. Build them and think. Do they look the same from the front?

(1) 

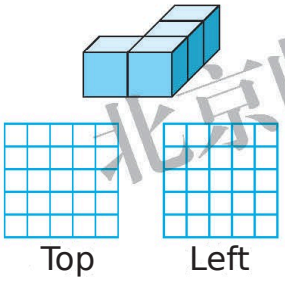
(2) 

2. Think. What shape do you see from the top? Match them.





3. Think and draw on the grid paper.

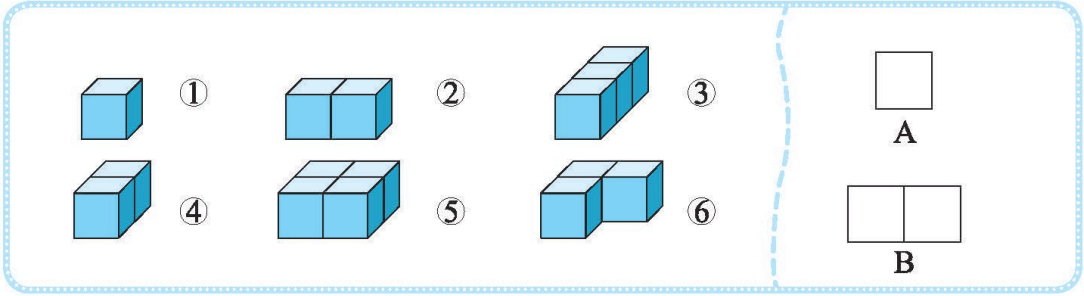


(1) What shape do you see from the top?

(2) What shape do you see from the left?

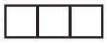
(3) Can you build other solid figures with these cubes? Try it and observe.

4. Think and fill in.



(1) Which look like Figure A from the front? _____

(2) Which look like Figure B from the front? _____

5. Build with 4 cubes so the front view is  Find 3 ways.

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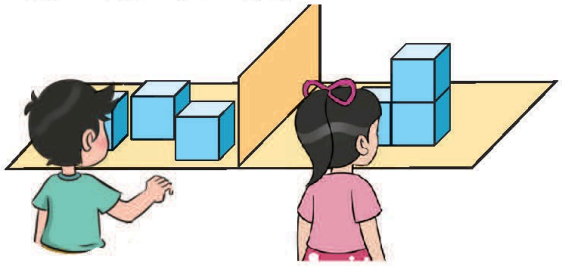
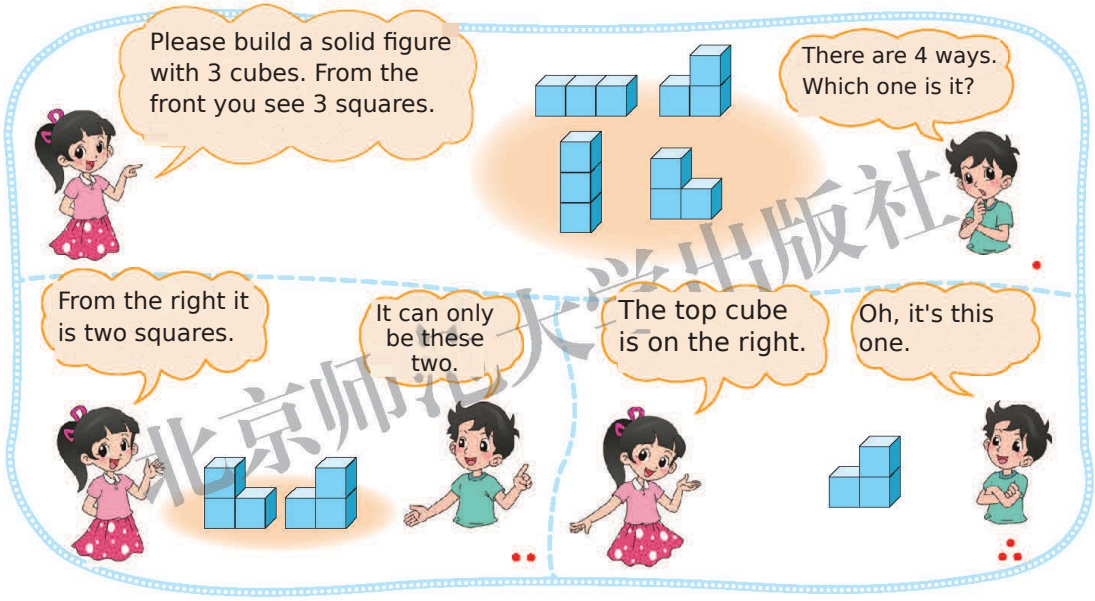
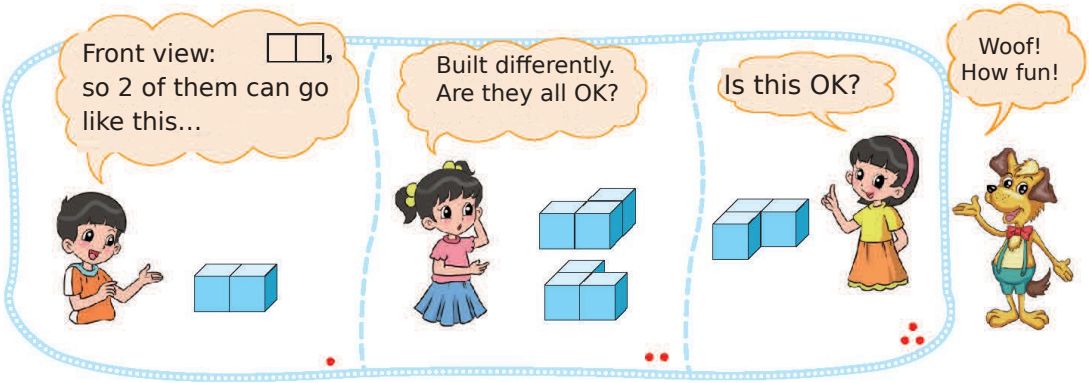
I Say, You Build

Xiaoxiao built a solid figure, and Taoqi built the same one by following Xiaoxiao's instructions. Look, talk and do.





Taoqi used 3 cubes to build another solid figure. From the front it is two squares side by side. Where can the third cube be placed?



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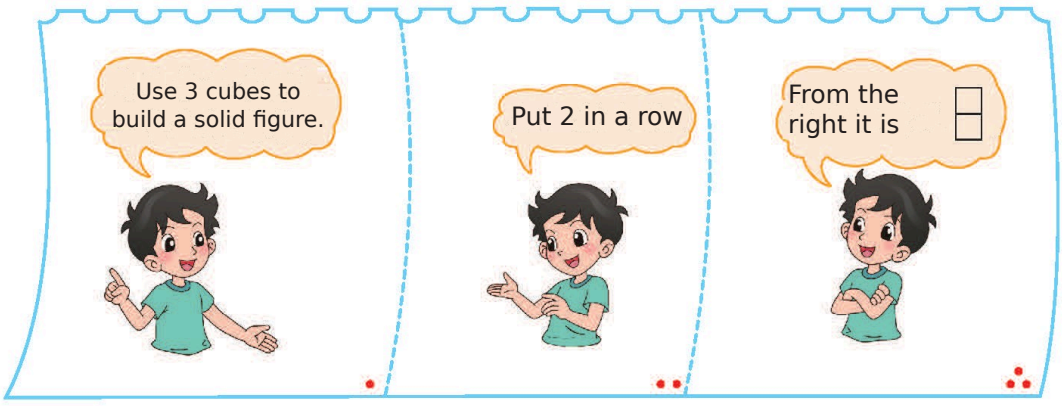
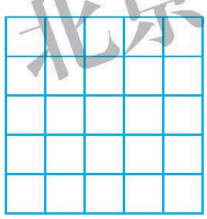
Practice

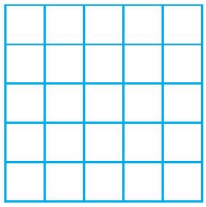
1. Build it following Taoqi's instructions.



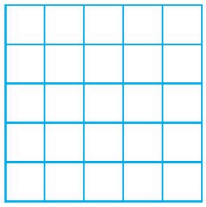
2. Xiaoxiao used 3 cubes to build a solid figure. From the front it looks like 2 squares. Try to build it in two ways, and draw the shapes you see from the front, right and top.



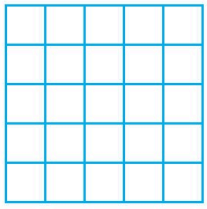
Front



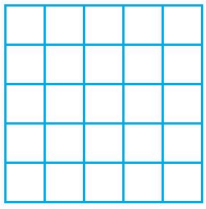
Right



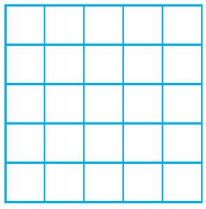
Top



Front



Right



Top

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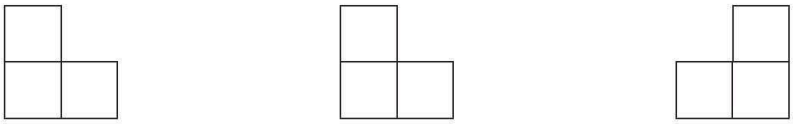
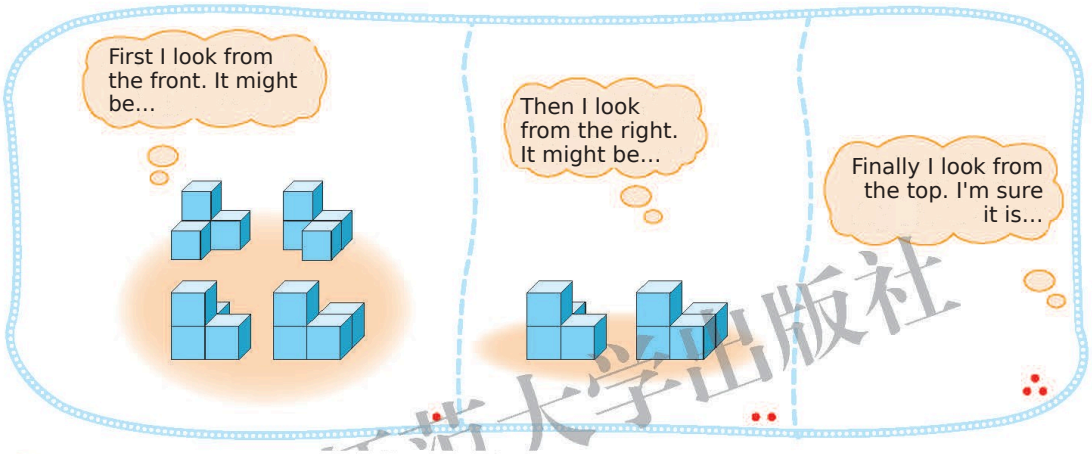
Build It

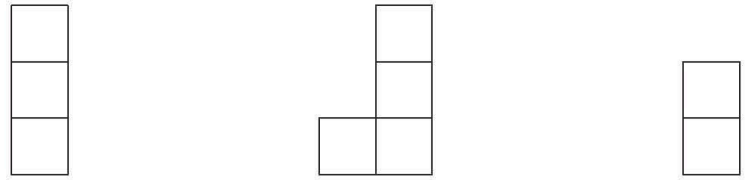
Taoqi used 4 cubes to build a solid figure. The shapes seen from the front, the right and the top are shown below. Can you build this solid figure?



Front Right Top

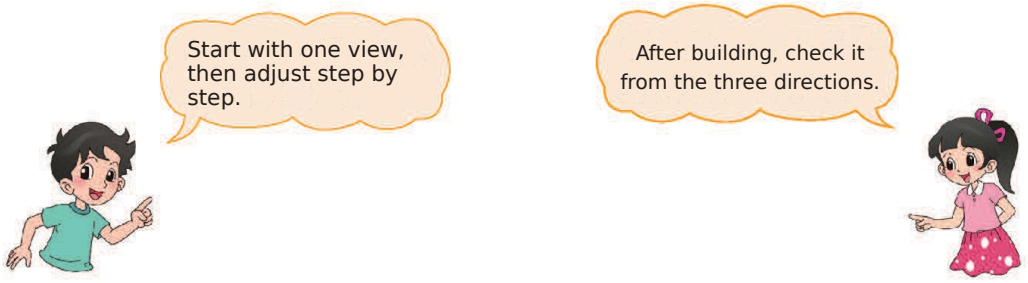


Xiaoxiao used 4 cubes to build a solid figure. The shapes seen from the front, the right and the top are shown below. Please build it.



Front Right Top

Share your good methods with a partner.



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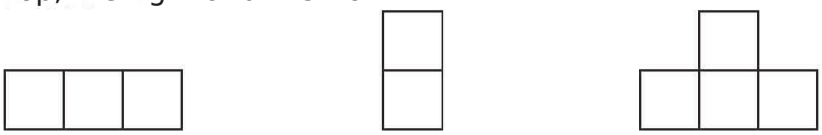
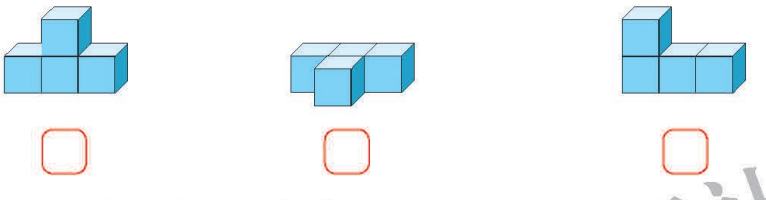
Practice

1. Xiaoxiao used 4 cubes to build a solid figure. Here are the shapes seen from the top, the right and the front.



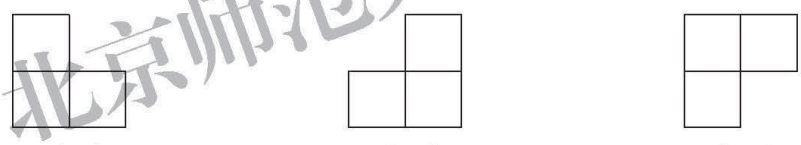
Top Right Front

(1) Which one is it? Put a ✓ under it.



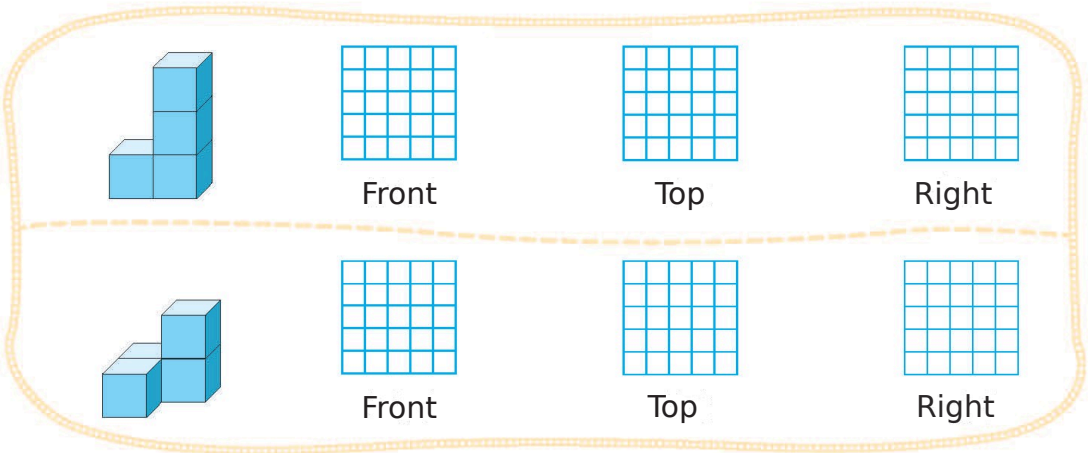
(2) Build it and have a look. Did you choose correctly?

2. Taoqi used 4 cubes to build a solid figure. Here are the shapes seen from the front, the right and the top. Can you build it?



Front Right Top

3. Draw the shapes of each solid seen from the front, the top and the right, then look at the real solid to check.



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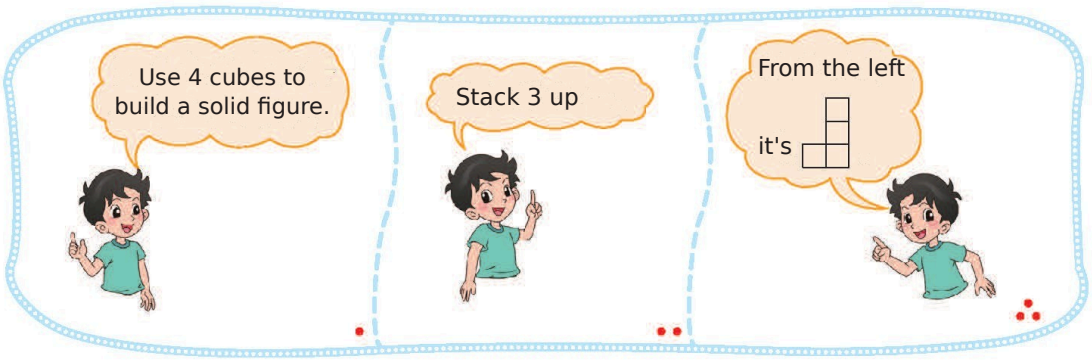
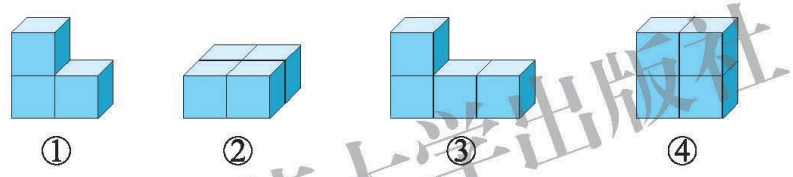
Exercise 4

1. Build it following Taoqi's instructions.



2. What shape is each solid figure below seen from above? Build them and match.



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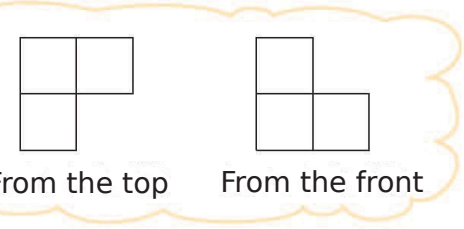
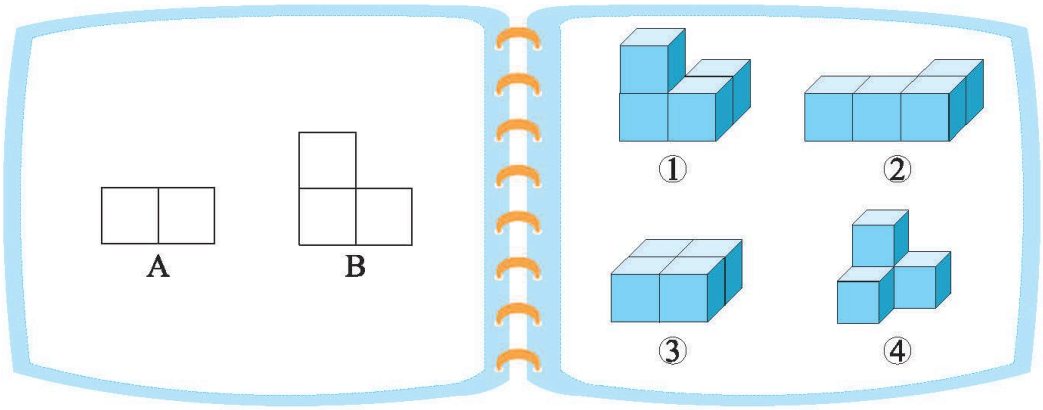
Solids whose right view is : ( ) of them. They are ( )

3. Build it.



Which one is built correctly? Put a ✓ by the correct one.

4.

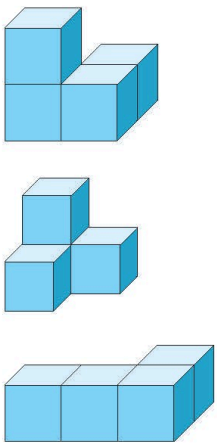


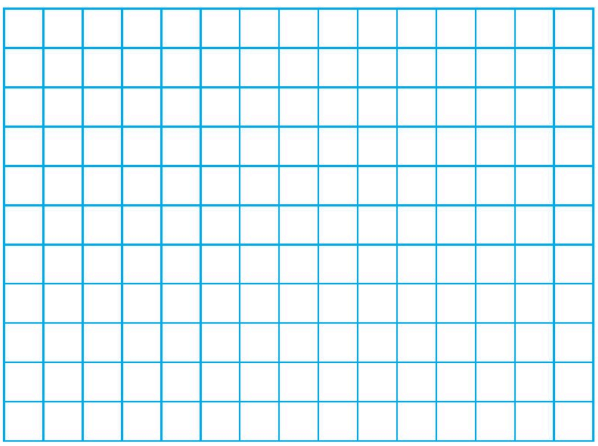
(1) Left view is Figure A: _____

(2) Front view is Figure B:_____

(3) What other solids look like Figure A from the left? What other solids look like Figure B from the front? Think and build them.

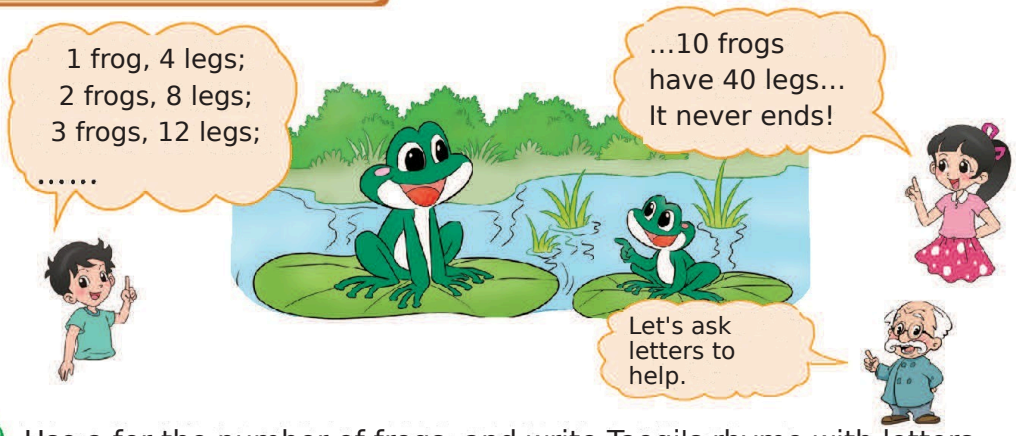
5. What shapes do you see when you look at the solids below from the front, the top and the right? Draw them on the grid paper.





Unit 5 Getting to Know Equations

Letters for Numbers

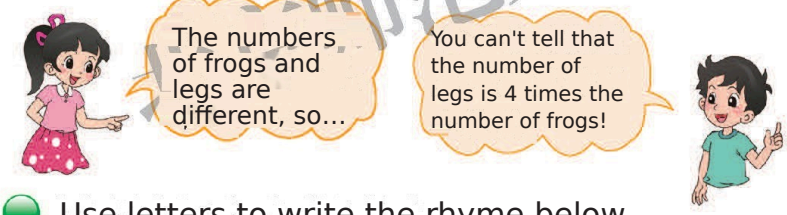


- Use a for the number of frogs, and write Taoqi's rhyme with letters.
- Do you agree with the ideas below? Give your reasons.

a frogs a legs.

a frogs b legs.

a frogs 4 × a legs



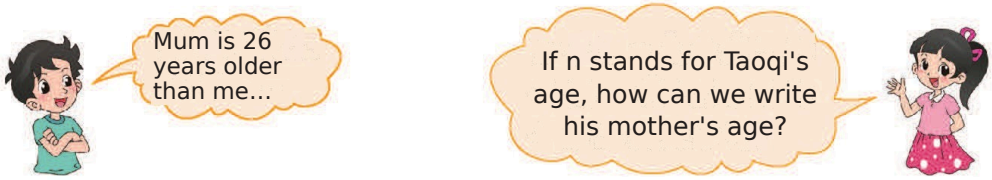
- Use letters to write the rhyme below.

1 frog, 1 mouth,
2 eyes, 4 legs;
2 frogs, 2 mouths,
4 eyes, 8 legs;
.....

a frogs have a mouths, 2 × a eyes and 4 × a legs.

x frogs have x mouths, 2 × x eyes and 4 × x legs.

- Talk about when else letters stand for numbers in daily life.



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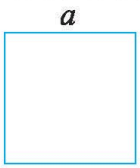
Try It

How do you calculate the perimeter of a square? Can you write it with letters?



C stands for perimeter:

C=4 × a.

4 × a can be written as 4·a or 4a; the number is usually written before the letter.

What other everyday problems can be shown by 4a?

1 table has 4 legs; a tables have 4a legs.

1 book costs ¥a; 4 books cost ¥4a.

Use letters to write the formulas and laws of operations you have learned.

Area of a square = side × side

S= a × a

a × a can be written as a^2.

Practice

1. Can you sum up each rhyme below in one sentence? Share your ideas with a partner.

1 plum blossom has 5 petals;

2 plum blossoms have 10 petals;

3 plum blossoms have 15 petals;

……

1 week has 7 days;

2 weeks have 14 days;

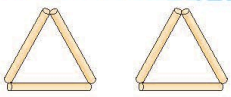
3 weeks have 21 days;

……

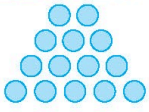
2. Fill in.

1 hand has 5 fingers, 2 hands have 10 fingers, n hands have ___ fingers.

Halley's Comet can be seen only once every 76 years. It appeared in the year s, so it will next appear in the year ___.



Making 1 triangle needs 3 sticks, making 2 triangles needs 6 sticks, making a triangles needs ___ sticks.



There are 2 counters in the 1st row, 3 counters in the 2nd row and ___ counters in the nth row.

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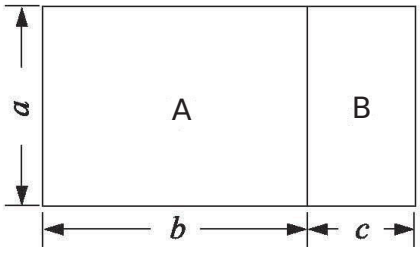
3. Fill in.

(1) An ostrich runs
___ km in 2 h,
___ km in 3.5 h,
___ km in t h.

An ostrich runs at 70 km/h

(2)



Perimeter of rectangle A: _____,

Perimeter of rectangle B: _____.

(3) Xiaoxiao has ¥20 and spends ¥a on a schoolbag. She has ¥___ left.

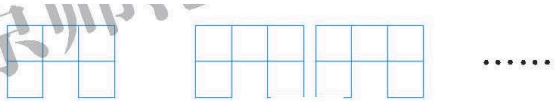
(4) A rectangle is 80 cm wide and x cm long. Its area is ___ cm^2.

(5) Pick an expression with letters from above and give 2 real-life examples it can represent.

"cm^2" means square centimetres.

4. Arrange shapes.

(1) Taoqi makes gates with small squares.



1 gate 2 gates .....

1 gate needs ___ small squares, 2 gates need ___ small squares, and n gates need ___ small squares.

(2) Xiaoxiao makes rectangles with small squares.

1st 2nd 3rd .....

Rectangle	1st	2nd	3rd	.....	nth	.....
Squares needed	2	4		...		...

(3) Make shapes with small squares following a pattern, and use a letter to show the number of small squares needed.

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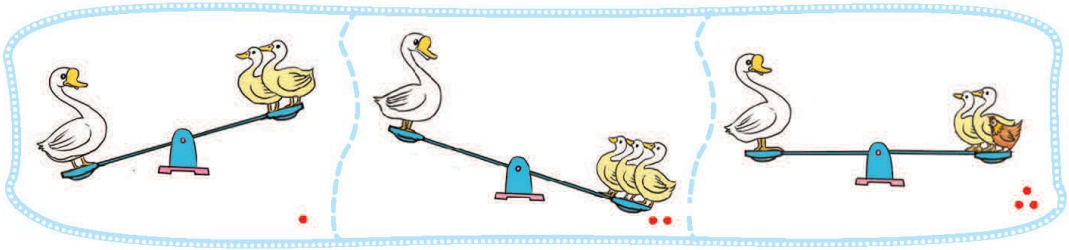
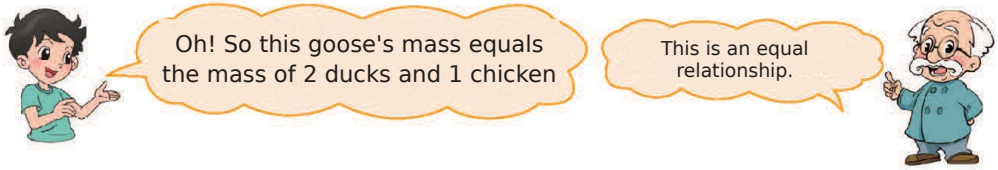
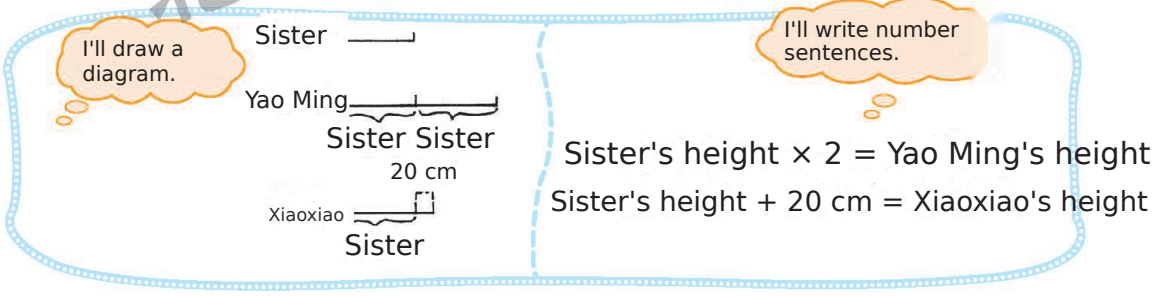
Equal Relationships

Discuss: when are they equal?





Show how Sister's height relates to Yao Ming's and Xiaoxiao's heights.



They also found equal relationships like these. Can you understand them?

Yao Ming's height ÷ 2 = Sister's height

Xiaoxiao's height − 20 cm = Sister's height

Yao Ming's height ÷ 2 = Xiaoxiao's height − 20 cm

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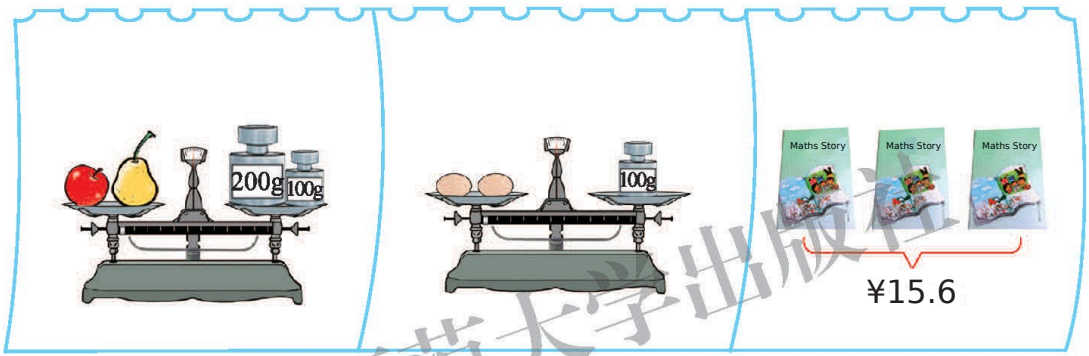
Practice

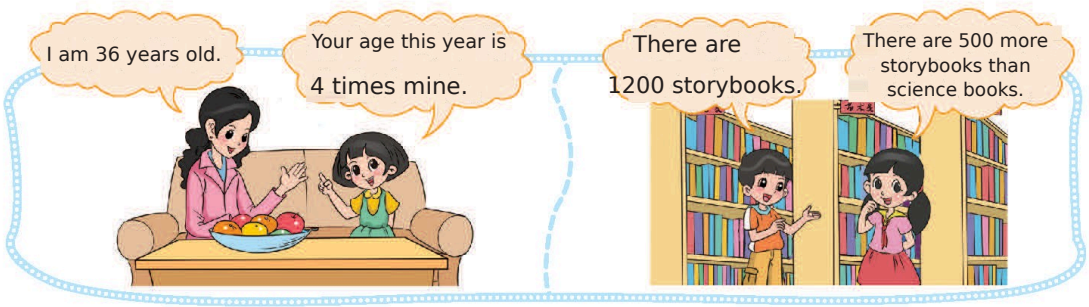
1. When are they equal? Can you state the equal relationships?

2. Show the equal relationships between the quantities below.

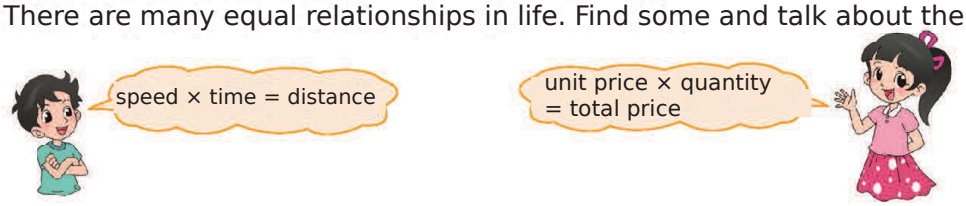


3. The length, width, perimeter and area of a rectangle are shown by a, b, C and S. What equal relationships can you write?

4. Talk about the equal relationships between quantities in the situations below.



5. There are many equal relationships in life. Find some and talk about them.



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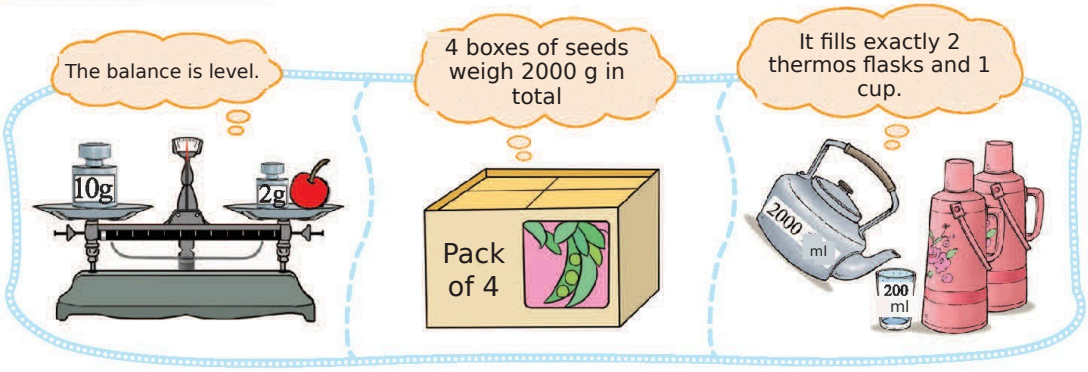
Equations



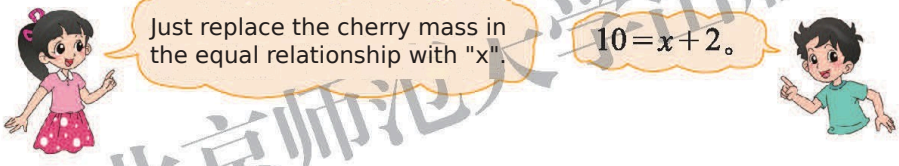
Talk about the equal relationships in each picture above.

10 g = mass of the cherries + 2 g

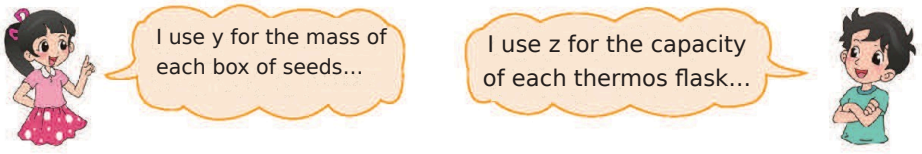
mass of 1 box of seeds × 4 = 2000 g

2000 ml = capacity of each flask ×2+200 ml

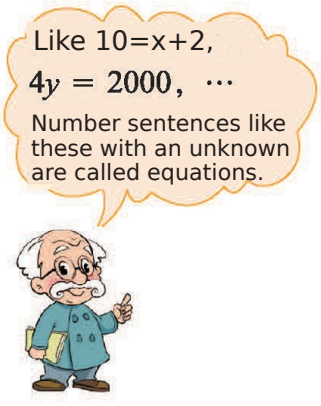
Use x for the cherries' mass and write the balance's equal relationship.



As above, show the equal relationships in the other situations.



What do the equations above have in common? Discuss with your partner.

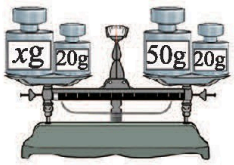


Around 250 CE, the ancient Greek mathematician Diophantus wrote a great mathematical work, Arithmetica, introducing the concept of the unknown and using symbols to represent unknowns. This was an important event in the history of mathematics, opening the door to symbolic algebra. Therefore, people often call Diophantus the "father of algebra".

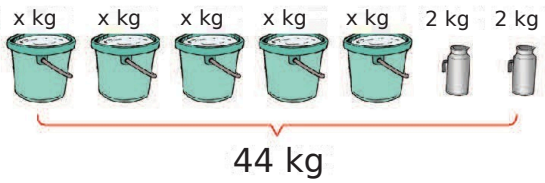
Practice

1. First talk about the equal relationship in each picture, then write an equation.

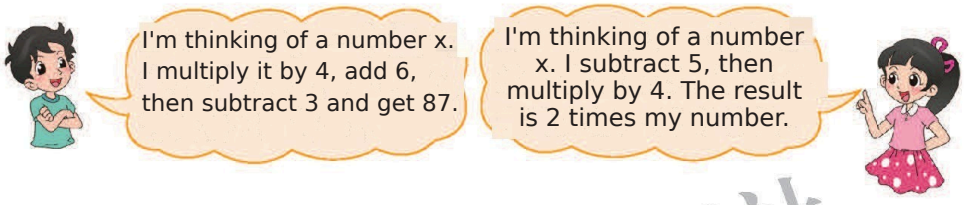
(1)



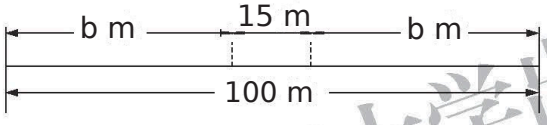
(2)



(3)



(4)



2. For each problem, first state the equal relationship, then write the equation.

(1) When a bus arrives at a stop, 5 people get off and 8 people get on. There are now 15 people on the bus. There were originally x people on the bus, so _____.

(2) Remember the shape game on page 63? If 95 small squares make x gates, then _____.

3. Calendar patterns.

Look carefully at how the middle number in the shaded box relates to the other four numbers.

Calendar

		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

(1) The middle number is y. The number on the left is ___, the number on the right is ___, the number above is ___ and the number below is ___.

(2) How is the sum of the 5 numbers related to the middle one?

(3) When the sum of the 5 numbers is 115, what is the middle number?

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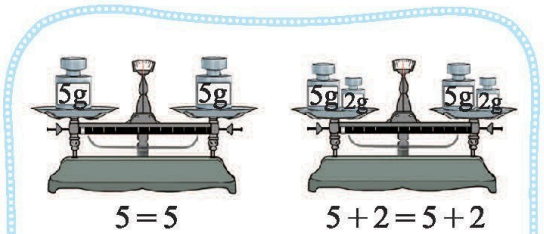
Solving Equations (1)

Look at the pictures below. What pattern do you notice?

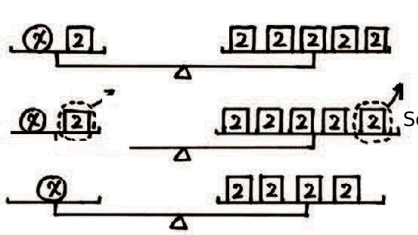


Add the same mass to both sides and the balance stays level.

Take the same mass off both sides and the balance stays level.

Add (or subtract) the same number on both sides, and the equation still holds.

Can you use the pattern you found to solve the equation we wrote earlier?



Solution: $x+2=10$

$x+2-2=10-2$

$x=8$

Subtract 2 from both sides

8+2=10, so x=8 is correct.

Solve the equations.

$y-7=12$

Solution: y-7+7=12 ◯ ☐

$y=$ ☐

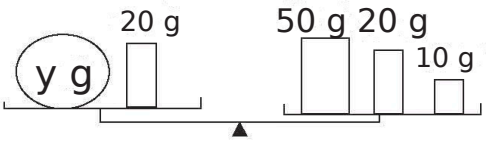
$23+x=45$

Solution:

Practice

1. Draw a diagram or give an example to explain the meaning of the following sentence: if the same number is added to (or subtracted from) both sides of an equation, the equation still holds.

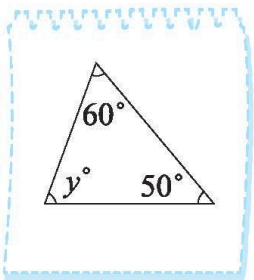
2. Write an equation from the picture and solve it.



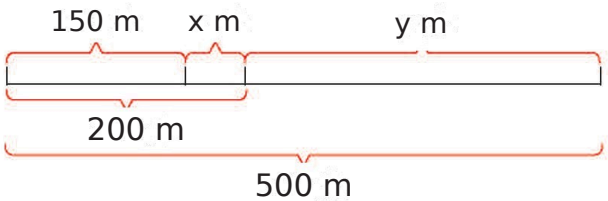
3. Solve the equations.

$x-19=2$ $x-12.3=3.8$

4. Write an equation from each picture and solve it.



5. Look at the picture and answer the questions.



(1) Which part of the line is (200 − x) m long?

(2) Which part of the line is (200+y) m long?

(3) From the figure above, can you write two different equations?

6. Grow a pot of garlic sprouts at home or at school. Measure their height every 2 days and record the data.

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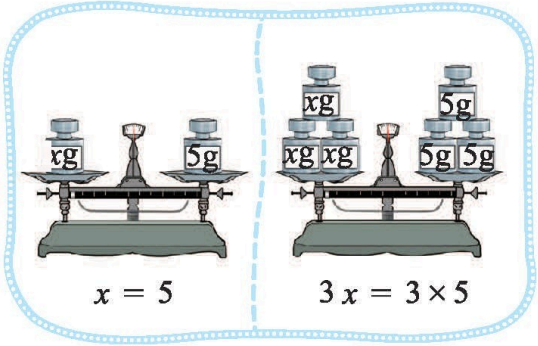
Solving Equations (2)

If both sides of an equation are multiplied by the same number (or divided by the same number that is not 0), does the equation still hold? Share your ideas with a partner.

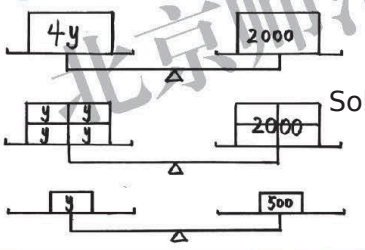


$x = 5$ $\quad 3x = 3 \times 5$ $\quad 2x = 20$ $\quad 2x \div 2 = 20 \div 2$

If both sides of an equation are multiplied by the same number, the equation holds.

If both sides of an equation are divided by the same number (not 0), the equation holds.

Use the pattern you found to solve the equation we wrote earlier.



$4y = 2000$

Solution: $4y \div 4 = 2000 \div 4$

$y = 500$

Divide both sides of the equation by 4.

$4 \times 500 = 2000$, y=500 is correct

Solve the equations.

$x \div 3 = 9$

Solution: $x \div 3 \times 3 = 9$ ○ □

$x =$ □

$7y = 28$

Solution:

Are the solutions below correct? Discuss with a partner.

$x - 19 = 19$

Solution:
x-19+19=19-19

$x = 0$

$3x = 36$

$= 36 \div 3$

$x = 12$

Both sides of the equal sign must be equal.

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Practice

1. Draw a picture or give an example to explain what the sentence below means: if both sides of an equation are multiplied by the same number (or divided by the same number that is not 0), the equation still holds.

2. Forest Doctor

$7x=35=35\div7=5$

$x-5=8$

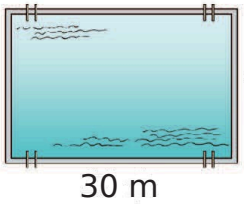
Solution: x-5-5=8-5

$x=3$

3. Solve the equations.

$6x=156$	$3x=630$	$x-47=47$
$59+x=120$	$x\div52=7$	$x\div28=0$

4. A rectangular swimming pool has an area of 600 m² and is 30 m long. How many metres wide is it?



5.

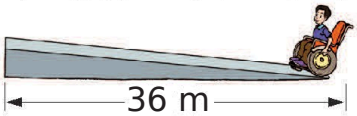
(1) What is the side length of this square flower bed? Write an equation and solve it.

(2) If the bed is changed to a rectangle with the same perimeter and a width of 4 m, how long is it? Write an equation and solve it.

6. To make it easier for wheelchair users to get around, a city passed a rule about the height of ramps in front of buildings: for every 1 m of ramp height, at least 12 m of horizontal length is needed.

(1) What is the least horizontal length for a ramp 2 m high? And for 4 m or x m?

(2) The open space in front of a building is 36 m long. What is the greatest possible height of the ramp here?



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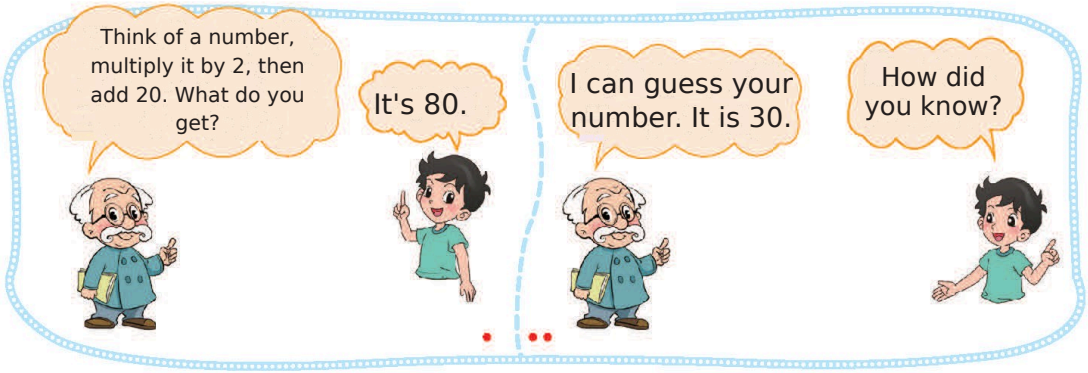
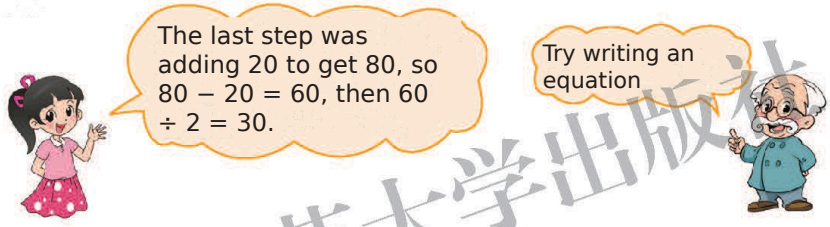
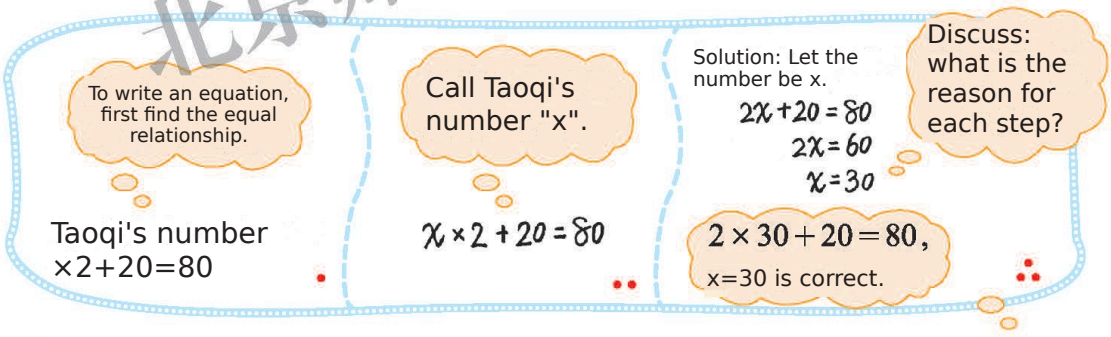
Guess the Number



- Do you understand this game? How did the Wise Old Man guess the answer? Think about it and discuss.



- Try solving this problem with an equation.



- Play the number-guessing game with a partner, write an equation and solve it.

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Practice

1. Play the number-guessing game with a partner using Xiaoxiao's rule, and write an equation to solve it.

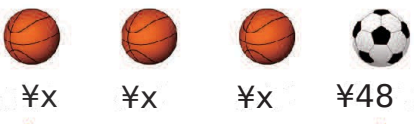
Think of a number, multiply it by 4, then add 10. What do you get?

2. Solve the equations.

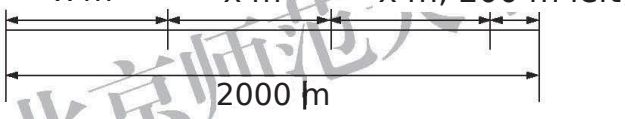
$3x+6=15$ $4x-2=26$ $8+x=20$

3. Look at the picture, discuss the equal relationship, then write an equation to solve the problem.

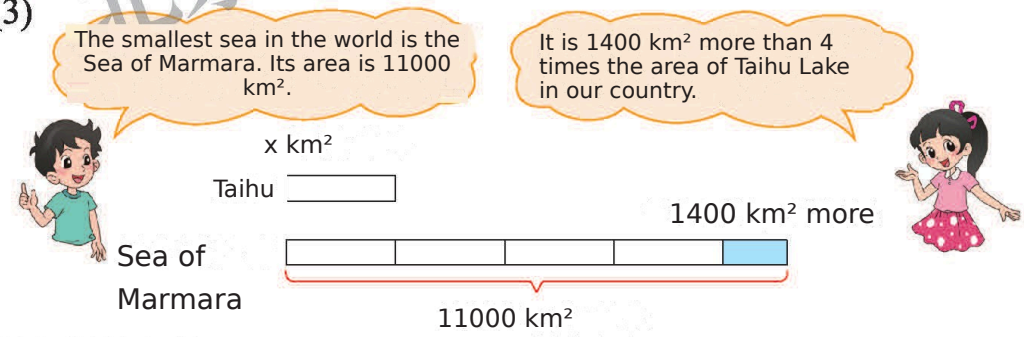
(1)



(2)



(3)



4. Write an equation and solve it.

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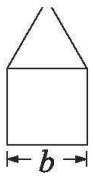
Exercise 5

1. Fill in.

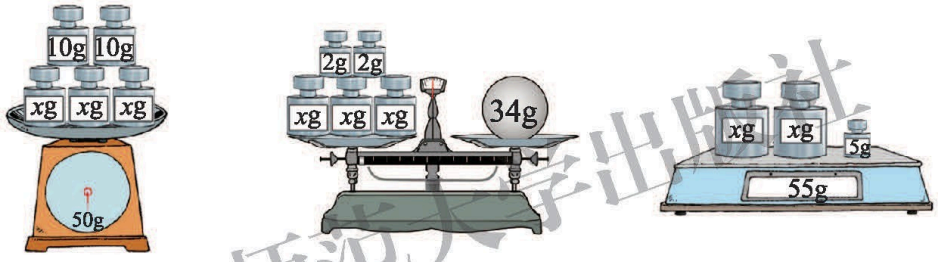
(1) An adult's foot is about 1/7 of their height. If an adult's height is a m, their foot is about ( ) m long.

(2) The figure on the right is made of an equilateral triangle and a square. Its perimeter is ( ).



(3) Apples cost ¥7.2 per kg and pears ¥5 per kg. Buying x kg of apples and y kg of pears costs ¥( ) altogether.

2. State the equal relationship in each picture, then match it to an equation.



$3x+4=34$　　$2x+5=55$　　$2x+20=50$　　$3x+20=50$

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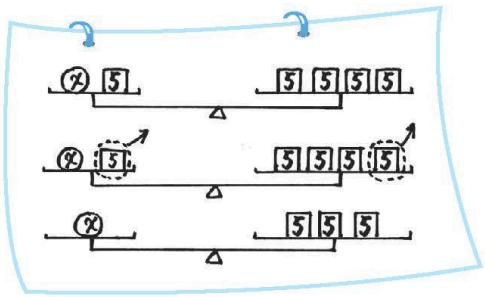
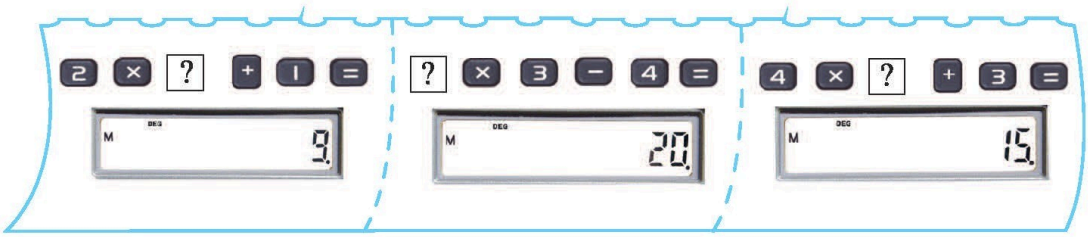
3. Below is how Taoqi solved x + 5 = 20 by drawing.



(1) Can you understand it? Discuss with a partner.

(2) Solve 3x=18 and discuss with a partner the reason for each step.

4. “ ? ” stands for what number?

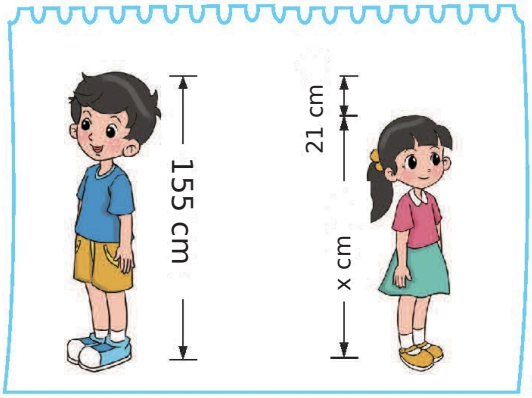


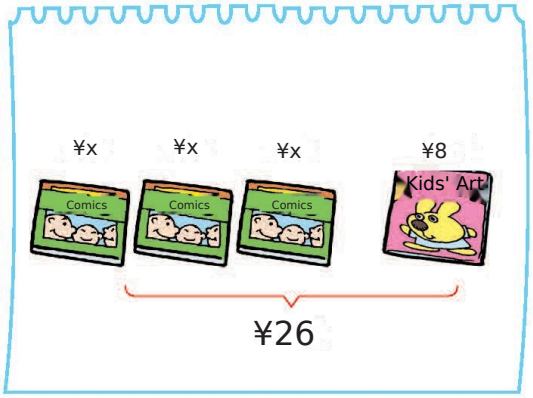
5. Solve the equations.

$x+23.5=45$ $\qquad$ $4x+9=249$

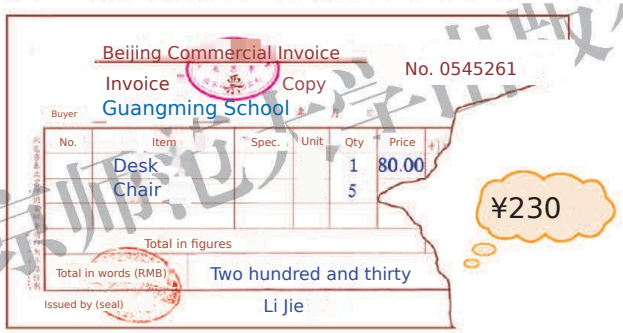
$36+2x=48$ $\qquad$ $m\div 0.6=0.45$

6. First state the equal relationship, then set up an equation and solve it.

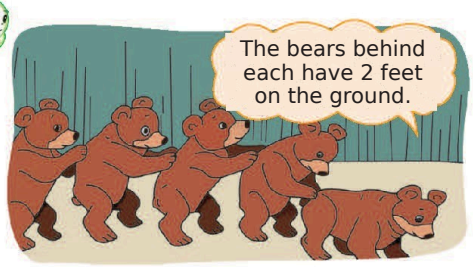




7. Below is a receipt with a torn corner. Can you calculate the unit price of each chair?



8.



(1) If 2, 3, 4, ..., n bears are performing, how many feet are on the ground in each case?

(2) If there are 26 feet on the ground in total, how many bears are performing?

Did You Know?

More than 3600 years ago, an ancient Egyptian papyrus already had problems that amount to solving simple equations. In China, The Nine Chapters on the Mathematical Art, written around the 1st century AD, records solving problems with sets of equations, so it already had the idea of equations. Around the 12th century, Chinese mathematicians began to use special symbols for unknowns.

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Tiling

Floors and walls are often tiled. Here are some common tiling patterns.

Task

Can triangles tessellate? Can quadrilaterals?

Design Plan

1. What main steps are needed to solve this problem?

2. How do you want to solve the problem? (Work independently / Work in groups) If you work in groups, how will you divide the work?

3. Write down the main steps and the division of work.

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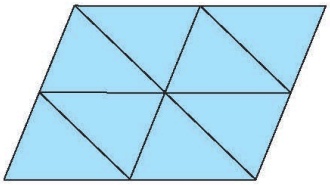
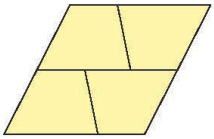
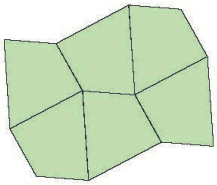
Hands-on

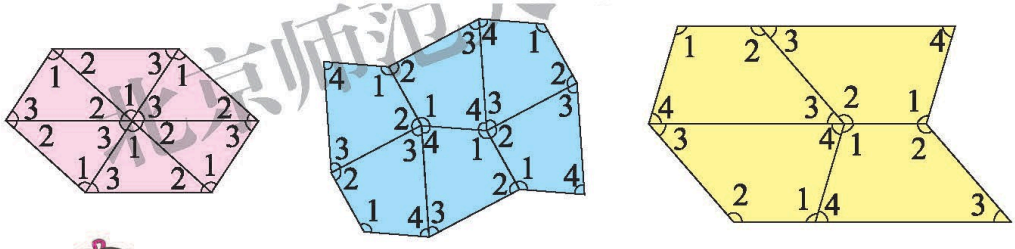
1. Following your design plan, fit and arrange the cut-out triangles or quadrilaterals.

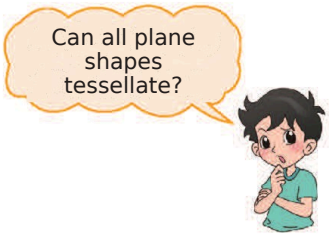
2. Share your tessellations with the class. Can triangles tessellate? Quadrilaterals?

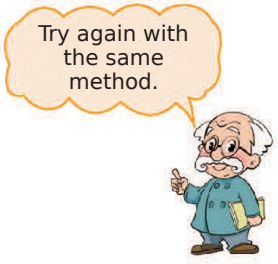
Reflection

1. Try the method below. What do you find?



2. What did you learn from the activity above? What other questions would you like to explore further?





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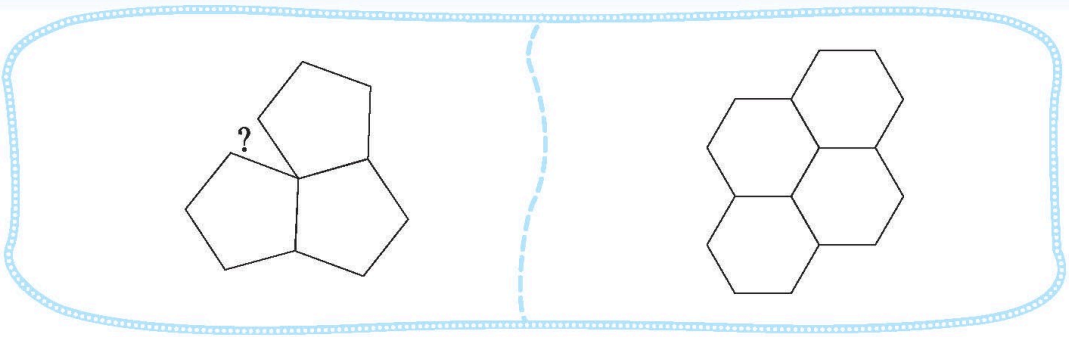
3. Not all plane figures can tessellate. Look and try it.



4. Look at the tessellation patterns below and think about how they are formed.

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Self-Check

In this activity, my performance was (please colour the ☆ after each item; all 5 coloured means the best):

Can plan a sensible way to solve the problem.	☆ ☆ ☆ ☆ ☆
Can cut out the shapes needed and try tessellating.	☆ ☆ ☆ ☆ ☆
Can link tessellating to shapes I have learned.	☆ ☆ ☆ ☆ ☆
Can cooperate and communicate with partners.	☆ ☆ ☆ ☆ ☆
Can find tessellations in everyday life.	☆ ☆ ☆ ☆ ☆

Maths in the Olympics

2004 Athens Olympics
Gold Medal Table

Rank	Country	Gold
1	USA	35
2	China	32
3	Russia	27

2008 Beijing Olympics
Gold Medal Table

Rank	Country	Gold
1	China	51
2	USA	36
3	Russia	23

2012 London Olympics
Gold Medal Table

Rank	Country	Gold
1	USA	46
2	China	38
3	UK	29

Athletics.

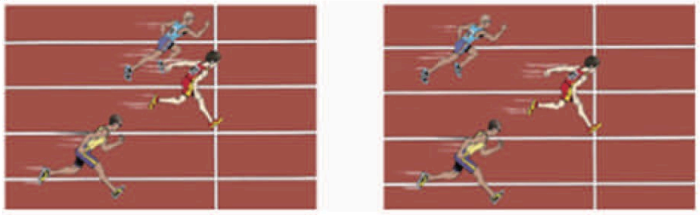
At the 2004 Olympics, Chinese athlete Liu Xiang won the men's 110 m hurdles, breaking the Olympic record for the event at the time and equalling the world record.

1. The table below shows the times of the 1st, 2nd and 3rd place athletes. How many seconds apart are they?

Men's 110 m hurdles final results

1	Liu Xiang	12.91 s
2	Trammell	13.18 s
3	García	13.20 s

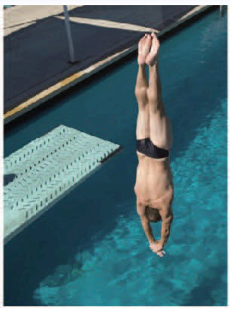
2. Using the data in the table above, decide which of the two pictures below shows the sprint finish of the final.



3. At that time the Olympic record for the men's 110 m hurdles was 12.95 s. How many seconds faster was Liu Xiang?

Diving.

In the men's 3 m springboard at the 2008 Olympics, He Chong went into the final dive 32.45 points ahead of second-placed Despatie, while Qin Kai was 7.65 points behind Despatie in third place. Below are the three athletes' scores for the final dive.



He Chong: 100.70 points; Despatie: 96.90 points; Qin Kai: 98.00 points

1. Before the last dive, how many points was Qin Kai behind He Chong?

2. Who came first, who came second and who came third?

Shooting.

In the women's 10 m air pistol final at the 2012 Olympics, after 7 shots Chinese shooter Guo Wenjun's total was 0.2 points behind French shooter Goberville's. Below are their scores for the 8th and 9th shots.

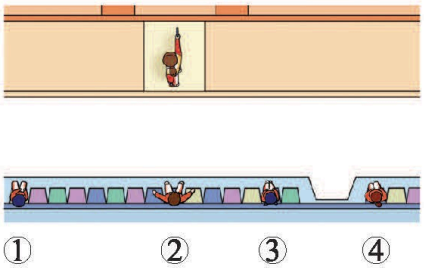
Guo Wenjun	9.8 points	10.4 points
Goberville	10.4 points	10.1 points

The highest score for one shot is 10.9 points.

1. Guo Wenjun scored 10.8 points on the 10th shot. At least how many points must Goberville score to win the gold medal?

2. Goberville scored 8.8 points on the 10th shot. By how many points do their total scores differ?

3. The picture on the left below shows a shooting competition seen from above. From which position, ①, ②, ③ or ④, was each of the four pictures on the right seen?



Optimise

Making tea.

Boil water: 8 min

Wash kettle: 1 min

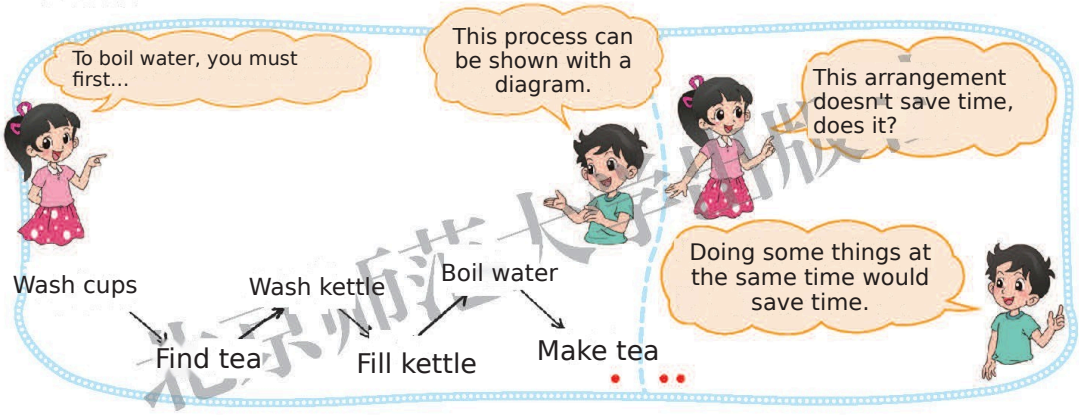
Wash cups: 2 min

Fill kettle: 1 min

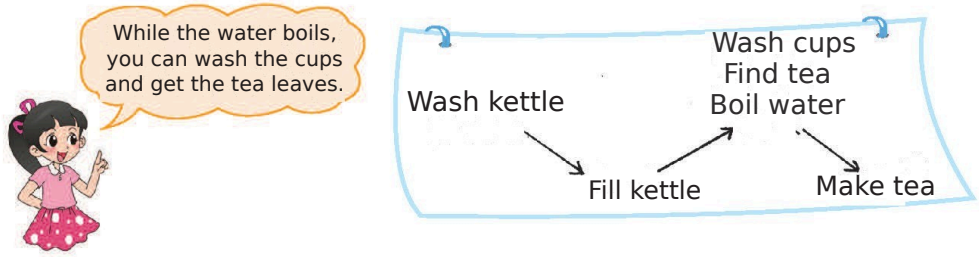
Find tea: 1 min

Make tea: 1 min

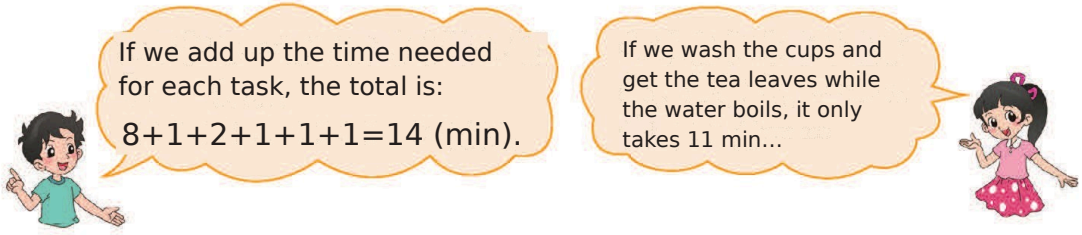
1. To boil water and make Mum a cup of tea, how should you arrange the tasks to save time? Share your ideas.



2. Which things can be done at the same time when making tea? Think and talk about it.



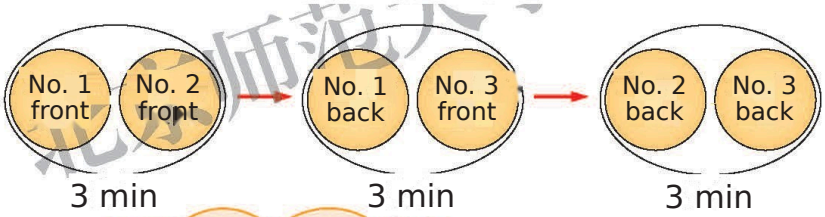
3. How long does it take? Think and discuss.

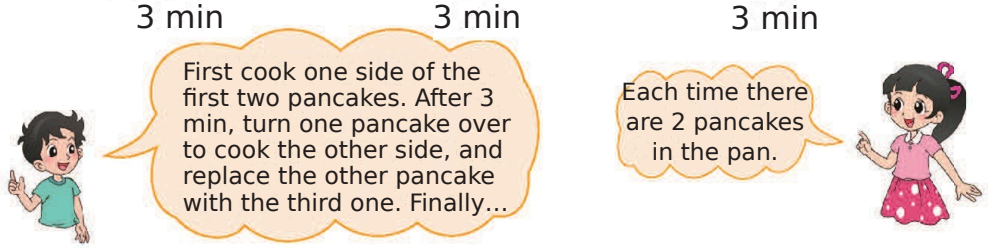


Pancakes.

1. How can we get to eat the pancakes as soon as possible? Share your ideas with your partner.

2. This is how Mum did it. Can you understand it? Talk about it and try it.





3. What if we need to cook 4 pancakes or 5 pancakes? Share your ideas with your partner.

Unit 6 Representing and Analysing Data

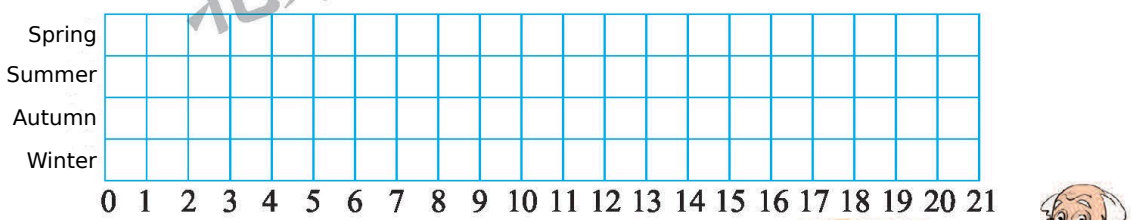
Birthday

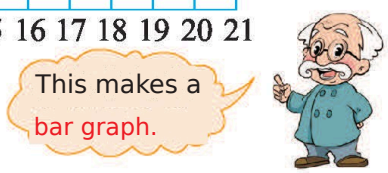
In which seasons are our classmates' birthdays? Let's do a survey together!

For convenience, we agree that:
March to May: spring,
June to August: summer,
September to November: autumn,
December to February: winter.

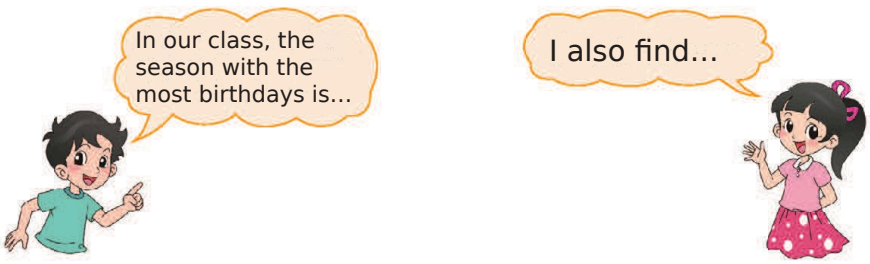
Tell us how you plan to do the survey.

1 square represents 1 person. Colour the squares.





Talk about what you find from the chart above.



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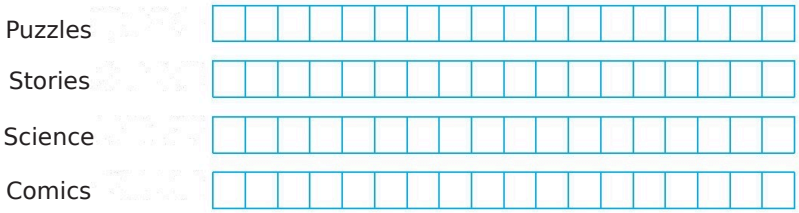
Practice

1. Grade 4 is going to hold a special sports day, and we need to survey which special sports events the students like best. Talk with your classmates about how you would do the survey.

2. Our class will set up a reading corner. Survey which kind of books your classmates like best.

(1) Colour in the graph below.



(2) Fill in.

Type	Puzzles	Stories	Science	Comics
Pupils				

(3) _____ is liked by the most students.

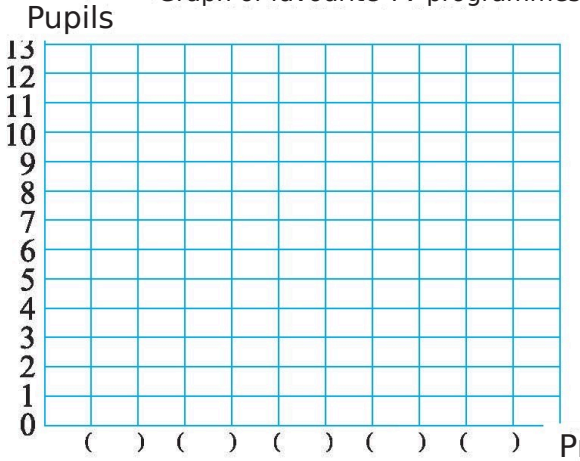
Puzzles and Science are liked by a total of _____ students.

(4) If you were buying books for the class, what would you buy? Share your ideas.

3. Taoqi surveyed the class's favourite TV programmes. Can you complete the chart below?

Programme	Children's	Cartoons	Sports	Films	Military
Pupils	8	12	6	7	3

Graph of favourite TV programmes



(1) The _____ programme is the most popular.

(2) Taoqi's class has _____ students.

(3) Survey your class and compare with the data above.

Growing Garlic Sprouts (1)

How are the garlic sprouts we planted 14 days ago growing?

Share each person's record of the garlic sprouts' growth in your group, and say how you plan to organise the data.

Let's put everyone's garlic sprout heights together and have a look.

We can also use the growth records from day 1 to day 14...

Record the heights of your group's garlic sprouts after 14 days in the table.

____ Group: Table of Garlic Sprout Heights after 14 Days

Name						
Sprout height/cm						

Complete the bar graph below.

____ Group: Bar Graph of Garlic Sprout Heights after 14 Days

Sprout height/cm

( )
( )
( )
( )
( )
( )
( )
( )
0

( ) ( ) ( ) ( ) ( ) ( ) Name

1 square for 1 cm is not enough to draw it. What should we do? Add more squares?

I use 1 square for 2 cm.

I use 1 square to show...

What have you found from the graph above?

The tallest garlic sprout in the group is...

The tallest and shortest garlic sprouts in the group differ in height by...

The garlic sprouts in the group are all about...

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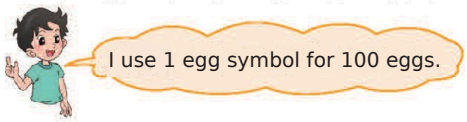
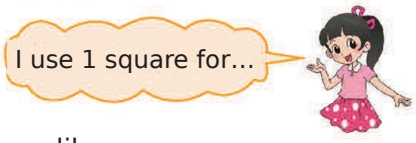
Try It

Eggs sold at Benben Chicken Farm last week

Time	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Number of eggs	300	600	500	500	700	1200	1000

Can you make a graph from the data in the table above?

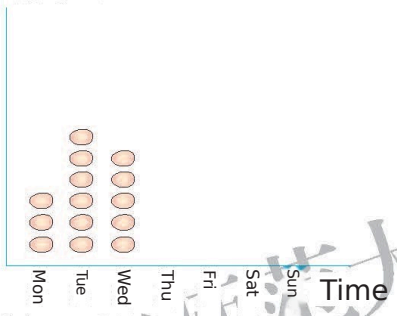




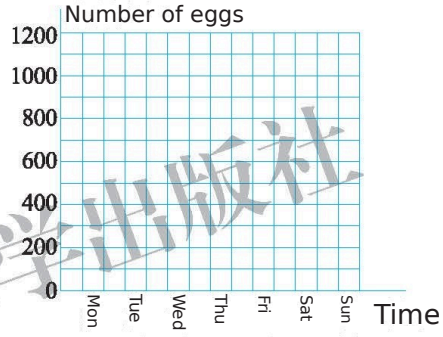
Complete the graph below in the way you like.

Graph of eggs sold last week

Number of eggs



Graph of eggs sold last week



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Practice

1. Do you often help with jobs at home that you are able to do?

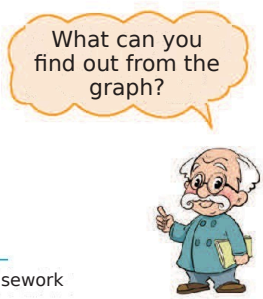
(1) Discuss with a partner how to collect and sort data on classmates' housework.

(2) The table below shows the housework done by pupils in Class 4 (1).

Housework / Number of people	Wash socks	Wash dishes	Wipe tables	Sweep
	2	4	16	18

Complete the bar graph below. Say how many pupils 1 square stands for.

Graph of housework done by Class 4 (1)



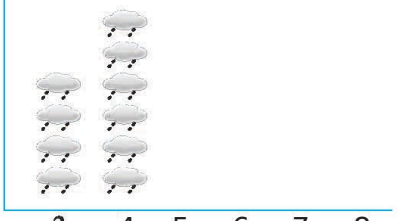
2. The table below shows the average monthly rainfall in a city in 2012, from month 3 to month 8.

Month	3	4	5	6	7	8
Rainfall/mm	40	60	150	110	80	130

Choose a way you like and make a graph from the data in the table above.

Average monthly rainfall, March to August

Rainfall/mm



(1) In the graph on the left, how many millimetres of rainfall does 1 rain cloud stand for?

(2) Complete the graph on the left.

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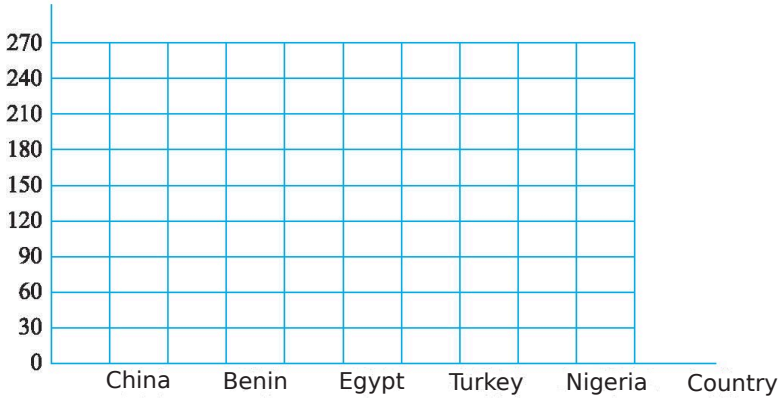
3. Complete the bar graph, then discuss with a partner. What maths questions can you ask?

Table of cotton growing periods

Country	China	Benin	Egypt	Turkey	Nigeria
Growing days	180	120	210	150	240

Graph of cotton growing periods

Growing period/days



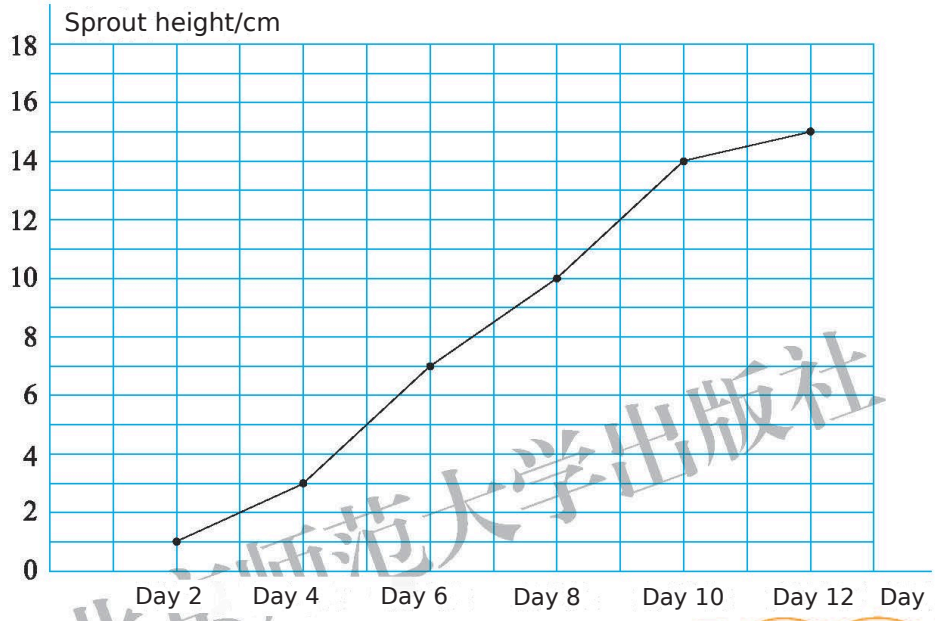
4. What things in life can be shown by data? Work in groups, choose a topic to survey, and make a graph of the data in Figure 1 on page 4 of the appendix.

Garlic Sprouts (2)

Growth of Xiaoxiao's garlic sprouts

Day	Day 2	Day 4	Day 6	Day 8	Day 10	Day 12
Sprout height/cm	1	3	7	10	14	15

Look at the graph below. Talk about how it was made.



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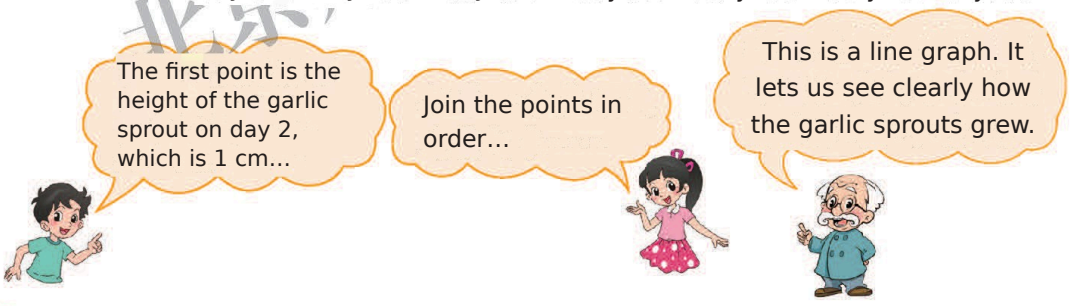


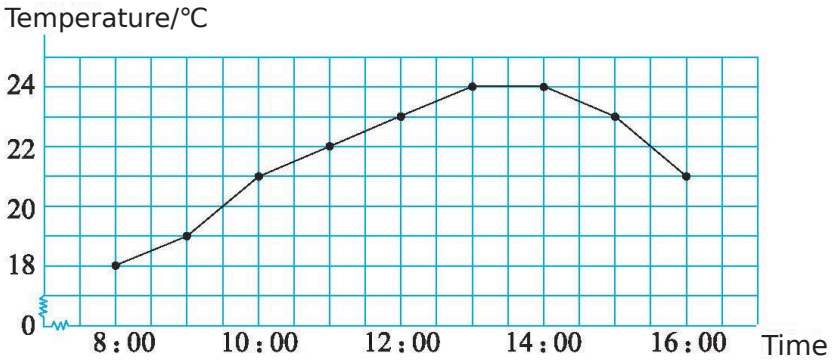
Look at the line graph. Describe how the sprouts grew overall over these days.

Think: during which period did the garlic sprouts grow fastest?

Based on the growth of your own garlic sprouts, draw a line graph on the grid paper in Figure 2 on page 4 of the appendix, and talk it over with your partner.

Practice

1. Xiaoxiao drew the graph below to show how the indoor temperature changed.



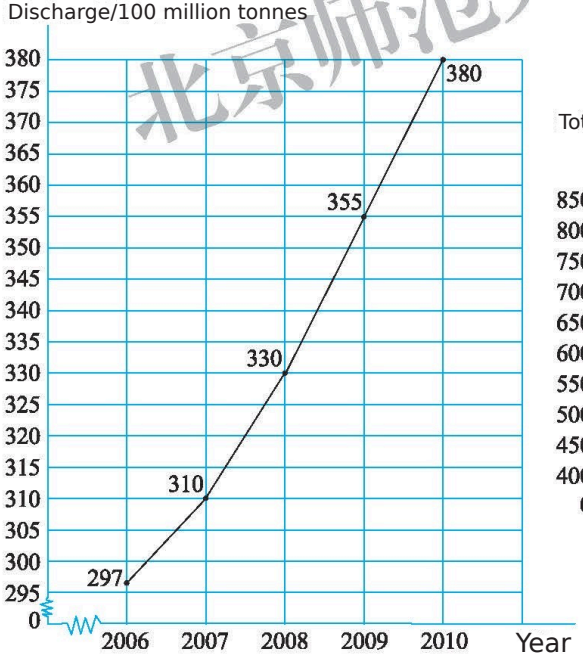
(1) Xiaoxiao measured the temperature every ___ hours.

(2) How did the temperature change overall on this day from 8:00 to 16:00?

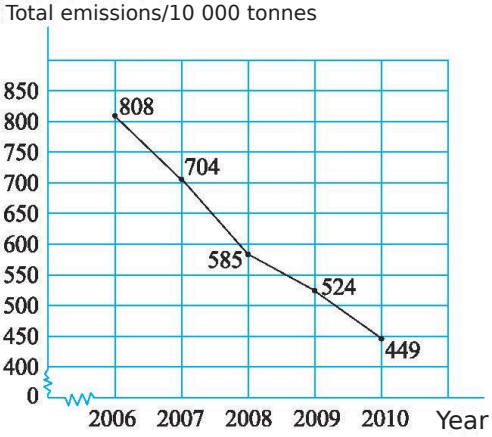
(3) Ask another maths question and try to answer it.

2. Environmental survey.

Total urban domestic sewage discharge in China, 2006~2010



Change in total industrial dust emissions in China, 2006~2010



Are you satisfied with the environment today? Give your reasons and try to make some suggestions.

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Mean

10 digits are shown for 3 seconds. How many can you remember each time?

Digits Taoqi remembered in 5 trials

Trial	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
Digits remembered	5	4	7	5	9

How many digits can Taoqi remember?

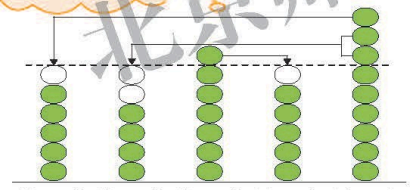
It's different every time. How should we answer?

5 appeared 2 times, so let's say 5 digits.

Taoqi remembers 6 digits on average each time.

How do we get an average of 6 digits each time?

Move some from the tall ones to the short ones.


Trial 1 Trial 2 Trial 3 Trial 4 Trial 5

You can calculate it.

The mean number of digits remembered each time is:

$(5+4+7+5+9)\div 5=6$ (digits)

Can you help solve Clever Dog's problem?

Taoqi never remembered 6 digits in any trial! What's going on?

Evening out the trials gives "6 digits".

The mean represents the average level of a set of data.

Talk about where you have seen the mean in everyday life.

The mean height of our class...

The mean temperature this month...

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Try It

Below is a table of tickets sold by the science museum in one week.

Time	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Tickets sold	700	640	910	990	1300	[redacted]	[redacted]

(1) Estimate about how many tickets were sold per day on average in the first 5 days.

It must be more than 640 and less than 1300.

About 900 tickets.

(2) 1700 tickets were sold on Saturday and 1460 on Sunday. How does the mean number of tickets sold this week change compared with the mean of the first 5 days?

Little Bear's cold drinks shop needs to order more ice lollies, so Little Bear looked at the record of ice lollies sold in the first 3 weeks of this month.

Practice

1. In a 1-minute basketball shooting contest, Qisi made 7, 7, 6 and 8 baskets in 4 rounds. What number can represent the number of baskets Qisi made?

2. The table shows one week's temperatures in a certain place. Work out the mean of this week's highest temperatures and the mean of its lowest temperatures.

Time	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mean
Lowest/°C	8	10	11	12	11	12	13	
Highest/°C	18	18	20	22	24	20	25	

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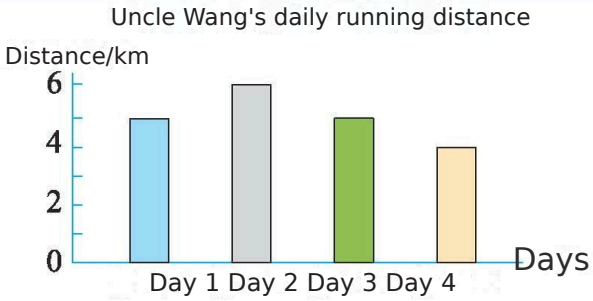
3. Uncle Wang runs along a circular track every day and records the distance he runs.



If Uncle Wang did not run on day 5, how does the mean daily distance of the first 5 days change compared with the mean daily distance of the first 4 days? Calculate it.

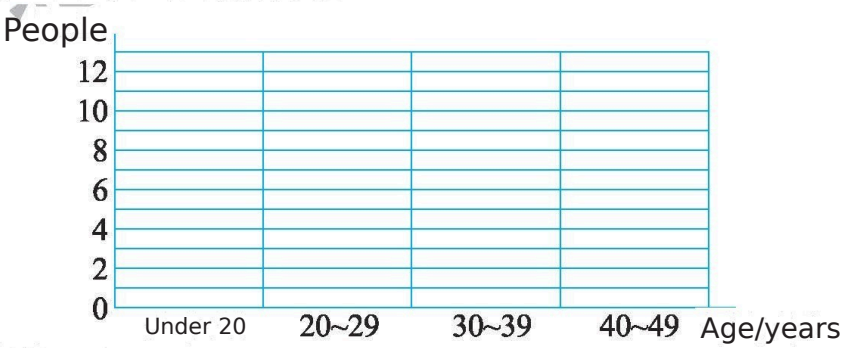
4. Below are the ages of some Red Army veterans on the Long March (1934).

Name	Age/years	Name	Age/years	Name	Age/years	Name	Age/years
Mao Zedong	41	Luo Ruiqing	28	Peng Dehuai	36	Yang Shangkun	27
Wang Jiaxiang	28	Nie Rongzhen	35	Zhang Wentian	34	Wang Shoudao	28
Yang Dezhi	23	Zhu De	48	Song Renqiong	25	Chen Yun	29
Zhou Enlai	36	Hu Yaobang	19	Liu Bocheng	42	Liu Shaoqi	36
Yang Yong	21	Ye Jianying	37	Deng Xiaoping	30	Lu Dingyi	28

(1) Use 正 marks to tally the people in each age group in the table above.

Age/years	Under 20	20~29	30~39	40~49
Tally				
People				

(2) Draw a graph from the table.



(3) Answer the questions.

① The oldest of these Red Army veterans was ____, aged ___ at the time.

___ people were under 30. Fraction of the total: $\frac{(\quad)}{(\quad)}$

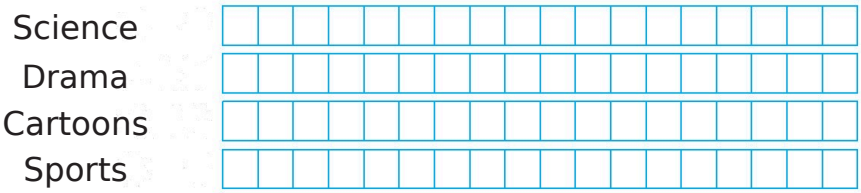
② Without calculating, can you estimate roughly what the mean age of these Red Army veterans was back then?

Exercise 6

1. A TV station plans to show pupils' favourite programmes during the holidays. The table below shows the results of Qisi's survey of 200 pupils' favourite TV programmes.

Programme	Science	Drama	Cartoons	Sports
Pupils	35	10	90	65

(1) Colour in the squares using the table. How many people does 1 square stand for?



(2) What can you find out from the graph above?

2. Below is the score table of the "New Seedling Cup" children's singing contest.

Judge / Singer	Ms Chen	Ms Wang	Ms Li	Mean score
No. 1	89	85	96	
No. 2	91	96	86	
No. 3	92	95	92	

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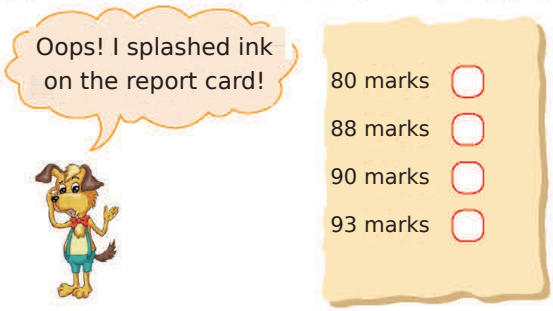
(1) How do you understand the mean score? Share your ideas with your partner.

(2) Please complete this score table.

(3) Rank the three contestants and share your method with your partner.

3. Can you choose the right answer without calculating? (Tick "√") Lele's mean mark in maths, Chinese and English is 90. What is her English mark?

Subject	Maths	Chinese	English
Mark	89	88	



4. Below are the standing long jump test results of the boys in Class 4(1).

m

No.	Result	No.	Result	No.	Result	No.	Result	No.	Result
1	1.58	6	1.57	11	1.62	16	1.62	21	1.74
2	1.50	7	1.58	12	1.28	17	1.56	22	1.54
3	1.56	8	1.47	13	1.62	18	1.67	23	1.46
4	1.48	9	1.35	14	1.59	19	1.77	24	1.35
5	1.62	10	1.75	15	1.48	20	1.20	25	1.65

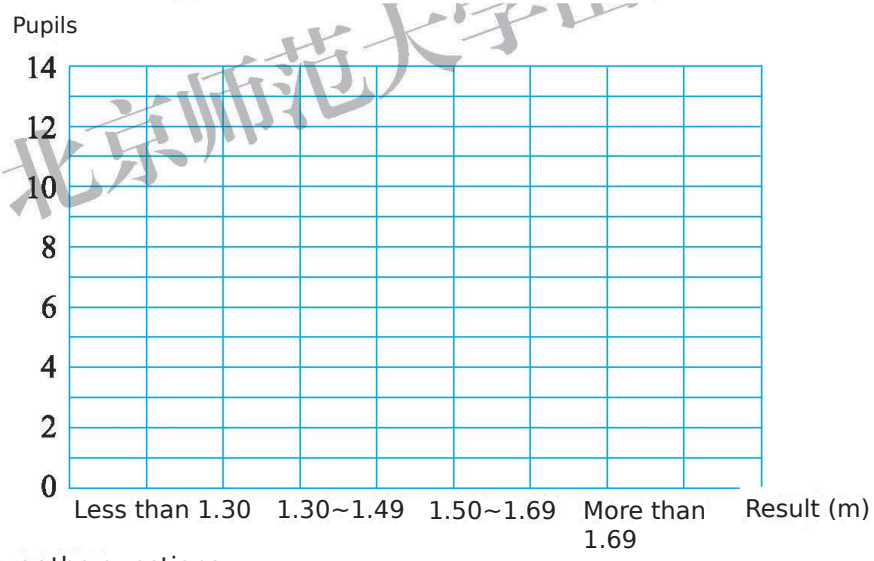
(1) Use the table and tally marks to count the boys in each range.

Standing Long Jump Results of Boys in Class 4(1)

Result (m)	Less than 1.30	1.30~1.49	1.50~1.69	More than 1.69
Tally				
Pupils				

(2) Draw a graph from the table.

Standing Long Jump Results of Boys in Class 4(1)



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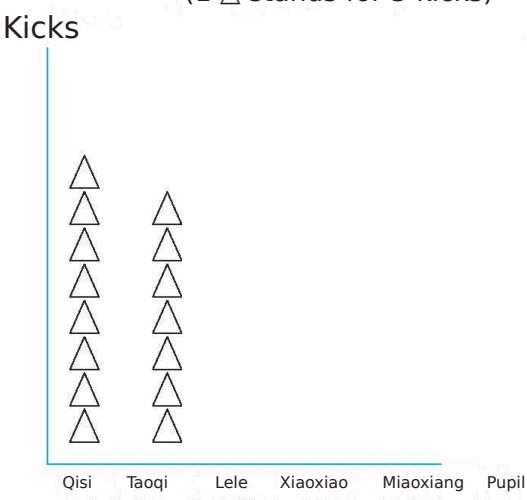
(3) Answer the questions.

① The best jumper among the boys of Class 4(1) is No. _____ who jumped _____ m.

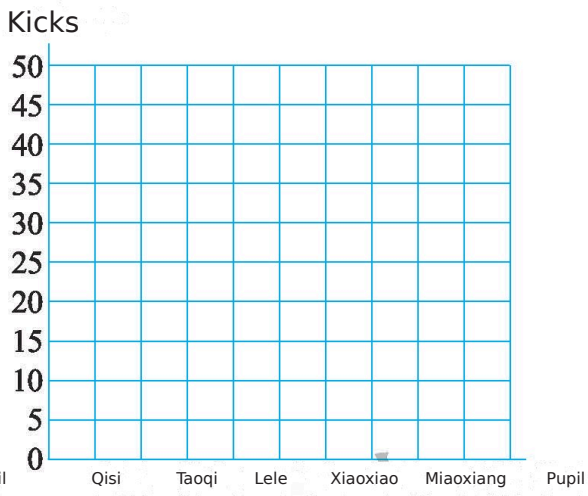
② The pass mark for the standing long jump is 1.30 m. Boys in Class 4(1) who passed: _____. Fraction of all the boys: $\frac{(\quad)}{(\quad)}$

5. Qisi, Taoqi, Lele, Xiaoxiao and Miaoxiang kicked the shuttlecock 40, 35, 25, 50 and 45 times. Draw a chart to show their results.

Shuttlecock Kicking Chart
(1 △ stands for 5 kicks)



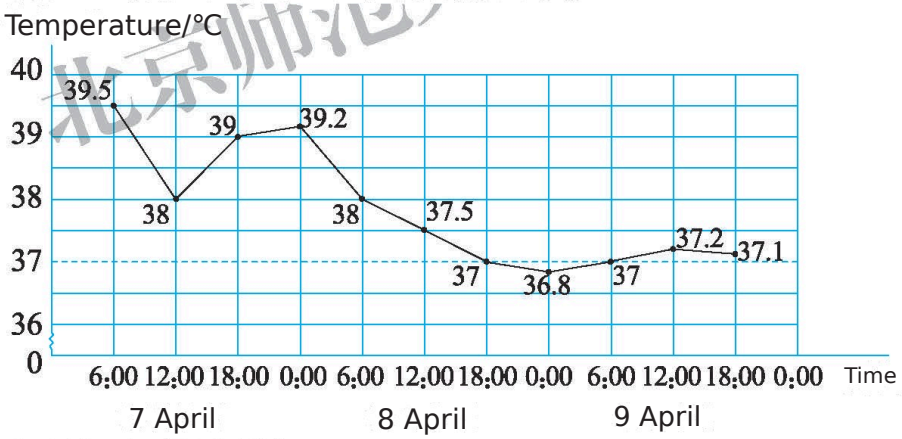
Shuttlecock Kicking Chart



(1) Explain how you drew the bar graph.

(2) What information can you get from the chart? Discuss with your partner.

6. Below is a line graph of a patient's temperature record.



Look at the graph and answer the questions below.

(1) How often did the nurse measure the patient's temperature?

(2) What was the patient's temperature in °C at 12:00 on 8 April?

(3) When did the patient's temperature fall fastest? When was it fairly steady?

(4) What does the horizontal dashed line on the graph show?

(5) Judging from the temperature, how did the patient's condition change?

Final Review

Review and Share

Number and Algebra

On your own

1. Use examples to explain how decimals, whole numbers and fractions are related.

2. Find decimals in everyday life and explain their meaning.

3. Calculate, then explain how you did it.

3.5×1.2 $(7.5+2.5) \times 0.25$ $2.25+1.25 \times 0.18$

4. Collect the questions you often get wrong and say what to watch out for in decimal calculations.

5. Can you explain why they calculated it this way?

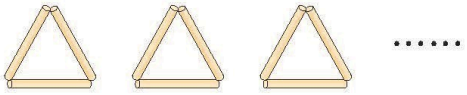
Factor	36	3.6	36	3.6	0.36
Factor	24	2.4	2.4	24	2.4
Product	864	8.64	86.4	86.4	0.864

6 - 2.4 = 3.6

6.0
− 2.4
3.6

2.8 × 5 + 7.2 × 5
= (2.8 + 7.2) × 5
= 10 × 5
= 50

6. Taoqi made the pattern on the right with sticks.



(1) How many sticks do n triangles need?

(2) What patterns in everyday life can be shown by 3n?

7. Use letters to write two formulas or rules, and say why using letters for numbers is helpful.

8. Give examples of how you set up and solve equations.

Share ideas

6 dm can be written as the fraction $\frac{6}{10}$ m or as the decimal 0.6 m. In whole numbers and decimals, each place is worth 10 times the next place to its right.

36 × 24 = 864

10 times as large

10 times as large

1/100 as large

3.6 × 2.4 = 8.64

To write an equation, it is important to find what is equal. To solve an equation, we can use...

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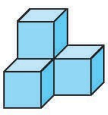
Shapes and Geometry

On your own

1. Which shapes have we learned? Can you organise them?

2. Describe the features of each 2D shape you have learned, how these features were found, and give examples of where they are used in everyday life.

3. Match the 3D shape below to its views from the front, top and right.



Front

Top

Right

Share ideas

The sum of the interior angles of a triangle is 180°. I measured ....

Organise the shapes we have learned by sorting them...

Using the stability of triangles...

Statistics and Probability

On your own

1. Give an example of how data is used to solve a problem.

2. Choose the bar chart and the line chart you made that you like best,

and say how each was made and what information you can get from it.

3. Use an example to explain the mean. How is the mean used in everyday life?

Share ideas

To observe how garlic sprouts grow, I first did an experiment and collected data, then showed the data in a line chart. From the chart I can tell...

The mean is the average of a set of bigger and smaller numbers, found by taking from the bigger to top up the smaller...

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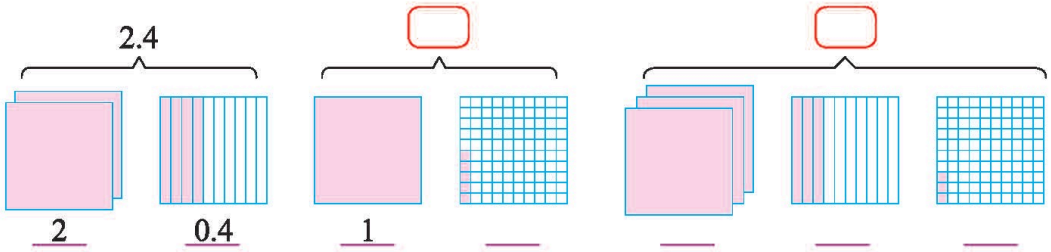
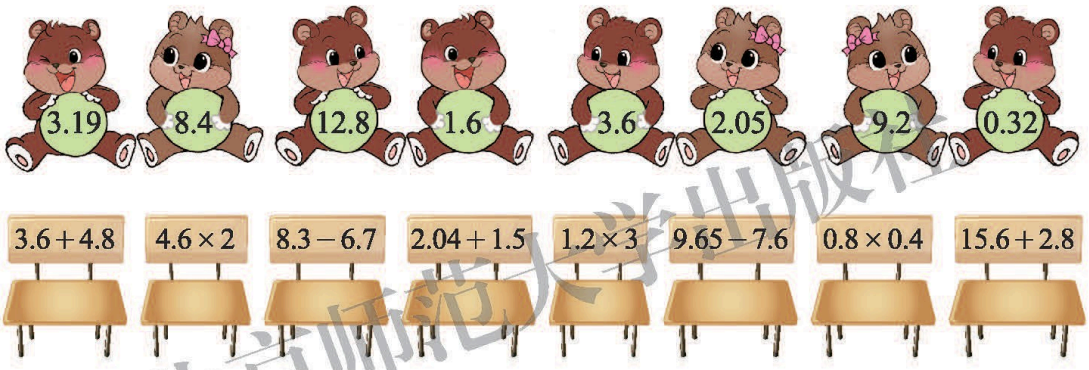
Practice

Number and Algebra

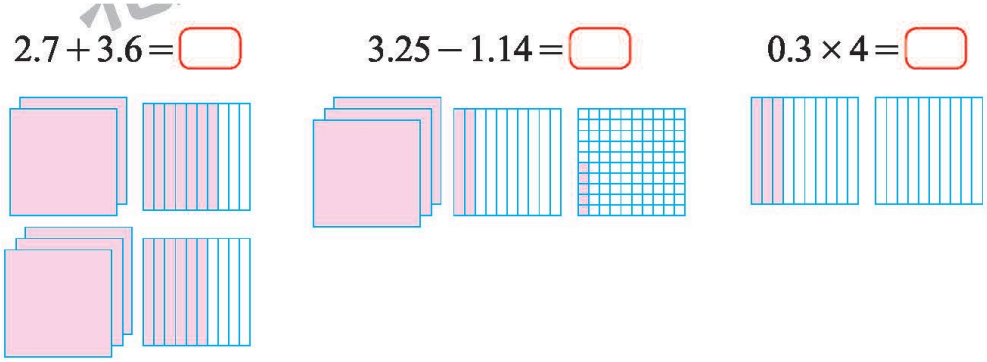
1. Use numbers to show the coloured parts below.



2. Find each bear's seat.



3. Draw, then calculate.



4. The class committee bought 25 prizes at ¥1.8 each and 15 prizes at ¥2.5 each.

(1) How much did the two kinds of prizes cost altogether?

(2) They took ¥100. How much change should they get?

(3) Is the money left enough to buy 20 more prizes at ¥0.8 each?

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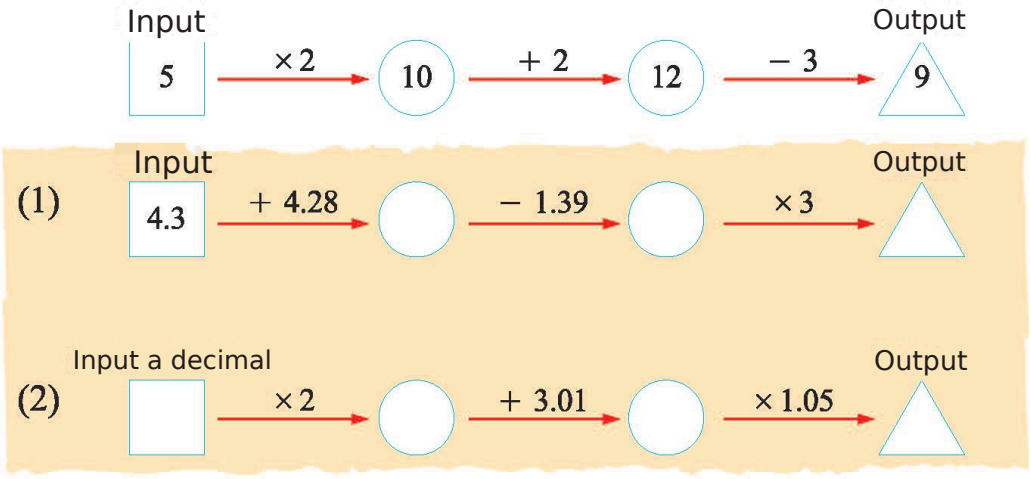
5. A number machine takes a number as input and does calculations on it. As shown below, when 5 is input, the output is 9. Can you complete the calculations below?



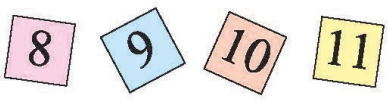
6. A ribbon is cut into 1.4 dm pieces, and each piece makes 1 bow. 27 bows are made in all, with 1.2 dm left over. How many metres long was the ribbon?

7. A butterfly flies 7.8 km in 1 hour. A bee flies 2.4 times as fast as this butterfly. How many kilometres does the bee fly per hour?

8.

$7.8+4.56+2.2$	$0.65\times6.4-0.65\times5.4$
$1.3+4.7\times0.9$	$5.01-1.9+4.99$
$0.4\times99+0.4$	$7.5-7.5\times0.8$

9. Which numbers can go in the brackets?



$15.4\times(\quad)>150$

10. Make up a problem for each number sentence below, and solve it.

$9.6-1.64$ $\quad 8\times1.57$ $\quad 62.4+7.62$ $\quad 6.7\times2.4$

11. Fill in.

(1) Lele had 5 storybooks and bought a more. Now he has ( ) books.

(2) Grade 4 has x pupils. Grade 3 has 15 fewer, so it has ( ) pupils.

(3) A lorry carries n tonnes per trip. In 6 trips it carries ( ) tonnes.

(4) 16 cartons of milk cost ¥y. Each carton costs ¥( ) on average.

12. Solve.

$4x=36$　　$2.5+x=10$

$x-18=40$　　$3x-3.9=5.1$

$2y+7=91$　　$8m\div 2=24$

13. Basketball shooting contest.

How many points did Xiaoxiao and Taoqi each score?

14. 1 pen costs ¥0.6 more than 6 pencils. Each pencil costs ¥0.85. How much does a pen cost?

15. I can guess the number you are thinking of! Try it and work it out.

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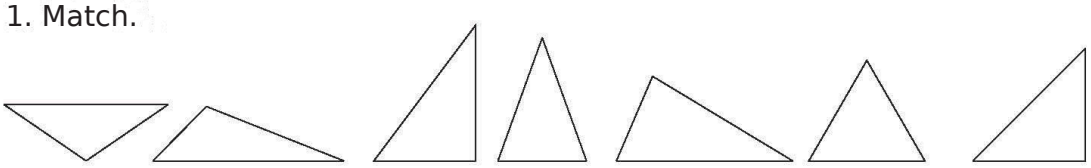
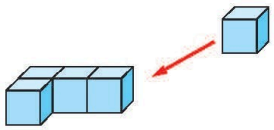
Shapes and Geometry

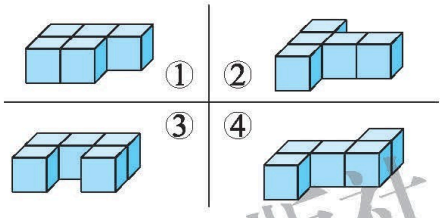
1. Match.



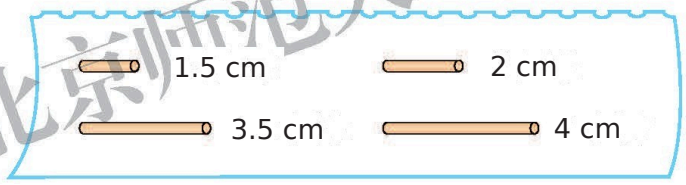
obtuse | right-angled | acute | equilateral | isosceles

2. Choose. Add one more cube to the figure below. In which arrangements do the views from the front and the left stay the same?

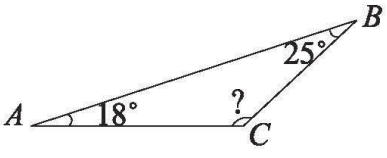




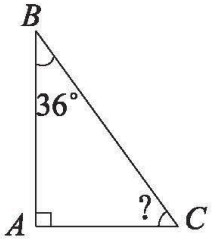
3. Choose 3 of the 4 sticks below to make a triangle. Its side lengths could be _____, _____, _____. (Give one possible answer.)



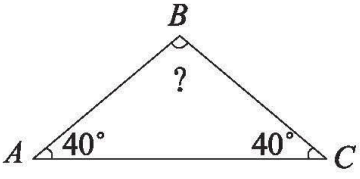
4. Find the size of each angle.



$\angle C =$ _____

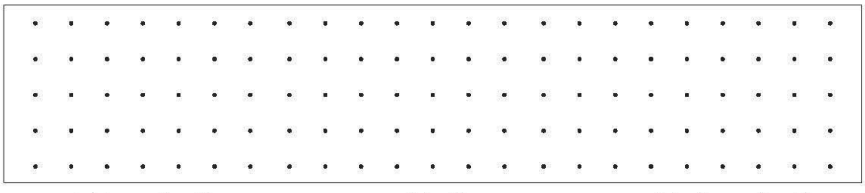


$\angle C =$ _____



$\angle B =$ _____

5. Draw the figures on the dot grid as required.



parallelogram | trapezium | obtuse triangle

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Statistics and Probability

1. Below is Taoqi and Xiaoxiao's record of the vehicles that passed the school gate in 10 minutes.

car	正正正正正正
bicycle	正正正正正正正正正正
bus	正正
truck	正正正正

Show their results in a bar graph, and talk with a partner: what did you find?

2. Below is a record of the waste newspaper collected by Class 4 (1).

Group	Group 1	Group 2	Group 3	Group 4	Group 5	Group 6
Mass/kg	25	28	30	18	24	25

(1) The whole class collected ___ kg of waste newspaper in total.

(2) On average, each group collected ___ kg.

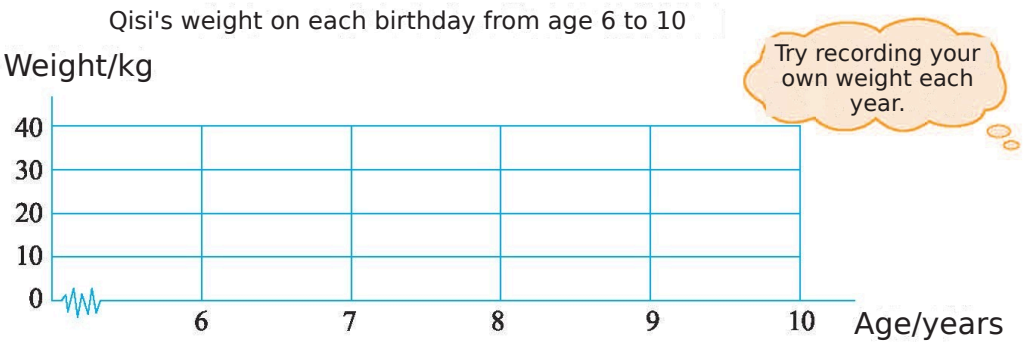
(3) At 6 jiao per kg, the newspaper collected is worth ¥___ in total.

(4) What other maths questions can you ask?

3. The table below shows Qisi's weight on each birthday from age 6 to 10.

Age/years	6	7	8	9	10
Weight/kg	17	19	22	25	30

Use the data in the table to complete the line graph below, and talk with a partner: what did you find?



What Have You Learned This Term?

1. _____

2. _____

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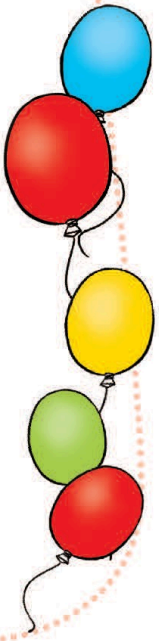
3. _____



Question Bank

What maths problems have you found in your daily life and studies? Write them down. Can you solve them?

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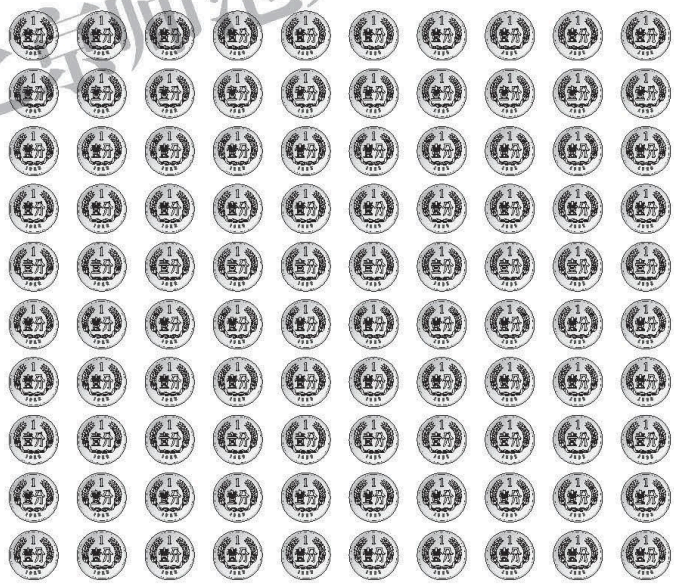
Appendix 1



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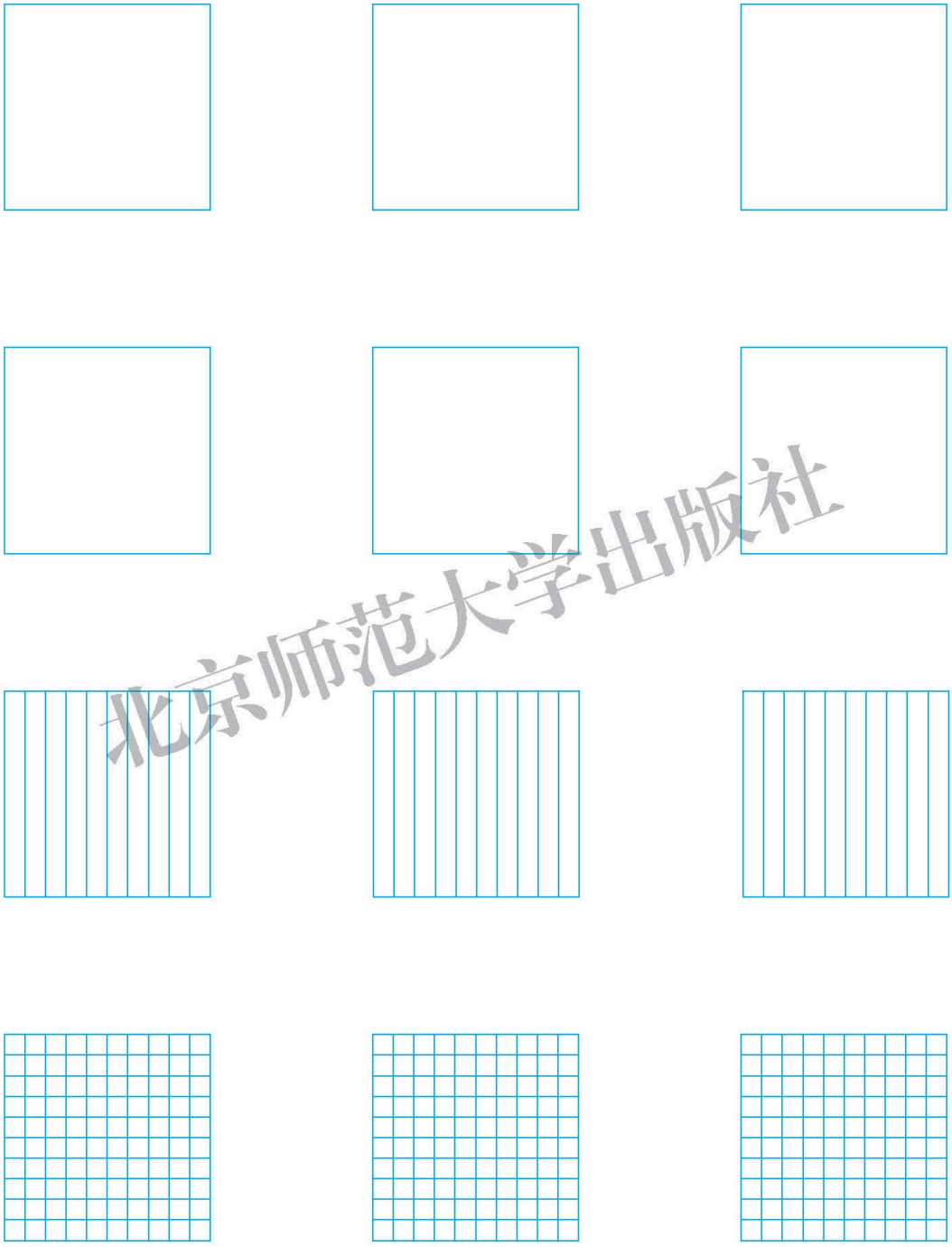
Appendix 2



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Appendix 3

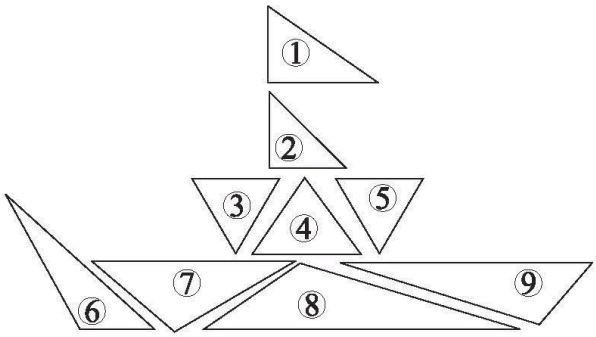

Figure 1

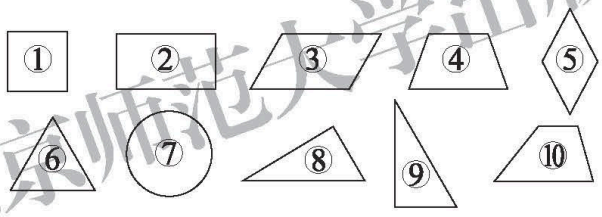

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Figure 2

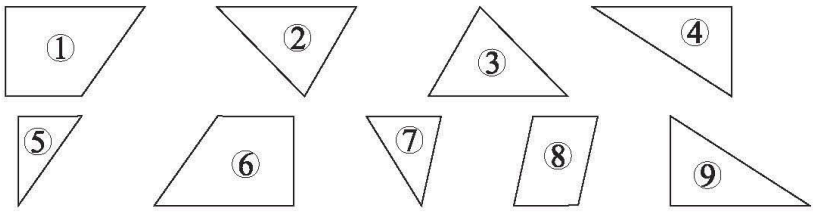

Figure 3

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Appendix 4

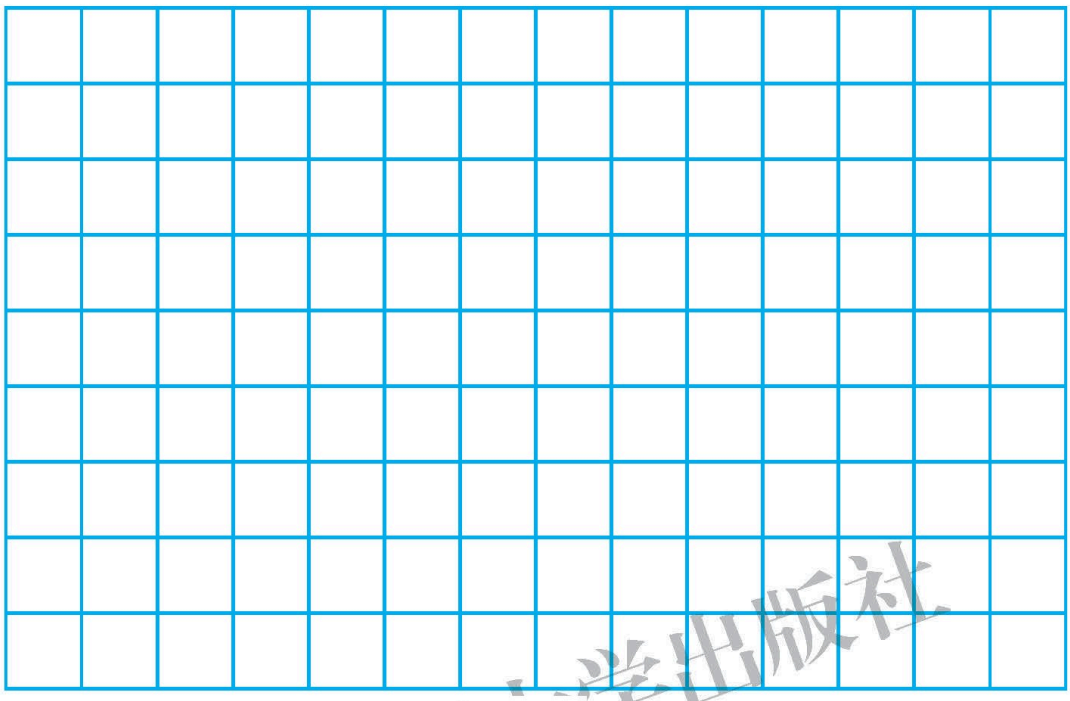

Figure 1

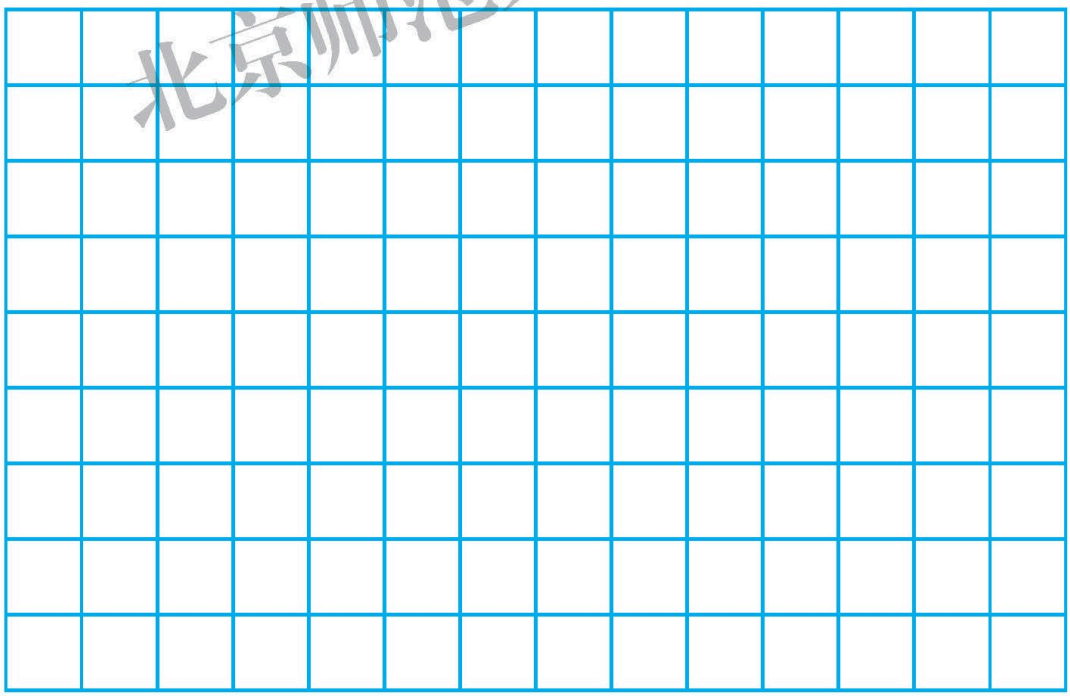

Figure 2

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Afterword

The Compulsory Education Textbooks (Beijing Normal University Edition) were written by many heads and core members of the groups that drew up the national curriculum standards for basic education, together with subject experts, education experts, psychologists and master teachers. They rest on solid research and advanced educational ideas, and offer high-quality writing and professional service. The textbooks strive to reflect the spirit of the national curriculum standards for basic education, make use of many kinds of information resources, show the latest progress in each subject where appropriate, and stress the use of knowledge, skills and ways of thinking in real life. They stay close to students' lives, pay attention to how students learn, meet their different learning needs and promote the all-round development of every student.

The Compulsory Education Textbooks (Beijing Normal University Edition) · Mathematics (Grades 1–6), also known as the New Century Primary School Mathematics Textbooks, fully embody the basic ideas of the new century's reform of the mathematics curriculum, with the healthy growth of children as their fundamental aim. This is the 4th edition of the New Century Primary School Mathematics Textbooks. Building on the strengths of the first three editions, it has been fully revised in accordance with the Mathematics Curriculum Standards for Compulsory Education (2011 Edition).

The new edition has the following distinctive features. Its basic form of presentation is "a situation plus a series of questions", which seeks to make the unfolding of the content, the students' learning, the teachers' teaching and the reaching of the curriculum goals one whole, good for learning and for teaching, so that students keep going through the process of thinking a problem through "from beginning to end". It seeks to give students realistic, interesting and challenging learning materials, to give them time and space to explore and to share their ideas, to show how mathematical knowledge is formed and used, and to meet the needs of different students.

The aim is for students to: appreciate the close links between mathematics, nature and human society; gain the basic knowledge, basic skills, basic ideas and basic experience of activities that they need and that suit their age; develop the ability to find and pose problems and to analyse and solve them; understand the value of mathematics, become more interested in learning it and more confident of learning it well; and gain a first sense of innovation and a scientific attitude.

The chief editors of the New Century Primary School Mathematics Textbooks are Liu Jian, Kong Qiping and Zhang Dan. The other members of the writing group are Hu Guangti, Liu Keqin, Wang Yong, Tao Wenzhong, Wang Mingming, Zhu Dejiang, Chen Xiaomei, Zhang Hong, Qian Shouwang, He Fengbo, Hua Yinglong, Huang Lihua, Zhao Yanhui, Li Tie'an, Wei Huinü, Zhu Yuhong, Ren Jingye and Hou Huiying.

The chief editors of this volume are Hua Yinglong, Huang Lihua and Zhao Yanhui. The main people who took part in revising and finalising it were Zhu Dejiang, Chen Xiaomei, Hou Huiying, Wang Mingming and Hu Qinzhu; the final text was compiled and approved by Liu Jian and Zhang Dan. Many local teaching researchers and front-line teachers gave valuable advice on this new round of revision, and we thank them all.

We hope that teachers and students will send us their valuable comments as they use the book, so that we can revise and improve it further.

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Beijing Normal University Press